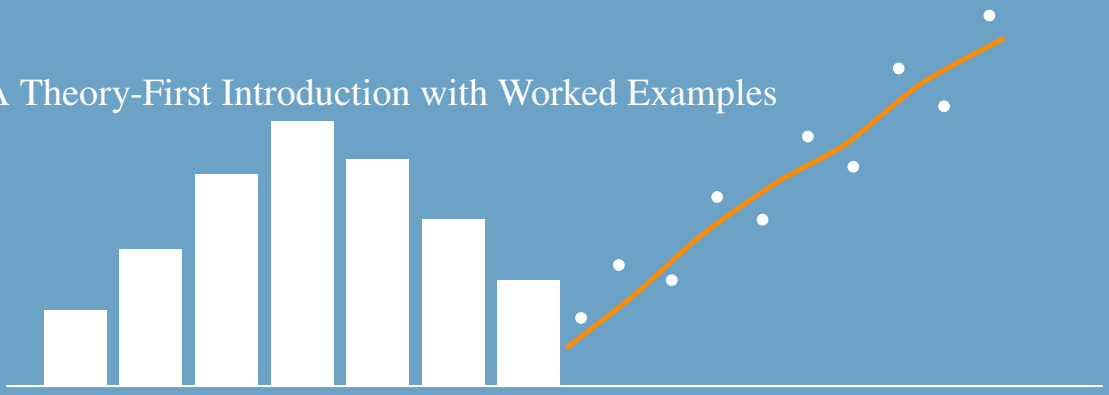
Statistics and Data Analysis for Engineers

A Theory-First Introduction with Worked Examples



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Part I

Data and Probability Models

Chapter 1 Descriptive Statistics

In software engineering, we constantly deal with data: performance metrics, user behavior patterns, system response times, and code complexity measures. To make sense of this information and extract meaningful insights, we need tools to summarize, organize, and visualize data. Descriptive statistics provides these essential tools, allowing us to understand the characteristics of our data without making inferences about larger populations.

Consider a software development team analyzing the performance of their web application. They collect response times for 1000 user requests and find values ranging from 50 milliseconds to 3.2 seconds. Without proper analysis, this raw data is overwhelming and uninformative. However, by applying descriptive statistics, they can determine that the average response time is 450 milliseconds, that 75% of requests complete within 600 milliseconds, and that there are a few unusually slow requests that might indicate performance issues.

This chapter introduces the fundamental concepts of descriptive statistics, focusing on measures that help us understand the central tendencies, variability, and distribution patterns in our data. We will explore how these techniques apply specifically to software engineering contexts, from analyzing algorithm performance to understanding user behavior patterns.

In practical data analysis, statistical software such as Python is normally used to construct frequency tables and graphs and to evaluate numerical summaries for realistic datasets. The definitions and formulas in this chapter remain essential because they specify what is being calculated and determine how the result should be interpreted. Software performs the arithmetic and drawing; it does not decide which summary or graph is appropriate for the data and the engineering question.

1.1 Introduction to Descriptive Statistics

Sometimes statistical work begins with existing data, such as precipitation records, unemployment rates, or GDP figures, which we then summarize and analyze. In other situations, data must be generated through an experiment or study. For example, to compare two teaching methods in an introductory programming course, an instructor might randomly divide students into two groups, apply a different method to each, and then compare test scores. Random assignment is essential: it ensures that differences between groups are not due to preexisting factors such as aptitude, but rather to the teaching method itself.

After the data are collected, they are summarized and visualized—for example, by reporting the average score for each group. This process is the essence of **descriptive statistics**: organising, presenting, and describing data in a meaningful way.

Population versus Sample

In statistical analysis, we distinguish between two key concepts:

Definition 1.1 (Population)

A **population** is the complete set of all possible observations or measurements of interest in a particular study. In software engineering contexts, this might include all possible execution times of an algorithm, all user sessions on a website, or all lines of code in a project.

Definition 1.2 (Sample)

A **sample** is a subset of the population that we actually observe or measure. Due to practical constraints, we often work with samples rather than entire populations.

In the teaching-method example, the students in the classroom form only a *sample* of the larger *population* of potential learners. Descriptive statistics helps us understand the sample itself, while inferential methods later allow us to use that sample to draw conclusions about the broader population. Inferential statistics lies beyond the scope of this chapter and is developed later in this book. We focus here on descriptive statistics.

Example 1.1 Software Performance Analysis

Consider a web application serving millions of users daily. The **population** would be the response times for all possible user requests. However, due to computational and storage limitations, we might only collect response times for a **sample** of 10,000 requests per day. This sample should be representative of the population to draw meaningful conclusions.

Types of Data

Data can be classified into different types, each requiring different statistical approaches:

Definition 1.3 (Qualitative Data)

Qualitative data (also called categorical data) consists of non-numerical information that can be categorized. Examples include programming languages used in a project, user satisfaction ratings (satisfied/neutral/dissatisfied), or bug severity levels (critical/high/medium/low).

The essential difference between **qualitative** and **quantitative** data lies in what they represent and how they are analyzed. Qualitative data refers to categories or attributes that cannot be expressed meaningfully as numbers. Such data are used to classify or label elements, for example, programming language or bug severity. Quantitative data, on the other hand, consists of numerical values that represent counts or measurements, such as response times or number of users. Because these values are numerical, they permit mathematical operations and statistical analysis. Recognizing the distinction between qualitative and quantitative data is fundamental, as it guides the choice of statistical methods and the types of visualizations that are appropriate for analysis.

Definition 1.4 (Quantitative Data)

Quantitative data consists of numerical measurements that can be ordered and subjected to mathematical operations. This can be further divided into:

- **Discrete**: Countable values (number of bugs, lines of code, user sessions)
- **Continuous**: Measurable values that can take any value within a range (response times, memory usage, CPU utilization)

Example 1.2

- **Qualitative**: Programming language (Java, C++, Rust), deployment environment (development, staging, production)
- **Quantitative Discrete**: Number of commits per day, lines of code, number of test cases
- **Quantitative Continuous**: Response time in milliseconds, memory usage in MB, CPU utilization percentage

Outliers and Extreme Data

In any dataset, most values tend to cluster around a central region, but occasionally, we encounter values that are much higher or lower than the rest. These are called **outliers** or **extreme values**. Outliers can arise for many reasons: measurement errors, unusual but valid events, or natural variability in the data.

For example, if most web requests complete in under 400 milliseconds, but one request takes 1200 milliseconds, that 1200ms value is an outlier. Outliers can have a strong influence on summary statistics like the mean, making the data appear more variable or shifting the average away from where most values lie.

It's important to look for outliers when analyzing data, as they can signal interesting phenomena (such as a rare performance bottleneck), data entry mistakes, or the need for further investigation. While there are formal methods to detect outliers, which we will discuss later, often a simple plot or a scan of the sorted data is enough to spot values that "stand out" from the rest.

In practice, understanding the context is key: sometimes outliers are errors to be corrected or removed, but other times they are the most important part of the story.

1.2 Describing Data Sets

When presenting numerical results, clarity and brevity are essential. Tables and graphs are particularly effective, as they allow the reader to quickly grasp the main characteristics of a dataset. Visual and tabular summaries often highlight important aspects such as the overall range, how tightly the values are concentrated, and whether the data appear symmetric or skewed.

Frequency Tables and Graphs

The first step in data analysis is often to organize raw data into a frequency distribution, which shows how often each value (or range of values) occurs.

Definition 1.5 (Frequency Distribution)

A **frequency distribution** is a table that shows the frequency (count) of each value or class of values in a dataset. It can be presented as:

- **Absolute frequency**: The actual count of occurrences
- **Relative frequency**: The proportion of total observations
- **Cumulative frequency**: The running total of frequencies



Example 1.3 A dataset with a relatively small number of distinct values can be conveniently presented in a

frequency table. For instance, consider a dataset consisting of the starting monthly salaries (to the nearest thousand Danish Kroner) of 42 recently graduated students with B.S. degrees in software engineering.

The raw data is as follows:

$$37, 37, 37, 37, 38, 39, 39, 39, 40, 40, 40, 40, 40, 41, 41, 41, 41, 41, 41, 41, 41,$$
$$42, 42, 42, 42, 42, 42, 42, 42, 42, 42, 44, 44, 44, 44, 44, 46, 46, 47, 47, 47, 50$$

From this data, we can construct a frequency table. Table 1.1 tells us, among other things, that the lowest starting salary of 37,000 DKK was received by four of the graduates, whereas the highest salary of 50,000 DKK was received by a single student. The most common starting salary was 42,000 DKK, received by 10 of the students.

Salary ($\times 1000$ **DKK**)	**Frequency**
37	4
38	1
39	3
40	5
41	8
42	10
44	5
46	2
47	3
50	1

Table 1.1: Frequency Distribution of Monthly Starting Salaries

Data from a frequency table can be graphically represented by a line graph that plots the distinct data values on the horizontal axis and indicates their frequencies by the heights of vertical lines. A line graph of the data presented in Table 1.1 is shown in Figure 1.1.

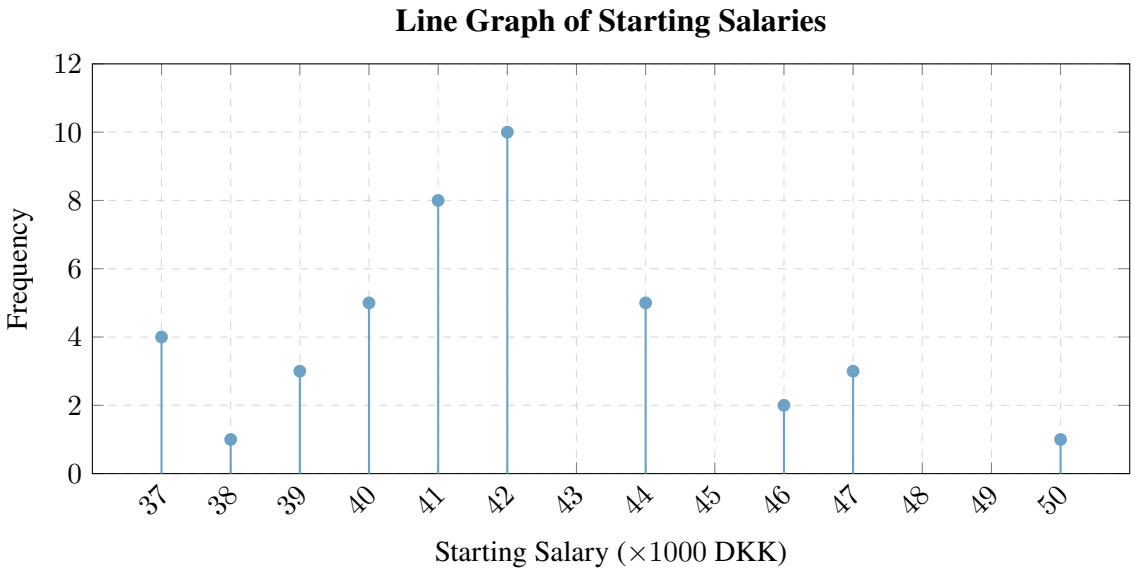


Figure 1.1: A line graph showing the frequency of different starting salaries.

When the lines in a line graph are given added thickness, the graph is called a bar graph. Figure 1.2 shows a bar graph for the same salary data.

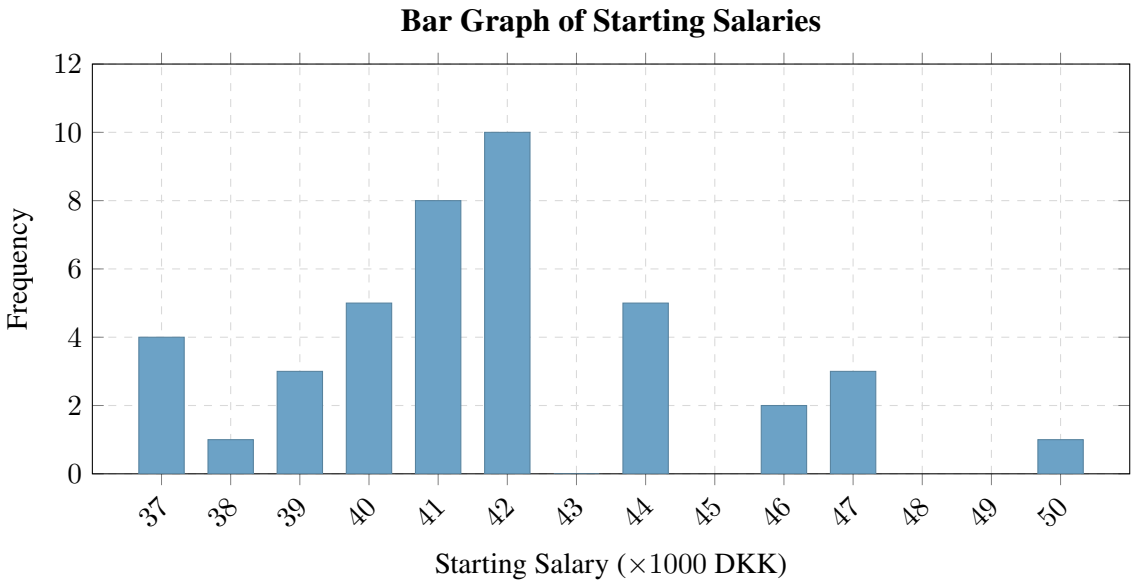


Figure 1.2: A bar graph of the salary data.

Another type of graph used to represent a frequency table is the frequency polygon, which plots the frequencies of the different data values on the vertical axis, and then connects the plotted points with straight lines. Figure 1.3 shows a frequency polygon for the same salary data.

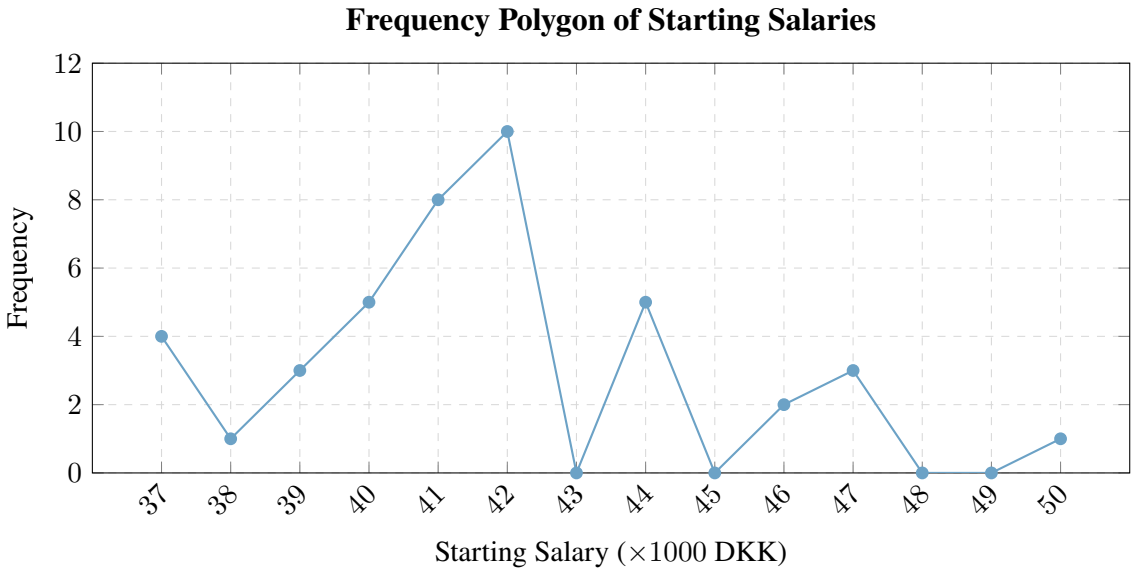


Figure 1.3: A frequency polygon of the salary data, connecting the points from the frequency table.

Sometimes we prefer the relative frequency over the absolute frequency. This is particularly useful when dealing with large datasets and when you want to compare values from datasets of different sizes. An extension of the relative frequency is the cumulative relative frequency, which is the running total of the relative frequencies. Table 1.2 shows the frequency, relative frequency, and cumulative relative frequency for the salary data.

Salary	Frequency	Relative Freq.	Cumulative Rel. Freq.
37	4	$4/42 \approx 0.095$	$4/42 \approx 0.095$
38	1	$1/42 \approx 0.024$	$5/42 \approx 0.119$
39	3	$3/42 \approx 0.071$	$8/42 \approx 0.190$
40	5	$5/42 \approx 0.119$	$13/42 \approx 0.310$
41	8	$8/42 \approx 0.190$	$21/42 = 0.500$
42	10	$10/42 \approx 0.238$	$31/42 \approx 0.738$
44	5	$5/42 \approx 0.119$	$36/42 \approx 0.857$
46	2	$2/42 \approx 0.048$	$38/42 \approx 0.905$
47	3	$3/42 \approx 0.071$	$41/42 \approx 0.976$
50	1	$1/42 \approx 0.024$	$42/42 = 1.000$

Table 1.2: Frequency, Relative Frequency, and Cumulative Relative Frequency of Monthly Starting Salaries.

Grouped Data and Histograms

As we saw in section 1.2, line and bar graphs are useful for showing frequencies of data values. When a dataset contains many distinct values, such as the lamp lifetimes in Table 1.3, creating a frequency table for each individual value becomes impractical. A more effective approach is to group the data into a set of **class intervals**, which are also commonly referred to as **bins**.

Item Lifetimes (in Hours)									
1067	919	1196	785	1126	936	918	1156	920	948
855	1092	1162	1170	929	950	905	972	1035	1045
1157	1195	1195	1340	1122	938	970	1237	956	1102
1022	978	832	1009	1157	1151	1009	765	958	902
923	1333	811	1217	1085	896	958	1311	1037	702
521	933	928	1153	946	858	1071	1069	830	1063
930	807	954	1063	1002	909	1077	1021	1062	1157
999	932	1035	944	1049	940	1122	1115	833	1320
901	1324	818	1250	1203	1078	890	1303	1011	1102
996	780	900	1106	704	621	854	1178	1138	951
1187	1067	1118	1037	958	760	1101	949	992	966
824	653	980	935	878	934	910	1058	730	980
844	814	1103	1000	788	1143	935	1069	1170	1067
1037	1151	863	990	1035	1112	931	970	932	904
1026	1147	883	867	990	1258	1192	922	1150	1091
1039	1083	1040	1289	699	1083	880	1029	658	912
1023	984	856	924	801	1122	1292	1116	880	1173
1134	932	938	1078	1180	1106	1184	954	824	529
998	996	1133	765	775	1105	1081	1171	705	1425
610	916	1001	895	709	860	1110	1149	972	1002

Table 1.3: Life in Hours of 200 Incandescent Lamps.

The choice of how many bins to use involves a trade-off. If there are too few bins, we risk losing important

information by grouping too many distinct values together. If there are too many bins, the frequencies within each bin may become too small to reveal any clear pattern in the data's distribution.

While the optimal number of bins is a subjective choice that depends on the dataset, a selection of 5 to 10 bins is typical. It is also customary, though not required, to use bins of equal width, as this often makes interpretation easier.

The endpoints of a bin are called its **class boundaries**. We will adopt the common **left-end inclusion convention**, which specifies that a bin contains its left-end boundary but not its right-end one. For example, the bin 700–800 would contain any value x such that $700 \leq x < 800$.

Following this approach, the raw data from Table 1.3 is summarized in Table 1.4 using bins of length 100, starting at 500.

Class Interval	Frequency	Relative Freq.	Cumulative Rel. Freq.
500–600	2	0.010	0.010
600–700	5	0.025	0.035
700–800	12	0.060	0.095
800–900	25	0.125	0.220
900–1000	58	0.290	0.510
1000–1100	41	0.205	0.715
1100–1200	43	0.215	0.930
1200–1300	7	0.035	0.965
1300–1400	6	0.030	0.995
1400–1500	1	0.005	1.000

Table 1.4: Frequency Distribution for the Lifetimes of 200 Incandescent Lamps.

A bar graph that plots the frequencies of data grouped into bins is called a **histogram**. A key feature of a histogram is that the bars are placed directly adjacent to one another, reflecting the continuous nature of the bins. The vertical axis can represent either the absolute frequency (the count of values in each interval) or the relative frequency. The former is called a **frequency histogram**, while the latter is a **relative frequency histogram**. Figure 1.4 shows a frequency histogram for the lamp lifetime data.

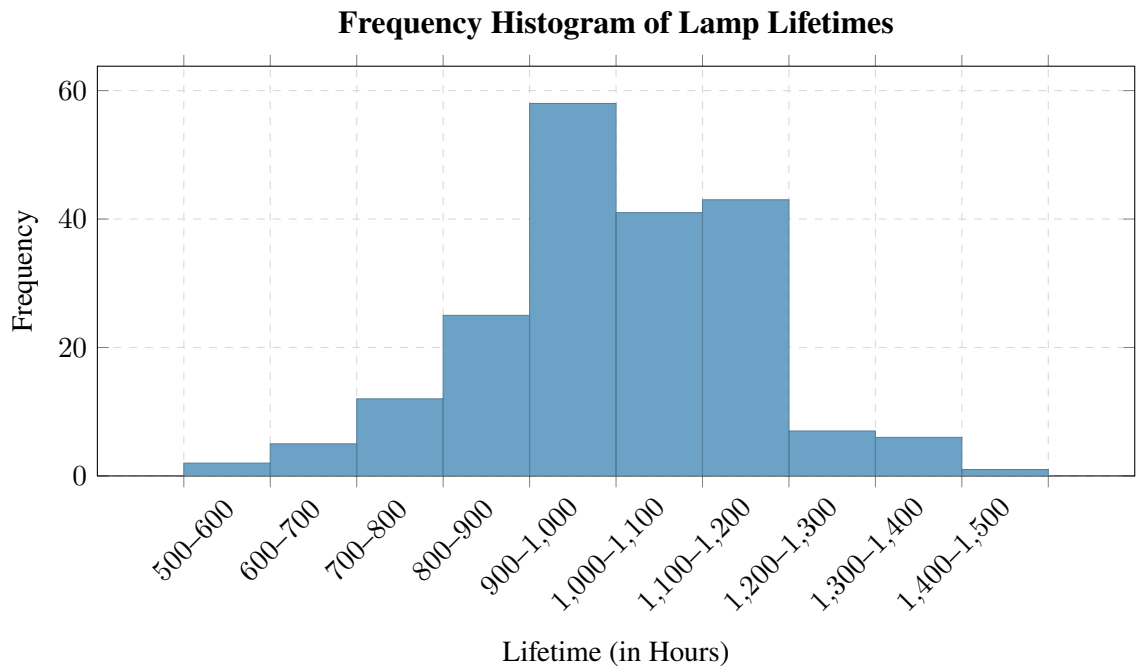


Figure 1.4: A frequency histogram for the lamp lifetime data. The adjacent bars show the distribution of data across continuous intervals.

Finally, we are sometimes interested in plotting the cumulative frequency or, more commonly, the cumulative relative frequency. Such a graph, often called an **ogive**, shows the number or proportion of data points that fall at or below a certain value.

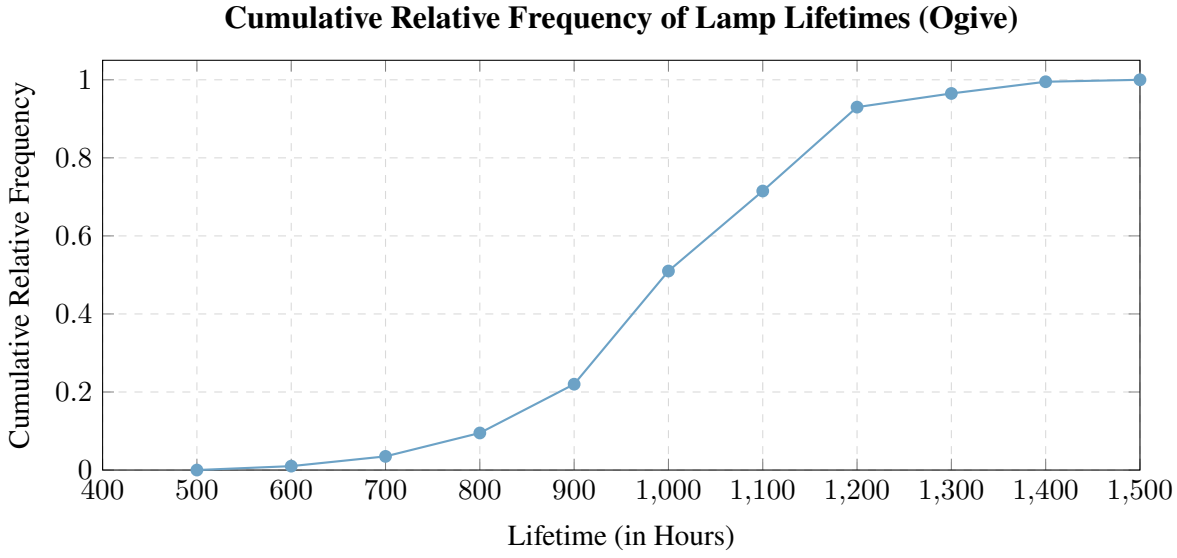


Figure 1.5: An ogive showing the cumulative relative frequency for the lamp lifetime data.

On an ogive, each point on the horizontal axis represents a data value, while the corresponding point on the vertical axis shows the proportion of the data that is less than or equal to that value. For example, Figure 1.5 presents the cumulative relative frequency plot for the lamp lifetime data. From this graph, we can quickly determine that approximately 50% of the lamps had a lifetime of less than 1000 hours, about 95% had a lifetime of less than 1200 hours (the actual value is 93%), and 100% of the lamps failed by 1500 hours.

1.3 Summarizing Data

Having seen how to visually and numerically summarize data distributions, we now focus on two fundamental aspects of describing a dataset: its *center* (or typical value) and its *dispersion* (or variability).

Measures of Central Tendency

The **mean**, often called the average, is the most commonly used measure of central tendency. Definition 1.6 shows how to calculate the mean of a dataset and Example 1.4 exemplifies the calculation.

Definition 1.6 (Arithmetic Mean)

For a dataset with n values $x_1, x_2, \ldots, x_n$, the **arithmetic mean** is defined as:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

♣

Example 1.4 Response Time Analysis

A web application recorded the following response times (in milliseconds) for 10 requests:

$$\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$$

The mean response time is:

$$\bar{x} = \frac{120 + 150 + 180 + 200 + 220 + 250 + 280 + 300 + 350 + 400}{10} = \frac{2450}{10} = 245 \text{ milliseconds}$$

The mean incorporates every value in a dataset, which makes it a powerful measure of central tendency. It is most appropriate when the data values are of similar magnitude and when the distribution is reasonably symmetric without extreme outliers. Under these conditions, the mean provides a reliable summary of the dataset.

For example, in the analysis of web performance, the mean response time offers an overall indication of how quickly a system responds on average. However, if a few response times are much slower than the rest, the mean may give a misleading impression of the typical user experience.

Beyond its role as a summary statistic, the mean also serves as a foundation for many statistical methods. In particular, it is central to the definitions of variance and standard deviation, which measure the dispersion of data around the mean.

Remark: As mentioned earlier, and as we will revisit, the mean is highly sensitive to extreme values or outliers. Even a single unusually large or small observation can shift the mean considerably, reducing its ability to reflect the typical value of the dataset. It is therefore important to consider the distribution of the data before relying solely on the mean to describe the center of a dataset.

The **median** is the middle value when data is arranged in ascending order. It is less sensitive to outliers than the mean.

Definition 1.7 (Median)

For a dataset with n values arranged in ascending order, the median is defined as:

$$\text{median} = \begin{cases} x_{\frac{n+1}{2}} & \text{if } n \text{ is odd} \\ \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2} & \text{if } n \text{ is even} \end{cases}$$

♣

Example 1.5 Median Response Time

Using the same response time data from the previous example:

$$\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$$

Since $n = 10$ (even), the median is the average of the 5th and 6th values:

$$\text{Median} = \frac{220 + 250}{2} = 235 \text{ milliseconds}$$

Example 1.6

Consider the same data with an extreme value:

$$\{120, 150, 180, 200, 220, 250, 280, 300, 350, 2000\}$$

The mean becomes: $\bar{x} = \frac{4050}{10} = 405$ milliseconds (significantly affected by the outlier)

The median remains: $\frac{220+250}{2} = 235$ milliseconds (unaffected by the outlier)

Because the median divides a dataset into two equal halves, it serves as a robust measure of central tendency, particularly when the data contain outliers or are skewed. Unlike the mean, which can be drawn toward extreme values, the median remains largely unaffected, offering a more reliable indication of the "typical" value in such situations.

The **mode** is the most frequently occurring value in a dataset.

Definition 1.8 (Mode)

The **mode** is the value that appears most frequently in a dataset. A dataset can have:

- **No mode**: If all values appear with equal frequency
- **One mode (unimodal)**: If one value appears most frequently
- **Multiple modes (multimodal)**: If two or more values tie for the highest frequency

♣

Sometimes, when a data set has two modes, it is called a **bimodal** dataset.

Example 1.7

A software project has the following bug severity levels:

$$\{\text{High}, \text{Medium}, \text{Low}, \text{High}, \text{Critical}, \text{Medium}, \text{High}, \text{Low}, \text{High}, \text{Medium}\}$$

Counting frequencies:

- High: 4 occurrences
- Medium: 3 occurrences
- Low: 2 occurrences
- Critical: 1 occurrence

The mode is ”High” since it appears most frequently.

The mode is especially useful when analyzing categorical or discrete data, where calculating a mean or median may not make sense. For example, in software engineering, the mode can help identify the most common error code returned by an API, the most frequently used programming language in a codebase, or the most reported type of bug in an issue tracker.

When to Use Each Measure

Measure	Best For	Limitations
Mean	Continuous data, symmetric distributions	Sensitive to outliers
Median	Data with outliers, skewed distributions	Less informative for symmetric data
Mode	Categorical data, discrete data	May not exist or be unique

Table 1.5: Comparison of Central Tendency Measures

Measures of Dispersion

While measures of central tendency tell us about the typical value, **measures of dispersion** describe how spread out the data are.Understanding variability is crucial in software engineering for assessing consistency and reliability.

The **range** is the simplest measure of dispersion.

Definition 1.9 (Range)

The **range** of a dataset is the difference between the maximum and minimum values:

$$\text{Range} = x_{\max} - x_{\min}$$

♣

Example 1.8

For the response time data: $\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$

$$\text{Range} = 400 - 120 = 280 \text{ milliseconds}$$

Remark: The range is sensitive to outliers and doesn't provide information about the distribution of values between the extremes.

The **variance** and **standard deviation** are more sophisticated measures that consider all data points.

Definition 1.10 (Sample Variance and Standard Deviation)

For a sample with n values, the **sample variance**, denoted by s^2, is defined as:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

The **sample standard deviation**, denoted by s, is the square root of the variance:

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$



While the mean identifies the center of a dataset, the **variance** and **standard deviation** measure its variability or spread. A small standard deviation indicates that data points are clustered tightly around the mean, whereas a large standard deviation signifies that they are more spread out.

The sample variance is a scaled sum of the squared deviations from the mean, using the divisor $n-1$. Squaring each deviation serves two purposes: it makes all contributions non-negative, and it gives greater weight to larger deviations. However, this calculation leaves the variance in squared units (e.g., milliseconds squared), which are not intuitive to interpret.

To solve this, we use the standard deviation, which is simply the square root of the variance. This crucial step returns the measure to the original units of the data (e.g., milliseconds), making it directly interpretable. In short, the standard deviation represents the typical distance of a data point from the mean.

Example 1.9 Calculating Standard Deviation

For the response time data: $\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$ with $\bar{x} = 245$:

$$\begin{aligned}
s^2 &= \frac{1}{9}[(120-245)^2 + (150-245)^2 + \cdots + (400-245)^2] \\
&= \frac{1}{9}[(-125)^2 + (-95)^2 + \cdots + (155)^2] \\
&= \frac{1}{9}[15625 + 9025 + \cdots + 24025] \\
&= \frac{1}{9} \times 70850 \approx 7872.22
\end{aligned}$$

$$s = \sqrt{7872.22} \approx 88.73 \text{ milliseconds}$$

The following algebraic identity is often helpful when computing variance or standard deviation by hand:

We start from the definition:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Using the identity

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n}x_i^2 - \frac{\left(\sum_{i=1}^{n}x_i\right)^2}{n}$$

we can rewrite sample variance as

$$s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}\right) = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right)$$

This is known as the **computational formula for variance**, as it allows you to calculate s^2 directly from $\sum x_i$ and $\sum x_i^2$ (or equivalently, from $\sum x_i^2$ and $\bar{x}$), without needing to compute each individual deviation from the mean.

The **coefficient of variation** allows comparison of variability across datasets with different scales.

Definition 1.11 (Coefficient of Variation)

The **coefficient of variation** is the ratio of standard deviation to the mean:

$$CV = \frac{s}{\bar{x}} \times 100\%$$

♣

Example 1.10 Comparing Variability

Consider two systems:

- System A: Mean response time = 100ms, Standard deviation = 20ms
- System B: Mean response time = 500ms, Standard deviation = 100ms

$$CV_A = \frac{20}{100} \times 100\% = 20\%$$
$$CV_B = \frac{100}{500} \times 100\% = 20\%$$

Both systems have the same relative variability (20%), despite different absolute standard deviations.

Measures of Position

To introduce a measure of spread that is less sensitive to outliers, we first need to understand measures of position, which describe the relative standing of a data point within a dataset.

Definition 1.12 (Percentile)

The p-th **percentile** is a value at or below which approximately $p\%$ of the observations fall. Several conventions are used to calculate percentiles, especially for small samples, so the chosen method should always be stated.

♣

For example, if a response time is at the 95th percentile, it means that 95% of all response times are faster than this one.

Example 1.11 Percentile Calculation

For the response time data, use the nearest-rank method to find the 90th percentile:

1. Sort data: $\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$
2. Rank: $\lceil 0.9 \times 10 \rceil = 9$
3. 90th percentile = 350ms

Quartiles are specific, widely used percentiles that divide the data into four equal parts.

Definition 1.13 (Quartiles)

Quartiles divide an ordered dataset into four equal parts.

- **Q1 (First Quartile)**: The 25th percentile.
- **Q2 (Second Quartile)**: The 50th percentile. This is the **median** of the dataset.
- **Q3 (Third Quartile)**: The 75th percentile.

♣

Example 1.12 Calculating Quartiles

For the sorted response time data: $\{120, 150, 180, 200, 220, |250, 280, 300, 350, 400\}$

- The lower half is $\{120, 150, 180, 200, 220\}$. The median of this half is $Q1 = 180$ ms.
- The upper half is $\{250, 280, 300, 350, 400\}$. The median of this half is $Q3 = 300$ ms.

Example 1.13 Quartiles from an Ogive

Quartiles can be read directly from an ogive. As an example, we will again use the lamp lifetime data. Reading from the ogive in Figure 1.6, the approximate quartiles are $Q1 \approx 900$, $Q2$ (median) ≈ 1000, and $Q3 \approx 1100$ hours.

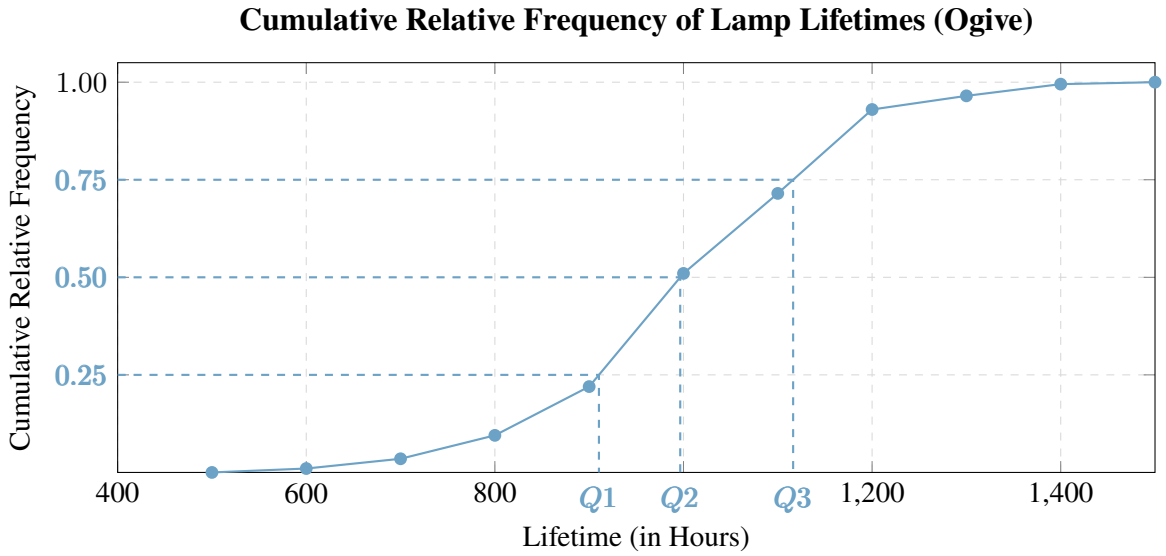


Figure 1.6: An ogive for the lamp lifetime data with quartiles $Q1$, $Q2$, and $Q3$ indicated by dashed lines.

Percentiles divide data into 100 equal parts, while quartiles divide data into four equal parts.

The **interquartile range** is a robust measure of dispersion that is generally much less sensitive to extreme observations than the full range or the standard deviation.

Definition 1.14 (Interquartile Range)

The **interquartile range (IQR)** measures the spread of the middle 50% of the data and is defined as

$$IQR = Q_3 - Q_1,$$

where Q_1 and Q_3 are the first and third quartiles. A larger IQR indicates that the central portion of the data is more widely dispersed, while a smaller IQR indicates that it is more tightly clustered.

♣

Example 1.14 IQR Calculation

For the response time data: $\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$

- Lower half: $\{120, 150, 180, 200, 220\} \rightarrow Q1 = 180$
- Upper half: $\{250, 280, 300, 350, 400\} \rightarrow Q3 = 300$
- $IQR = 300 - 180 = 120$ milliseconds

Summary Statistics

Before we move on to the next section, we will review the summary statistics covered in this section. Table 1.6 is a hypothetical example for a separate dataset of 1000 web response times. The underlying observations are not listed, so the values are included to illustrate the structure of a summary table rather than as a reproducible calculation from the preceding ten observations.

Example 1.15

For a dataset of 1000 web response times:

Measure	Value
Count	1000
Mean	245.3 ms
Median	238.0 ms
Mode	220.0 ms
Standard Deviation	95.6 ms
Variance	9139.4 ms^2
Minimum	45.0 ms
Maximum	1200.0 ms
Range	1155.0 ms
Q1	180.0 ms
Q3	300.0 ms
IQR	120.0 ms
Skewness	1.2 (positive skew)
Kurtosis	2.8 (leptokurtic)

Table 1.6: Complete Statistical Summary

This table provides a complete picture of the data and is often a first step in the analysis of a dataset. Note that

the skewness and kurtosis have not been discussed but will be explained in the next section. We have chosen to include them here for completeness as they are often used in the analysis of a dataset - although how to calculate them is beyond the scope of this book.

1.4 Understanding Data Distributions

Once we have visualized our data with histograms and box plots, we can begin to analyze the *shape* of the distribution. The shape tells us about the underlying patterns in our data and helps us choose the right statistical tools. In software engineering, understanding the distribution of metrics like response times or error rates is critical for setting performance baselines and detecting anomalies. First we review the concept of outliers.

Normal Data Sets

Many large data sets yield histograms with a distinctive overall shape: the frequencies peak near the sample median and decrease symmetrically on both sides, forming a characteristic **bell-shaped curve**. Data sets exhibiting this property are called **normal**, and their histograms are referred to as **normal histograms** (see Figure 1.7).

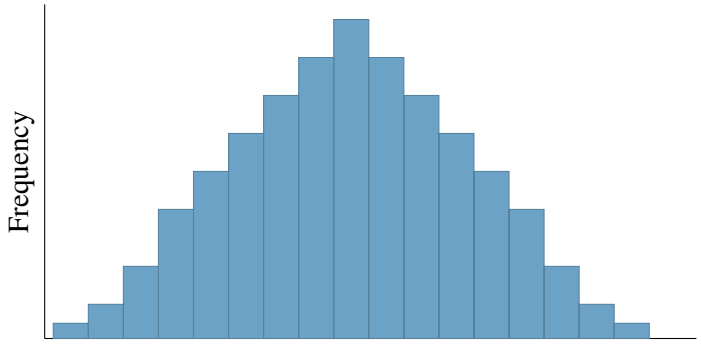


Figure 1.7: A histogram of a normal data set, showing the characteristic bell shape.

In software systems, perfectly normal distributions are rare, but some metrics can be **approximately normal**. For example, the latency of a highly optimized, internal microservice that performs a consistent task might be approximately normal. A histogram of an approximately normal data set looks like Figure 1.8.

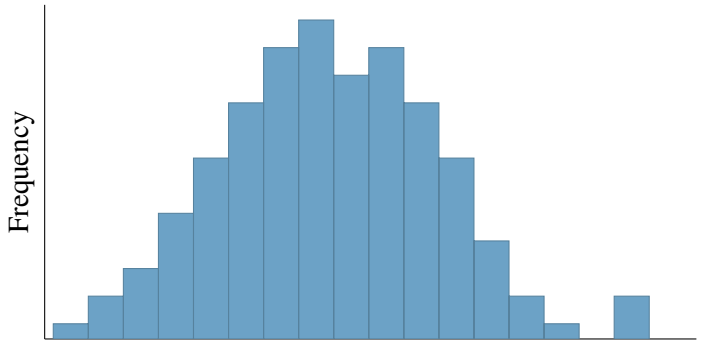


Figure 1.8: A histogram of a data set that is approximately normal. While not perfectly symmetric, it follows the general bell shape.

Many datasets observed in practice, from human height to measurement errors, follow this specific shape. Such data is said to be **normally distributed**.

Definition 1.15 (Normal Distribution)

A **normal distribution** is a symmetric, bell-shaped distribution where the mean, median, and mode are all equal and located at the center. Its shape is determined entirely by its mean (μ) and standard deviation (σ).

A histogram estimates the shape of the distribution that generated the observations. With more data and an appropriate choice of bin width, it can give a more detailed picture of the underlying distribution, but it does not necessarily become normal. Whether a normal model is appropriate depends on the data-generating process.

The mathematical normal model, its density function, and probability calculations are developed in chapter 4. Its probability density has the following characteristic shape:

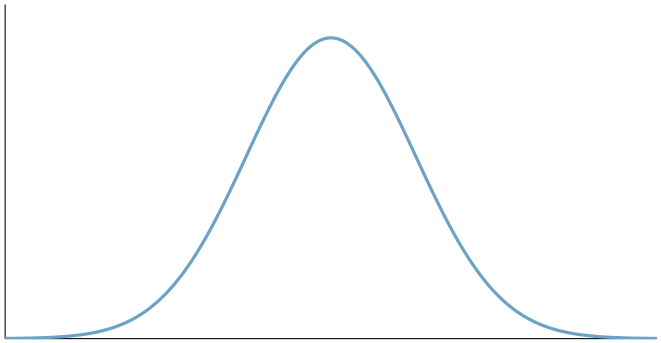


Figure 1.9: The normal distribution curve.

Skewed Data Sets

When a data set is not symmetric about its sample median, it is described as **skewed**. A long tail extending to the right indicates the distribution is **skewed to the right**; a long tail to the left means it is **skewed to the left**. Thus, the histogram in Figure 1.10 is skewed left, while the one in Figure 1.11 is skewed right.

Definition 1.16 (Skewness)

Skewness measures the asymmetry of data distribution:

- **Positive skew (skewed to the right)**: Tail extends to the right (mean ¿ median)
- **Negative skew (skewed to the left)**: Tail extends to the left (mean ¡ median)
- **Symmetric**: Mean $\approx$ median

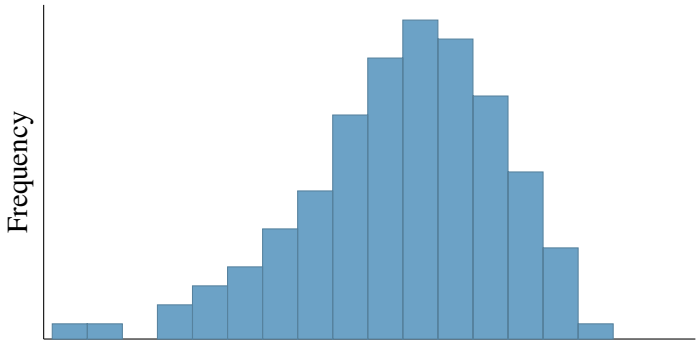


Figure 1.10: A histogram of a data set that is skewed to the left (negatively skewed). The tail of the distribution is on the left.

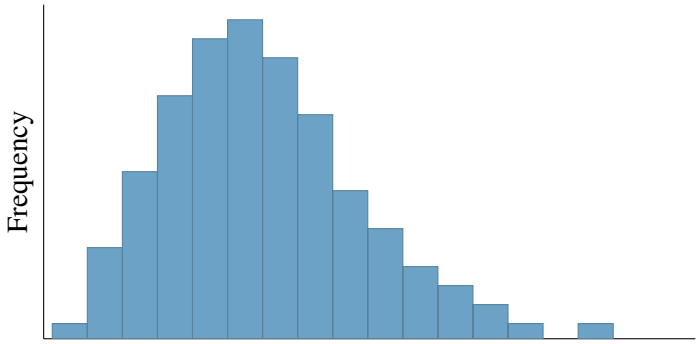


Figure 1.11: A histogram of a data set that is skewed to the right (positively skewed). The tail of the distribution is on the right.

Figure 1.12 illustrates common patterns for the mean, median, and mode in unimodal skewed distributions. These orderings are useful heuristics, not definitions of skewness and not universal rules.

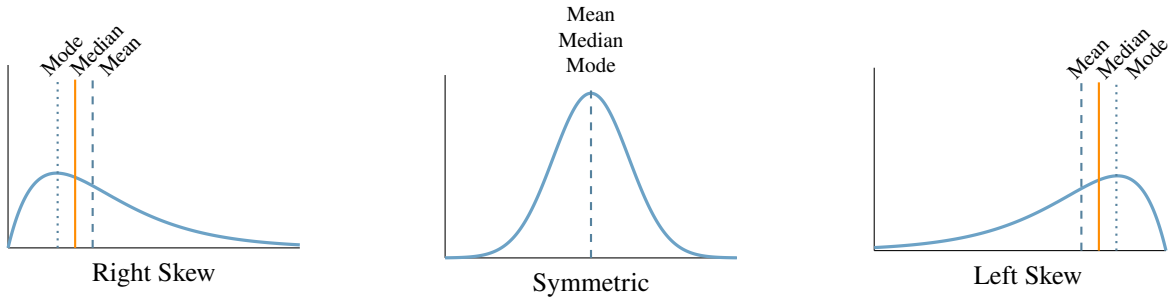


Figure 1.12: Relationship between mean, median, and mode for positively skewed, symmetric, and negatively skewed distributions.

Outlier Detection

As mentioned in section 1.1, outliers are data points that are significantly different from the rest of the data. They can arise due to measurement errors, data entry errors, or genuine variability in the data. Detecting outliers is important because they can distort statistical analyses.

A common method for identifying outliers is to use the interquartile range (IQR):

- **Mild outliers**: Data points outside the $1.5 \times IQR$ fences but not outside the $3 \times IQR$ fences.
- **Extreme outliers**: Data points less than $Q_1 - 3 \times IQR$ or greater than $Q_3 + 3 \times IQR$.

Example 1.16 Detecting Outliers

Given the response time data: $\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400\}$

- $Q_1 = 180$, $Q_3 = 300$, $IQR = 120$
- Lower fence: $Q_1 - 1.5 \times IQR = 180 - 180 = 0$
- Upper fence: $Q_3 + 1.5 \times IQR = 300 + 180 = 480$

All data points are between 0 and 480, so there are no mild outliers in this dataset.

Example 1.17

Suppose we have the following set of response times (in milliseconds) for a web application:

$\{120, 150, 180, 200, 220, 250, 280, 300, 350, 400, 1200\}$.

Let's identify if there is an upper outlier:

- Sorted data: $120, 150, 180, 200, 220, 250, 280, 300, 350, 400, 1200$
- $n = 11$
- Median (Q_2): 250 (6th value)
- Lower half: $120, 150, 180, 200, 220 \rightarrow Q_1 = 180$
- Upper half: $280, 300, 350, 400, 1200 \rightarrow Q_3 = 350$
- $IQR = Q_3 - Q_1 = 350 - 180 = 170$
- Upper fence: $Q_3 + 1.5 \times IQR = 350 + 1.5 \times 170 = 350 + 255 = 605$

The value 1200 is greater than 605, so it is an upper outlier.

Box Plots

Box plots provide a compact summary of data distribution, highlighting the median, quartiles, and outliers. Figure 1.13 shows an example of a box plot.

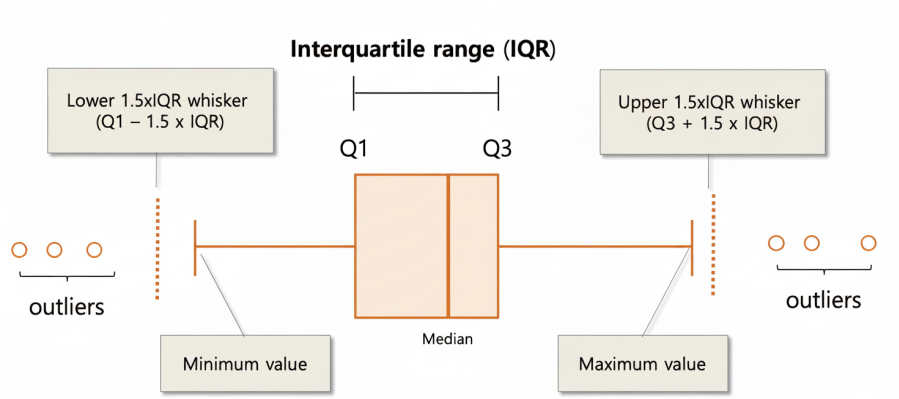


Figure 1.13: An example of a box plot, illustrating the median, quartiles, whiskers, and potential outliers.

Definition 1.17 (Box Plot Components)

A box plot is based on the five-number summary and may also display outliers:

- **Lower whisker**: Usually the smallest observation within the lower $1.5 \times IQR$ fence
- **Q1 (First Quartile)**: The 25th percentile
- **Median (Q2)**: The 50th percentile
- **Q3 (Third Quartile)**: The 75th percentile
- **Upper whisker**: Usually the largest observation within the upper $1.5 \times IQR$ fence
- **Outliers**: Points beyond $1.5 \times IQR$ from the box

A box plot provides a visual summary of the distribution of a dataset, making it easy to compare groups and spot outliers. Here's how to read a box plot.

- The **length of the box** shows the spread of the central half of the data. A longer box means more variability in the middle 50%.
- The **position of the median line** and the relative whisker lengths can suggest asymmetry, but a box plot alone does not determine the direction or degree of skewness.
- In many unimodal skewed distributions, the mean is pulled toward the longer tail. Consequently, one often sees mode $<$ median $<$ mean for right-skewed data and the reverse ordering for left-skewed data. These are common patterns rather than guaranteed identities.
- The **whiskers** show the range of the bulk of the data, excluding outliers.
- **Outliers** highlight unusually high or low values that may warrant further investigation.

Box plots are especially useful for comparing distributions across several groups or datasets, as they quickly reveal differences in medians, spreads, and the presence of outliers.

This concludes the foundational treatment of data visualization and descriptive summary statistics. The next chapter develops the probability concepts needed for statistical inference.

Chapter 2 Probability for Data Analysis

Probability provides a language for reasoning when the available information is incomplete. In engineering and data analysis, uncertainty appears whenever measurements vary, components can fail, users behave differently, or observations reveal only part of an underlying process. The purpose of a probability model is not to eliminate this uncertainty, but to describe it precisely enough that conclusions and decisions can be justified.

A famous illustration is the Monty Hall problem. Imagine a game show with three closed doors. A car is hidden behind one door and a goat behind each of the other two. You choose a door, but before it is opened, the host—who knows where the car is and always opens a door hiding a goat—opens one of the remaining doors. You are then offered the choice of staying with your original door or switching to the other unopened door. At first sight, the two unopened doors may appear equally likely. After all, only two possibilities remain.

The crucial point is that the host's action is not a random observation made without knowledge. The door that the host opens depends on both the location of the car and the door initially selected by the contestant. Consequently, the information supplied by the host must be interpreted conditionally. A careful probability model shows that the original choice still has probability (1/3) of hiding the car, while the switching strategy wins with probability (2/3). The puzzle is surprising because intuitive reasoning often ignores the mechanism that produced the observed information.

This is exactly the kind of problem for which conditional probability and Bayes' theorem are designed. Bayes' theorem allows probabilities to be revised when new evidence becomes available while taking account of how likely that evidence would be under each possible explanation. The same reasoning appears in diagnostic testing, fault detection, quality control, spam filtering, and many other data-driven systems. We build toward Bayes' theorem gradually: beginning with random experiments, sample spaces, and events, then developing the basic probability rules, conditional probability, total probability, and independence. The chapter concludes with Bayes' theorem in section 2.7.

2.1 Random Experiments, Sample Spaces, and Events

A **random experiment** is one that can lead to different outcomes, even when repeated under the same conditions. This randomness is a fundamental aspect of many engineering tasks.

Think of it this way: Let us say you are testing the speed of a website under different conditions. Sometimes it loads quickly, and other times it is slower. Even if you are using the same code and server, things like network traffic or server load make the results vary each time.

Definition 2.1 (Random Experiment)

A random experiment is one that can give different results, even if you do everything the same each time. ♣

Or, imagine you are measuring the signal strength in a wireless device. You might get slightly different readings each time because of things like interference or small changes in the environment.

This randomness shows up all over the place, from software performance tests to electrical engineering experiments. It is important to expect it and include it in your thinking. Otherwise, you might make decisions based

on incomplete or misleading data. When you account for random variation, you can make smarter predictions and designs.

To model and analyze a random experiment, it is crucial to understand the set of possible outcomes that can occur. In probability theory, this set is called the **sample space**, denoted by S. A sample space can be either **discrete** (consisting of a finite or countably infinite set of outcomes) or **continuous** (containing an interval of real numbers). The exact definition of a sample space often depends on the objectives of the analysis.

An **outcome** is a single possible result of the random experiment, and an **event** is any subset of the sample space, which may consist of one or more outcomes. Below are some examples to illustrate these concepts:

Example 2.1 Network Latency

Consider an experiment where you measure the latency of data packets in a network. The sample space can be defined based on the type of measurements:

- If latency is measured as a positive real number, the sample space is continuous:
  $S = \{x \mid x > 0\}$.
- If it is known that latency ranges between 10 and 100 milliseconds, the sample space can be refined to:
  $S = \{x \mid 10 \leq x \leq 100\}$.
- If the objective is to categorize latency as low, medium, or high, the sample space becomes discrete:
  $S = \{\text{low}, \text{medium}, \text{high}\}$.
- For a simple evaluation of whether the latency meets a standard threshold, the sample space can be reduced to:
  $S = \{\text{pass}, \text{fail}\}$.

Each outcome in these sample spaces represents a single possible latency measurement, and events can be defined as sets of outcomes, such as "latency is high."

Understanding the nature of sample spaces, outcomes, and events is fundamental in probability, as it allows us to define and work with probabilities of complex scenarios in various engineering contexts. Let us summarise these key concepts:

Definition 2.2 (Sample Spaces, Outcomes, and Events)

Sample Space: The set of all possible outcomes of a random experiment is called the sample space, denoted by S. Outcomes can be discrete or continuous, depending on the nature of the experiment.

Outcome: A single possible result of a random experiment.

Event: Any subset of the sample space, which may consist of one or more outcomes. ♣

Example 2.2 Software Release Testing

Imagine a software testing process where each test case can either pass or fail. The sample space for a single test case is discrete and can be represented as:

$$S = \{\text{pass}, \text{fail}\}.$$

If you run three test cases, the combined sample space for all possible outcomes is:

$$S = \{(\text{pass, pass, pass}), (\text{pass, pass, fail}), (\text{pass, fail, pass}),$$
$$(\text{pass, fail, fail}), (\text{fail, pass, pass}), (\text{fail, pass, fail}),$$
$$(\text{fail, fail, pass}), (\text{fail, fail, fail})\}.$$

This sample space includes all sequences of outcomes for the three tests.

- The total number of possible outcomes is $2^3 = 8$.
- An event could be defined as "at least one test fails," which would include outcomes like (fail, pass, pass), (pass, fail, fail), and others where at least one test fails.

Example 2.3 Component Quality in Manufacturing

A company manufactures electronic components, and each component is tested for compliance with quality standards. The test can return one of three outcomes: `pass`, `marginal`, or `fail`. The sample space is:

$$S = \{\text{pass}, \text{marginal}, \text{fail}\}.$$

Event: Suppose we are interested in the event that a component does not pass the quality test. This event is a set of outcomes:

$$E = \{\text{marginal}, \text{fail}\}.$$

This means that all the operations we have defined on sets translate directly into operations on events:

- $A \cup B$: the event that at least one of A or B occurs.
- $A \cap B$: the event that both A and B occur together.
- $\overline{A}$ or A^c: the event that A does not occur.
- $A \setminus B$ or $A - B$: the event that A occurs but B does not.
- $A \Delta B$ or $A \oplus B$: the event that either A or B occurs, but not both (symmetric difference).

Thus, probability theory builds directly on set theory: probability assigns a numerical measure to these subsets of the sample space. Later in this chapter, we will see how this measure is defined and used to reason about uncertainty.

We can summarise the considerations of this section in the following table:

Operation	Boolean Algebra	Logic	Set Theory
`NOT`	$\overline{x}$	$\neg x$	A^c or A'
`OR`	$+$	$\vee$	$\cup$
`AND`	$\cdot$	$\wedge$	$\cap$
`NAND`	$\overline{x \cdot y}$	$\neg(x \wedge y)$	$(A \cap B)^c$ or $\overline{A \cap B}$
`NOR`	$\overline{x + y}$	$\neg(x \vee y)$	$(A \cup B)^c$ or $\overline{A \cup B}$
`XOR` (Symmetric Difference)	$x \oplus y$	$(x \wedge \neg y) \vee (\neg x \wedge y)$	$A \Delta B$ or $A \oplus B$
Difference	$x \cdot \overline{y}$	$x \wedge \neg y$	$A - B$ or $A \setminus B$

Table 2.1: Comparison of Operators in Boolean Algebra, Logic, and Set Theory

In this book, we will use the notation A^c to denote the complement of the set A, which includes all elements not in A.

2.2 Basic Probability

Probability quantifies the likelihood or chance of events associated with a random experiment. For instance, when you hear, "The chance of rain today is 30%," it expresses uncertainty about the event that it rains. Probabilities range from 0 to 1 (or equivalently, from 0% to 100%). An impossible event has probability 0, and the entire sample space has probability 1. In general probability models, however, an event can have probability 0 without being empty, or probability 1 without containing every possible outcome; such events are called almost impossible and almost sure, respectively.

Probabilities can be interpreted in different ways:

- **Objective (or Classical) Probability**: Often referred to as classical probability, this approach is used when outcomes are equally likely, such as in rolling a fair die or flipping a coin. Probabilities are assigned based on the assumption that each outcome has an equal chance of occurring. For example, when rolling a fair six-sided die, the probability of rolling a 3 is $\frac{1}{6}$ because there are 6 equally likely outcomes (1, 2, 3, 4, 5, 6), and only one of them is a 3. The probability is the same for all observers.
- **Relative Frequency (Empirical Probability)**: Empirical probability is based on observations from experiments rather than theoretical calculations. For example, if a software tester runs a stress test on a server 100 times, and it crashes 7 times, the empirical probability of a crash is $\frac{7}{100} = 0.07$. This approach relies on actual data rather than assumptions or intuition.
- **Subjective Probability**: This reflects our personal belief or degree of confidence in an outcome. Different people might assign different probabilities to the same event based on their knowledge or perspective. You and your friends discuss Denmark's chances of winning the World Cup. Based on recent performance and team strength, you estimate a 10% chance. However, a more optimistic friend assigns a 20% chance, while another gives only 5%, considering stronger competitors. This illustrates subjective probability, where each person's estimate varies based on personal beliefs and biases rather than objective data.

When assigning probabilities, it's essential that the sum of all probabilities in an experiment equals 1, ensuring consistency with the relative frequency interpretation.

We start by establishing the Axioms of Probability, which lay the foundation for how probabilities are assigned to events. These axioms define the basic properties that every probability measure must satisfy.

Axiom 2.1 (Axioms of Probability)

- **Axiom 1:** For any event A, $0 \leq P(A) \leq 1$.
- **Axiom 2:** Probability of the sample space S is $P(S) = 1$.
- **Axiom 3:** If $A_1, A_2, A_3, \cdots$ are disjoint events, then $P\left(A_1 \cup A_2 \cup A_3 \cdots\right) = P\left(A_1\right) + P\left(A_2\right) + P\left(A_3\right) + \cdots$

♡

The property that $0 \leq P(A) \leq 1$ is equivalent to the requirement that a relative frequency must be between 0 and 1. The property that $P(S) = 1$ is a consequence of the fact that an outcome from the sample space occurs on every trial of an experiment. Consequently, the relative frequency of S is 1. Property 3 implies that if the events A_1 and A_2 have no outcomes in common, the relative frequency of outcomes in $A_1 \cup A_2$ is the sum of

the relative frequencies of the outcomes in A_1 and A_2.

In the next sections we will see more about the probability of events and how to calculate them.

Probability of an Event

The probability of an event is a measure of the likelihood that the event will occur. It is denoted by $P(A)$, where A is the event, and it ranges from 0 to 1. The empty event has probability 0, and the sample space has probability 1.

Definition 2.3 (Classical Probability for Equally Likely Outcomes)

If a finite sample space consists of equally likely outcomes, the probability of an event A is the ratio of the number of outcomes in A to the total number of outcomes in the sample space:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$



Example 2.4 Suppose you are testing a software module with 10 different test cases. Out of these, 3 test cases are known to fail due to a bug. If you randomly select one test case to run, what is the probability that the selected test case will fail?

Solution: Here, the event A is "the test case fails."

- Number of favorable outcomes (failing test cases) $= 3$
- Total number of outcomes (total test cases) $= 10$

Using the formula:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} = \frac{3}{10} = 0.3$$

Therefore, the probability that a randomly selected test case will fail is 0.3 , indicating that there is a 30% chance of failure. ◀

Example 2.5 Imagine a software development environment where you have 50 files, consisting of 20 JavaScript files, 15 Java files, and 15 configuration files. If you randomly select one file to edit, what is the probability that the file is a JavaScript file?

Solution: Here, the event A is "the selected file is a JavaScript file."

- Number of favorable outcomes (JavaScript files) $= 20$
- Total number of outcomes (total files) $= 50$

Using the formula:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} = \frac{20}{50} = 0.4$$

Thus, the probability of selecting a JavaScript file is 0.4 , meaning there is a 40% chance of choosing a JavaScript file from the set. ◀

2.3 Probability of Joint Events and Set Operations

Joint events are formed by applying basic set operations to individual events. Commonly, we encounter unions of events, such as $A \cup B$; intersections of events, such as $A \cap B$; and complements of events, such as A^c. These combined events are often of particular interest, and their probabilities can frequently be derived from the probabilities of the individual events that compose them. Understanding these set operations is essential for accurately calculating the probability of joint events. In this section, we will explore how unions of events and other set operations can be used to determine the probabilities of more complex events.

When dealing with events, the intersection represents `AND` while the union represents `OR` The probability of the intersection of events A and B, denoted as $P(A \cap B)$, can also be expressed as $P(A, B)$ or $P(AB)$.

From the axioms of probability, we can derive the following rules of probabilities:

Theorem 2.1 (Rules of Probability)

- **Complement Rule:** The probability of the complement of event A is
  $$P(A^c) = 1 - P(A)$$
- **Empty Set Rule:** The probability of the empty set is 0, i.e.,
  $$P(\emptyset) = 0$$
- **Addition Rule:** For any two events A and B, the probability of the union of events A and B is given by
  $$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
- **Difference Rule:** The probability of the difference between events A and B is given by
  $$P(A - B) = P(A) - P(A \cap B)$$
- **Subset Rule:** If A is a subset of B ($A \subset B$), then
  $$P(A) \leq P(B)$$

.



We can obtain the Complement Rule by noting:

$$\begin{aligned} 1 &= P(S) && \text{(axiom 2)} \\ &= P(A \cup A^c) && \text{(definition of complement)} \\ &= P(A) + P(A^c) && \text{(since } A \text{ and } A^c \text{ are disjoint)} \end{aligned}$$

Since $\emptyset = S^c$, we can apply part the Complement Rule to deduce that $P(\emptyset) = 1 - P(S) = 0$. This is intuitive because, by definition, an event occurs when the outcome of the random experiment is part of that event. However, since the empty set contains no elements, no outcome of the experiment can ever belong to it, making its probability zero.

The Difference Rule can be obtained by showing that $P(A) = P(A \cap B) + P(A - B)$. Note that the two sets $A \cap B$ and $A - B$ are disjoint and their union is A. Thus, by the third axiom of probability

$$\begin{aligned} P(A) &= P\left((A \cap B) \cup (A - B)\right) && \text{(since } A = (A \cap B) \cup (A - B)\text{)} \\ &= P(A \cap B) + P(A - B) && \text{(since } A \cap B \text{ and } A - B \text{ are disjoint)} \end{aligned}$$

The Addition Rule we obtain by noting that A and $B - A$ are disjoint sets and their union is $A \cup B$. Thus,

$$\begin{aligned} P(A \cup B) &= P(A \cup (B - A)) && \text{(since } A \cup B = A \cup (B - A)\text{)} \\ &= P(A) + P(B - A) && \text{(since } A \text{ and } B - A \text{ are disjoint)} \\ &= P(A) + P(B) - P(A \cap B) && \text{(by part the Difference Rule)} \end{aligned}$$

And finally the Subset Rule is a direct consequence of the fact that if $A \subset B$, then B can be written as the union of A and $B - A$. Since A and $B - A$ are disjoint, we have $P(B) = P(A) + P(B - A) \geq P(A)$.

We conclude this section with a few examples illustrating the application of these rules to calculate probabilities of joint events.

Example 2.6 A company has bid on two large construction projects. The company president believes that the probability of winning the first contract is 0.6, the probability of winning the second contract is 0.4, and the probability of winning both contracts is 0.2.

(a) What is the probability that the company wins at least one contract?
(b) What is the probability that the company wins the first contract but not the second contract?
(c) What is the probability that the company wins neither contract?
(d) What is the probability that the company wins exactly one contract?

Solution: Let A be the event that the company wins the first contract, and B be the event that the company wins the second contract. Given:

- $P(A) = 0.6$
- $P(B) = 0.4$
- $P(A \cap B) = 0.2$

(a) The probability that the company wins at least one contract is the probability of the union of events A and B. Using the Addition Rule:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.6 + 0.4 - 0.2 \\ &= 0.8 \end{aligned}$$

Therefore, the probability that the company wins at least one contract is 0.8.

(b) The probability that the company wins the first contract but not the second contract is the probability of the difference between events A and B. Using the Difference Rule:

$$\begin{aligned} P(A - B) &= P(A) - P(A \cap B) \\ &= 0.6 - 0.2 \\ &= 0.4 \end{aligned}$$

Therefore, the probability that the company wins the first contract but not the second contract is 0.4.

(c) The probability that the company wins neither contract is the probability of the complement of the union

of events A and B. Using the Complement Rule:

$$\begin{aligned} P((A \cup B)^c) &= 1 - P(A \cup B) \\ &= 1 - 0.8 \\ &= 0.2 \end{aligned}$$

Therefore, the probability that the company wins neither contract is 0.2.

(d) The probability that the company wins exactly one contract is the probability of the difference between the union of events A and B and the intersection of events A and B. Using the Difference Rule:

$$\begin{aligned} P((A \cup B) - (A \cap B)) &= P(A \cup B) - P(A \cap B) \\ &= 0.8 - 0.2 \\ &= 0.6 \end{aligned}$$

So, the probability that the company wins exactly one contract is 0.6.

◀

Example 2.7

- There is a 60 percent chance that it will rain today.
- There is a 50 percent chance that it will rain tomorrow.
- There is a 30 percent chance that it does not rain either day.

(a) The probability that it will rain today or tomorrow

(b) The probability that it will rain today and tomorrow.

(c) The probability that it will rain today but not tomorrow.

(d) The probability that it either will rain today or tomorrow, but not both.

Solution:

(a) Let A be the event that it rains today, and B be the event that it rains tomorrow.
Given:

$$\begin{aligned} P(A \cup B) &= 1 - P\left((A \cup B)^c\right) && \text{by the Complement Rule} \\ &= 1 - P\left(A^c \cap B^c\right) && \text{by De Morgan's Law} \\ &= 1 - 0.3 \\ &= 0.7 \end{aligned}$$

Therefore, the probability that it will rain today or tomorrow is 0.7.

(b) The probability that it will rain today and tomorrow: this is $P(A \cap B)$. To find this we note that

$$\begin{aligned} P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\ &= 0.6 + 0.5 - 0.7 \\ &= 0.4 \end{aligned}$$

(c) The probability that it will rain today but not tomorrow: this is $P\left(A \cap B^c\right)$.

$$\begin{aligned} P(A \cap B^c) &= P(A - B) \\ &= P(A) - P(A \cap B) \\ &= 0.6 - 0.4 \\ &= 0.2 \end{aligned}$$

(d) The probability that it either will rain today or tomorrow but not both: this is $P(A - B) + P(B - A)$. We have already found $P(A - B) = .2$. Similarly, we can find $P(B - A)$:

$$\begin{aligned} P(B - A) &= P(B) - P(B \cap A) \\ &= 0.5 - 0.4 \\ &= 0.1 \end{aligned}$$

Thus,

$$\begin{aligned} P(A - B) + P(B - A) &= 0.2 + 0.1 \\ &= 0.3 \end{aligned}$$

◀

2.4 Conditional Probability

Consider the following scenario: in a particular population, 5% of individuals have a specific medical condition. Therefore, the probability that a randomly selected person has the condition is 5%:

$$P(D) = 0.05,$$

where D represents the event that a person has the disease. This probability $P(D)$ is known as the *prior probability*, as it reflects our initial belief about the likelihood of the event before any additional information is obtained.

Now, imagine selecting a random person and being informed that they have tested positive for the disease. With this additional information, how should we update the probability that the person actually has the disease? In other words, what is the probability that a person has the disease given that they tested positive? Let T denote the event that a person tests positive. This conditional probability is expressed as:

$$P(D \mid T),$$

which represents the probability of D occurring given that T has occurred. This updated probability $P(D \mid T)$ is known as the *posterior probability*, as it reflects our revised belief about the likelihood of the event after taking the new evidence into account. Intuitively, it is reasonable to expect that $P(D \mid T)$ is greater than the prior probability $P(D)$. However, what is the exact value of $P(D \mid T)$? Before introducing a general formula, let us consider a simple example.

Example 2.8 I roll a fair die. Let A be the event that the outcome is a prime number, i.e., $A = \{2, 3, 5\}$. Also, let B be the event that the outcome is greater than or equal to 3, i.e., $B = \{3, 4, 5\}$.

(a) What is the probability of A, $P(A)$?

(b) What is the probability of A given B, $P(A \mid B)$?

Solution:

(a) The probability of A is the number of outcomes in A divided by the total number of outcomes. Since there are 3 prime numbers and 6 possible outcomes, we have:

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

(b) The probability of A given B is the number of outcomes in the intersection of A and B divided by the number of outcomes in B. Since the outcomes in the intersection of A and B are $\{3, 5\}$ and the outcomes in B are $\{3, 4, 5\}$, we have:

$$P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{|\{3, 5\}|}{|\{3, 4, 5\}|} = \frac{2}{3}$$

◀

Having understood the basic example, we can now generalize the approach to derive a more universal formula for conditional probability. Starting from the specific case, we can manipulate the expression by dividing both the numerator and the denominator by the total number of possible outcomes, $|S|$, as shown below:

$$P(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{\frac{|A \cap B|}{|S|}}{\frac{|B|}{|S|}} = \frac{P(A \cap B)}{P(B)}$$

Explanation:

- **Numerator:** $|A \cap B|$ represents the number of outcomes where both events A and B occur. Dividing by $|S|$ converts this count into a probability, $P(A \cap B)$.
- **Denominator:** $|B|$ is the number of outcomes where event B occurs. Similarly, dividing by $|S|$ yields the probability of event B, denoted as $P(B)$.

Thus, the conditional probability $P(A \mid B)$ can be expressed as the ratio of the joint probability of A and B to the probability of B.

While the above derivation assumes a finite sample space with equally likely outcomes, the resulting formula is remarkably general. It holds true regardless of whether the sample space is finite or infinite, and whether the outcomes are equally likely or not. This universality makes the formula a cornerstone of probability theory.

Definition 2.4 (Conditional Probability)

For any two events A and B with $P(B) > 0$, the conditional probability of A given B is defined as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0$$

♣

Here is the intuition behind the formula. When we know that event B has occurred, we effectively eliminate all outcomes that are not part of B. This reduction transforms our original sample space into the subset B.

Within this new, restricted sample space B, the only way for event A to occur is if the outcome lies in the intersection of A and B, denoted by $A \cap B$. To determine the conditional probability $P(A \mid B)$, we calculate

the ratio of the probability of both A and B occurring to the probability of B. Mathematically, this is expressed as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

This formulation ensures that the probabilities within the new sample space B are normalized, meaning the total probability sums to 1. For any event B with $P(B) > 0$, consider the conditional probability of B given B:

$$P(B \mid B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

This result intuitively confirms that if B has occurred, the probability of B occurring is certain, i.e., 1.

Note that the elementary formula for $P(A \mid B)$ is undefined when $P(B) = 0$, because it would require division by zero. A probability-zero event need not be empty in a continuous probability model, but conditioning on such events requires more advanced definitions beyond the scope of this chapter.

For a fixed conditioning event B with $P(B) > 0$, the conditional probability measure $P(\cdot \mid B)$ satisfies the following axioms:

Axiom 2.2 (Axioms of Conditional Probability)

- **Axiom 1:** For any event A, $0 \leq P(A \mid B) \leq 1$.
- **Axiom 2:** Conditional probability of B given B is 1, i.e., $P(B \mid B) = 1$.
- **Axiom 3:** If $A_1, A_2, A_3, \cdots$ are pairwise disjoint events, then
$$P\left(A_1 \cup A_2 \cup A_3 \cup \cdots \mid B\right) = P\left(A_1 \mid B\right) + P\left(A_2 \mid B\right) + P\left(A_3 \mid B\right) + \cdots$$

♡

And we are also able to derive other rules of conditional probability from these axioms:

Theorem 2.2 (Rules of Conditional Probability)

For any events A, B, and C with $P(C) > 0$:

- **Complement Rule:** The probability of the complement of event A given C is
$$P\left(A^c \mid C\right) = 1 - P(A \mid C)$$
- **Empty Set Rule:** The probability of the empty set given some event C is 0, i.e.,
$$P(\emptyset \mid C) = 0$$
- **Addition Rule:** For any two events A and B, the probability of the union of events A and B given another event C is
$$P(A \cup B \mid C) = P(A \mid C) + P(B \mid C) - P(A \cap B \mid C)$$
- **Difference Rule:** The probability of the difference between events A and B given another event C is
$$P(A - B \mid C) = P(A \mid C) - P(A \cap B \mid C)$$
- **Subset Rule:** If $A \subset B$, then
$$P(A \mid C) \leq P(B \mid C)$$

♡

The following example illustrates the application of conditional probability and introduces the concept of **contingency tables** as well as **false positives** and **false negatives**.

Example 2.9 A researcher aims to assess the effectiveness of a diagnostic test designed to detect renal disease in patients with high blood pressure. To achieve this, she conducts the test on a sample of 137 patients, categorized

as follows:

- **67 patients** with a confirmed diagnosis of renal disease.
- **70 patients** who are known to be healthy (i.e., do not have renal disease).

The diagnostic test produces one of two possible outcomes for each patient:

- **Positive**: Indicates that the patient has renal disease.
- **Negative**: Indicates that the patient does not have renal disease.

The findings are summarised in the following contingency table:

	Test Results		
Truth	Positive	Negative	Total
Renal Disease	44	23	67
Healthy	10	60	70
Total	54	83	137

In this experiment:

- **True Positives**: Patients who have renal disease and tested positive.
- **False Negatives**: Patients who have renal disease but tested negative.
- **False Positives**: Healthy patients who tested positive.
- **True Negatives**: Healthy patients who tested negative.

Determine the following probabilities:

(a) The Probability of having renal disease, $P(D)$.
(b) The Probability of a positive test, $P(T^+)$.
(c) The Probability of a negative test, $P(T^-)$.
(d) If a person has renal disease, what is the probability that they test positive for the disease?
(e) Determine the probability that a patient has renal disease given a positive test result, $P(D \mid T^+)$.
(f) Determine the probability that a patient does not have renal disease given a negative test result, $P(\text{Healthy} \mid T^-)$.
(g) Assess the overall accuracy of the diagnostic test.

Solution: Using the contingency table, we can calculate the following probabilities:

(a) **Probability of Renal Disease,** $P(D)$:

$$P(D) = \frac{\text{Number of patients with renal disease}}{\text{Total number of patients}}$$
$$= \frac{67}{137} \approx 0.4883 \text{ or } 48.83\%$$

(b) **Probability of a Positive Test,** $P(T^+)$:

$$P(T^+) = \frac{\text{Number of positive tests}}{\text{Total number of patients}}$$
$$= \frac{54}{137} \approx 0.3942 \text{ or } 39.42\%$$

(c) **Probability of a Negative Test,** $P(T^-)$:

$$P(T^-) = \frac{\text{Number of negative tests}}{\text{Total number of patients}} = \frac{83}{137} \approx 0.6058 \text{ or } 60.58\%$$

(d) **Probability of a Positive Test Given Renal Disease,** $P(T^+ \mid D)$:

$$P(T^+ \mid D) = \frac{44}{67} \approx 0.6567$$

(e) **Probability of Renal Disease Given a Positive Test,** $P(D \mid T^+)$:

$$P(D \mid T^+) = \frac{P(D \cap T^+)}{P(T^+)} = \frac{44}{54} = \frac{22}{27} \approx 0.8148$$

(f) **Probability of Being Healthy Given a Negative Test,** $P(\textbf{Healthy} \mid T^-)$:

$$P(\text{Healthy} \mid T^-) = \frac{P(\text{Healthy} \cap T^-)}{P(T^-)} = \frac{60}{83} \approx 0.7229$$

(g) **Overall Accuracy of the Diagnostic Test**: The overall accuracy of the diagnostic test is the proportion of correct diagnoses, i.e., the sum of true positives and true negatives divided by the total number of patients:

$$\text{Overall Accuracy} = \frac{44 + 60}{137} = \frac{104}{137} \approx 0.7591$$

The calculations describe several different performance measures. The sensitivity, which is the probability of a positive result among patients with renal disease, is $44/67 \approx 65.67\%$. The specificity, which is the probability of a negative result among healthy patients, is $60/70 \approx 85.71\%$. By contrast, $P(D \mid T^+) = 81.48\%$ is the positive predictive value, and $P(\text{Healthy} \mid T^-) = 72.29\%$ is the negative predictive value. The overall accuracy is 75.91%. ◀

Here's a classic probability puzzle known as the Two-Child Problem. This scenario has appeared in various forms in the literature, each with subtle twists that lead to different results. Before looking at the calculations, take a moment to make your own predictions — you might be surprised by the outcomes!

Example 2.10 Consider a family with two children, and we are interested in the possible biological gender combinations of the children. The sample space for this situation is:

$$S = \{(G, G), (G, B), (B, G), (B, B)\}$$

where G represents a biological girl (hence 'girl') and B represents a biological boy (hence 'boy'). For simplicity, we assume that all four outcomes are equally likely.

(a) What is the probability that both children are girls given that the first child is a girl?

(b) We ask the father: "Do you have at least one daughter?" He responds "Yes!" Given this extra information, what is the probability that both children are girls? In other words, what is the probability that both children are girls given that we know at least one of them is a girl?

Solution: Let A be the event that both children are girls, i.e., $A = \{(G, G)\}$. Let B be the event that the first child is a girl, i.e., $B = \{(G, G), (G, B)\}$. Finally, let C be the event that at least one of the children is a girl,

i.e., $C = \{(G,G),(G,B),(B,G)\}$. Since the outcomes are equally likely, we can write

$$P(A) = \frac{1}{4}$$
$$P(B) = \frac{2}{4} = \frac{1}{2}$$
$$P(C) = \frac{3}{4}$$

(a) What is the probability that both children are girls given that the first child is a girl? This is $P(A \mid B)$, thus we can write

$$\begin{aligned} P(A \mid B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(A)}{P(B)} \quad (\text{ since } A \subset B) \\ &= \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2} \end{aligned}$$

(b) What is the probability that both children are girls given that we know at least one of them is a girl? This is $P(A \mid C)$, thus we can write

$$\begin{aligned} P(A \mid C) &= \frac{P(A \cap C)}{P(C)} \\ &= \frac{P(A)}{P(C)} \quad (\text{ since } A \subset C) \\ &= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3} \end{aligned}$$

◀

Remark: Many people might intuitively guess that both $P(A \mid B)$ and $P(A \mid C)$ would be 50%. However, while $P(A \mid B) = 50\%$, $P(A \mid C)$ is only 33%. This illustrates how probability can be counterintuitive. The key is to recognize that event B is a subset of event C. Specifically, B excludes the outcome (B,G), which is included in C. As a result, C has more outcomes not in A than B, leading to a smaller $P(A \mid C)$ compared to $P(A \mid B)$.

The result in part (b) assumes that the information supplied is exactly the event C, "at least one child is a girl." If the father was selected or questioned using a different random procedure, the conditional probability can change. The information-generating procedure must therefore be specified when interpreting this puzzle.

2.5 Multiplication and Total Probability Rules

The multiplication rule is a fundamental principle in probability theory that allows us to calculate the probability of multiple events occurring in sequence. This rule is particularly useful when the events are dependent, meaning that the occurrence of one event influences the probability of the subsequent event.

Theorem 2.3 (Multiplication Rule)

If $P(A) > 0$ and $P(B) > 0$, the probability of both events occurring is given by:

$$P(A \cap B) = P(A)P(B \mid A) = P(B)P(A \mid B)$$

♡

Example 2.11 Suppose a software development team is testing a new feature. There are two stages of testing: Unit Testing (A) and Integration Testing (B). The probability that a bug is detected in Unit Testing is $P(A) = 0.3$. If a bug is detected during Unit Testing, the probability that it will also be detected in Integration Testing is $P(B \mid A) = 0.7$. What is the probability that a bug is detected in both stages of testing?

Solution: Using the Multiplication Rule:

$$P(A \cap B) = P(A) \cdot P(B \mid A) = 0.3 \times 0.7 = 0.21$$

Therefore, the probability that a bug is detected in both Unit Testing and Integration Testing is 0.21. ◀

Example 2.12 A company has implemented a two-layer security system to protect against network breaches: **Firewall (A)** and **Intrusion Detection System (IDS) (B)**. The probability that a breach attempt is detected by the Firewall is $P(A) = 0.4$. If the breach passes through the Firewall, the probability that it is detected by the IDS is $P(B \mid A^c) = 0.6$. Given that the Firewall detects the breach, the conditional probability that the IDS also detects it is $P(B \mid A) = 0.8$.

(a) What is the probability that a breach is detected by at least one of the security layers?
(b) What is the probability that a breach is detected by both security layers?

Solution:

(a) **Probability of Detection by at Least One Layer:**
To find the probability that the breach is detected by at least one layer, we calculate the probability of a breach passing through both layers undetected and subtract it from 1.

$$P(\text{Undetected}) = P(A^c) \cdot P(B^c \mid A^c) = (1 - 0.4) \cdot (1 - 0.6) = 0.6 \cdot 0.4 = 0.24$$

Therefore, the probability of detecting the breach with at least one layer is:

$$P(\text{Detected}) = 1 - P(\text{Undetected}) = 1 - 0.24 = 0.76$$

(b) **Probability of Detection by Both Layers:**
Using the Multiplication Rule:

$$P(A \cap B) = P(A) \cdot P(B \mid A) = 0.4 \cdot 0.8 = 0.32$$

The probability of detecting the breach with at least one layer is 0.76, while the probability of detecting the breach with by both the Firewall and the IDS is 0.32. This discrepancy highlights the importance of considering the dependencies between events when calculating probabilities.

◀

Remark: It might seem intuitive to some to calculate the joint probability of detection using $P(A \cap B) = P(A) \cdot P(B \mid A^c) = 0.4 \cdot 0.6 = 0.24$. However, $P(B \mid A^c)$ is the probability that the IDS detects the breach given that the Firewall did not detect it; it is neither $P(B)$ nor $P(B \mid A)$. For both layers to detect the breach, we must use $P(B \mid A)$, which properly accounts for the dependency between the two layers and yields the correct result of 0.32.

In some scenarios, the probability of an event depends on various conditions. By knowing the conditional

probabilities under these different scenarios, we can determine the overall probability of the event. For instance, consider semiconductor manufacturing, where we define:

- A as the event that a chip is highly contaminated
- B as the event that a product using the chip fails

The probability of failure for a non-contaminated chip is $P(B \mid A^c) = 0.005$. On the other hand, if the chip is subjected to high levels of contamination, the probability of failure is $P(B \mid A) = 0.10$. Suppose that in a given production run, 20% of the chips are highly contaminated ($P(A) = 0.20$). What is the probability that a product using one of these chips fails?

The event B can be decomposed according to whether the chip was exposed to high contamination or not. Its two mutually exclusive parts are the events $A \cap B$ and $A^c \cap B$. Mathematically, we can express this as:

$$B = (A \cap B) \cup (A^c \cap B)$$

This decomposition is visualized in the Venn diagram in Figure 2.1. Since A and A^c are mutually exclusive, their intersections with B are also mutually exclusive. Thus, using the rule for the probability of the union of mutually exclusive events and the multiplication rule, we can derive the total probability as follows:

$$P(B) = P(B \cap A) + P(B \cap A^c)$$

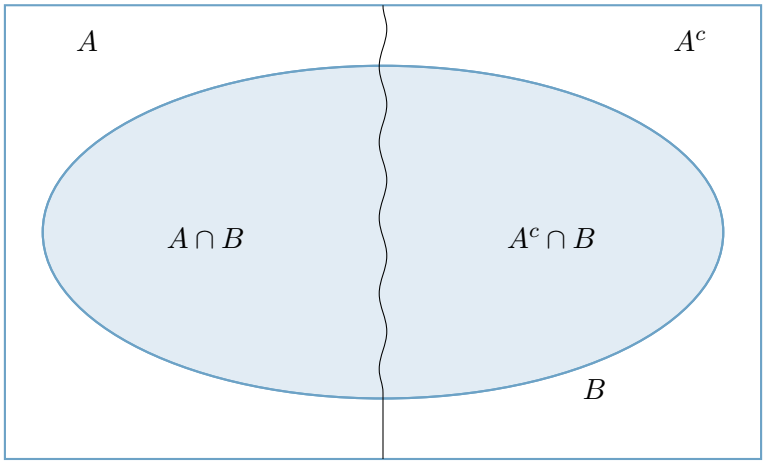


Figure 2.1: $P(B) = P(B \cap A) + P(B \cap A^c)$

We summarise this in the following theorem:

Theorem 2.4 (Law of Total Probability)

For any events A and B with $0 < P(B) < 1$, the total probability of event A is given by:

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B^c) \\ &= P(B)P(A \mid B) + P(B^c)P(A \mid B^c) \end{aligned}$$

For any event A and any finite partition $B_1, B_2, \ldots, B_n$ of the sample space S, with $P(B_i) > 0$ for every i, the total probability of event A is given by:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i)$$

♡

Let us consider some examples.

Example 2.13 A company produces two types of products: **Product A** and **Product B**. The probability that a product is defective is $P(D \mid A) = 0.05$ for Product A and $P(D \mid B) = 0.10$ for Product B. The company manufactures 60% of its products as Product A and 40% as Product B. What is the probability that a randomly selected product is defective?

Solution: Using the Law of Total Probability:

$$P(D) = P(D \mid A)P(A) + P(D \mid B)P(B)$$

Substituting the given values:

$$P(D) = 0.05 \times 0.60 + 0.10 \times 0.40 = 0.03 + 0.04 = 0.07$$

Therefore, the probability that a randomly selected product is defective is 0.07. ◀

A graphical display of a partition of the sample space into mutually exclusive and exhaustive events is shown in Figure 2.2. The events B_1, B_2, B_3, B_4, B_5 partition the sample space S. The total probability of event A is calculated as the sum, over the partition elements, of the conditional probability of A given B_i multiplied by $P(B_i)$.

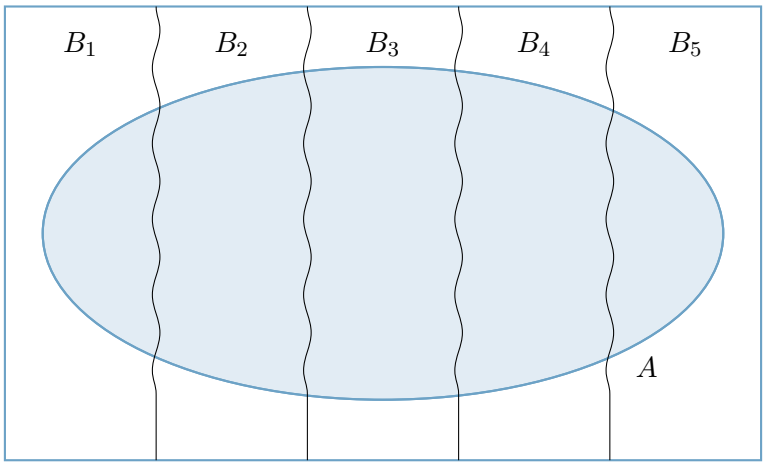


Figure 2.2: $P(A) = \sum_{i=1}^{5} P(A \mid B_i)P(B_i)$

Example 2.14 A university offers three types of courses: **Online**, **Hybrid**, and **In-Person**. The probability that a student fails a course is $P(F \mid \text{Online}) = 0.10$ for Online courses, $P(F \mid \text{Hybrid}) = 0.05$ for Hybrid courses, and $P(F \mid \text{In-Person}) = 0.02$ for In-Person courses. The university offers 50% of its courses as Online, 30% as Hybrid, and 20% as In-Person. What is the probability that a randomly selected student fails a course?

Solution: Using the Law of Total Probability:

$$P(F) = P(F \mid \text{Online})P(\text{Online}) + P(F \mid \text{Hybrid})P(\text{Hybrid}) + P(F \mid \text{In-Person})P(\text{In-Person})$$

Substituting the given values:

$$P(F) = 0.10 \times 0.50 + 0.05 \times 0.30 + 0.02 \times 0.20 = 0.05 + 0.015 + 0.004 = 0.069$$

Therefore, the probability that a randomly selected student fails a course is 0.069. ◀

2.6 Independence

Let A represent the event that it rains tomorrow, with $P(A) = \frac{1}{3}$. Additionally, suppose I toss a fair coin, and let B be the event that it lands heads up, so $P(B) = \frac{1}{2}$.

Now, consider the probability $P(A \mid B)$. What do you think it would be? You might intuitively guess that $P(A \mid B) = P(A) = \frac{1}{3}$, and you would be correct! The coin toss outcome has no influence on the weather forecast. This means that whether B occurs or not, the probability of A remains unchanged. This scenario illustrates the concept of independent events: two events are independent if the occurrence of one does not provide any information about the other.

Let's now formalize the definition of independence.

Definition 2.5 (Independence)

Two events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

When $P(B) > 0$, this is equivalent to $P(A \mid B) = P(A)$, and when $P(A) > 0$, it is equivalent to $P(B \mid A) = P(B)$.

♣

In summary, independence can be understood in two equivalent ways: "Independence means that the probability of the intersection of two events can be found by simply multiplying their individual probabilities," or, alternatively, "Independence means that the conditional probability of one event given the other is the same as the original, unconditioned probability."

Lemma 2.1

If A and B are independent then

- A and B^c are independent,
- A^c and B are independent,
- A^c and B^c are independent.

♡

When dealing with the probability of the union of multiple mutually independent events, $A_1, A_2, \ldots, A_n$, it is often easier to find the probability of their intersection than their union. In these situations, De Morgan's Law can be quite useful:

$$A_1 \cup A_2 \cup \cdots \cup A_n = (A_1^c \cap A_2^c \cap \cdots \cap A_n^c)^c$$

Using this relationship, we can express the probability of the union as:

$$\begin{aligned} P(A_1 \cup A_2 \cup \cdots \cup A_n) &= 1 - P(A_1^c \cap A_2^c \cap \cdots \cap A_n^c) \\ &= 1 - P(A_1^c) P(A_2^c) \cdots P(A_n^c) \quad \text{(since the } A_i\text{'s are mutually independent)} \\ &= 1 - (1 - P(A_1))(1 - P(A_2)) \cdots (1 - P(A_n)). \end{aligned}$$

Theorem 2.5 (Independence and DeMorgan's Law)

If $A_1, A_2, \cdots, A_n$ are mutually independent then

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = 1 - (1 - P(A_1))(1 - P(A_2)) \cdots (1 - P(A_n))$$

♡

Warning! A common misconception is to confuse independence with disjointness. However, these are fundamentally different concepts. Two events, A and B, are disjoint if $A \cap B = \emptyset$. If both events have positive probability, disjointness implies dependence: knowing that one occurred rules out the other. The exception is when at least one of the disjoint events has probability zero; in that case, the product criterion for independence is satisfied.

Concept	Description	Key Formulas
Disjoint	Events A and B cannot occur at the same time	$A \cap B = \emptyset$ $P(A \cup B) = P(A) + P(B)$
Independent	Occurrence of B gives no information about A	$P(A \mid B) = P(A)$ $P(B \mid A) = P(B)$ $P(A \cap B) = P(A) \cdot P(B)$

Table 2.2: Comparison of Disjoint and Independent Events.

We can extend the concept of independence to multiple events. A collection of events $A_1, A_2, \ldots, A_n$ is mutually independent if the occurrence of any subset of these events does not provide any information about the occurrence of the other events. Mathematically, this is expressed as:

Definition 2.6 (Independence of Multiple Events)

Events $A_1, A_2, \ldots, A_k$ are mutually independent if, for every subset $J \subseteq \{1, \ldots, k\}$ containing at least two indices,

$$P\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} P(A_j).$$

Requiring the equality only for the intersection of all k events is not sufficient.

Example 2.15 The Gambler's Fallacy

The **Gambler's Fallacy** is a common cognitive bias that arises when individuals believe that the outcome of a random event is influenced by previous outcomes. This fallacy is often observed in gambling scenarios, where individuals incorrectly assume that the probability of an event occurring is affected by past events.

A man tosses a fair coin eight times and observes whether the toss yields a head (H) or a tail (T) on each toss. Which of the following sequences of coin tosses is the man more likely to get a head (H) on his next toss? This one:

TTTTTTTT

or this one:

HHTHTTHH

The answer is neither as illustrated here:

$$P\left(H_9 \mid T_1T_2\ldots T_8\right) = \frac{P\left(H_9 \cap T_1 \cap T_2 \cap \cdots \cap T_8\right)}{P\left(T_1 \cap T_2 \cap \cdots \cap T_8\right)} = \frac{\left(\frac{1}{2}\right)^9}{\left(\frac{1}{2}\right)^8} = \frac{1}{2}$$

Try to avoid falling into the trap of the Gambler's Fallacy. For example, if a fair coin lands on tails eight times in a row, it's easy to think that a head is "due" on the next toss. However, each toss is independent, and the probability remains the same — there's still a 50% chance of heads or tails, regardless of past results.

2.7 Bayes' Theorem

We are now ready to introduce one of the most powerful tools in conditional probability: Bayes' rule (or theorem). This rule is particularly useful when we know $P(A \mid B)$ but want to find $P(B \mid A)$. Starting with the definition of conditional probability, we have:

$$P(A \mid B) \cdot P(B) = P(A \cap B) = P(B \mid A) \cdot P(A)$$

By dividing both sides by $P(A)$, we arrive at:

$$P(B \mid A) = \frac{P(A \mid B) \cdot P(B)}{P(A)}$$

This formula is famously known as Bayes' rule. In many cases, to calculate $P(A)$ in Bayes' rule, we need the law of total probability. Therefore, Bayes' rule is often presented in the form:

$$P(B \mid A) = \frac{P(A \mid B) \cdot P(B)}{P(A \mid B) \cdot P(B) + P(A \mid B^c) \cdot P(B^c)}$$

or more generally

$$P\left(B_j \mid A\right) = \frac{P\left(A \mid B_j\right) \cdot P\left(B_j\right)}{\sum_i P\left(A \mid B_i\right) \cdot P\left(B_i\right)}$$

where $B_1, B_2, \ldots, B_n$ form a partition of the sample space.

Theorem 2.6 (Bayes' Theorem)

For any two events A and B, where $P(A) \neq 0$, we have

$$P(B \mid A) = \frac{P(A \mid B) \cdot P(B)}{P(A)} = \frac{P(A \mid B) \cdot P(B)}{P(A \mid B) \cdot P(B) + P(A \mid B^c) \cdot P(B^c)}$$

If $B_1, B_2, B_3, \cdots$ form a partition of the sample space S, and A is any event with $P(A) \neq 0$, we have

$$P\left(B_j \mid A\right) = \frac{P\left(A \mid B_j\right) \cdot P\left(B_j\right)}{\sum_i P\left(A \mid B_i\right) \cdot P\left(B_i\right)}$$

♡

We start with an infamous example to highlight the importance of Bayes' Theorem.

Example 2.16 False Positive Paradox

Imagine a rare disease that affects about 1 in every 10,000 people. There is a test available to detect this disease, and while the test is highly accurate, it is not perfect. Specifically:

- The probability that the test shows a positive result (indicating the disease) when the person does not have the disease is 2%.
- The probability that the test shows a negative result (indicating no disease) when the person does have the disease is 1%.

Now, suppose a randomly selected person takes the test, and the result comes back positive. What is the probability that this person actually has the disease?

Solution: Let D be the event that the person has the disease, let T^+ be the event that the test result is positive, and let T^- be the event that it is negative. We are given:

$$P(D) = \frac{1}{10,000}$$
$$P\left(T^+ \mid D^c\right) = 0.02 \quad \text{(False Positive Rate)}$$
$$P\left(T^- \mid D\right) = 0.01 \quad \text{(False Negative Rate)}$$

We want to compute $P(D \mid T^+)$. First, we need $P(T^+ \mid D)$, which is the probability of a true positive. Since a person with the disease will either test positive or negative, we have:

$$P(T^+ \mid D) = 1 - P(T^- \mid D) = 1 - 0.01 = 0.99$$

Now, using Bayes' rule:

$$\begin{aligned} P(D \mid T^+) &= \frac{P(T^+ \mid D)P(D)}{P(T^+ \mid D)P(D) + P\left(T^+ \mid D^c\right)P\left(D^c\right)} \\ &= \frac{0.99 \times 0.0001}{0.99 \times 0.0001 + 0.02 \times (1 - 0.0001)} \\ &= \frac{0.000099}{0.000099 + 0.02 \times 0.9999} \\ &= \frac{0.000099}{0.000099 + 0.019998} \\ &\approx 0.0049 \end{aligned}$$

This result means there is less than half a percent chance that the person actually has the disease. Despite the positive test result, the low prevalence of the disease and the test's false positive rate contribute to this counterintuitive outcome. ◀

Example 2.17 Bayesian networks are commonly used on the websites of high-technology manufacturers to help customers quickly diagnose issues with their products. For instance, a printer manufacturer uses data from test results to identify potential causes of printer failures. Printer failures are primarily associated with three types of problems: hardware, software, and other issues (like connectors). The probabilities of these problems are as follows:

- Probability of a hardware issue: $P(H) = 0.1$
- Probability of a software issue: $P(S) = 0.6$
- Probability of another type of issue: $P(O) = 0.3$

The likelihood of a printer failing, given each type of problem, is:

- Probability of failure given a hardware issue: $P(F \mid H) = 0.9$
- Probability of failure given a software issue: $P(F \mid S) = 0.2$

- Probability of failure given another issue: $P(F \mid O) = 0.5$

Given that a customer experiences a printer failure and uses the manufacturer's website to diagnose the issue, what is the most likely cause of the problem?

Solution: To determine the most likely cause of the printer failure, we need to calculate the probability of each type of problem given that a failure has occurred. This involves using Bayes' theorem to compute the posterior probabilities.

Let F denote the event of a printer failure. We want to find:

$$P(H \mid F), \quad P(S \mid F), \quad P(O \mid F)$$

Step 1: Calculate the Total Probability of Failure

First, we use the law of total probability to find $P(F)$:

$$P(F) = P(F \mid H) \cdot P(H) + P(F \mid S) \cdot P(S) + P(F \mid O) \cdot P(O)$$

Substituting the given values:

$$P(F) = (0.9 \times 0.1) + (0.2 \times 0.6) + (0.5 \times 0.3)$$
$$P(F) = 0.09 + 0.12 + 0.15 = 0.36$$

Step 2: Apply Bayes' Theorem to Find the Posterior Probabilities

Now, apply Bayes' theorem for each problem type:

1. Probability of Hardware Problem Given Failure:

$$P(H \mid F) = \frac{P(F \mid H) \cdot P(H)}{P(F)} = \frac{0.9 \times 0.1}{0.36} = \frac{0.09}{0.36} = 0.25$$

2. Probability of Software Problem Given Failure:

$$P(S \mid F) = \frac{P(F \mid S) \cdot P(S)}{P(F)} = \frac{0.2 \times 0.6}{0.36} = \frac{0.12}{0.36} = 0.3333$$

3. Probability of Other Problems Given Failure:

$$P(O \mid F) = \frac{P(F \mid O) \cdot P(O)}{P(F)} = \frac{0.5 \times 0.3}{0.36} = \frac{0.15}{0.36} = 0.4167$$

Step 3: Interpret the Results

- Hardware Problem: 25% chance
- Software Problem: Approximately 33.33% chance
- Other Problem: Approximately 41.67% chance

Conclusion: Given a printer failure, the most likely cause is an Other Problem, such as connectors, with a probability of approximately 41.67%. ◀

Chapter 3 Random Variables and Discrete Distributions

In chapter 2, probabilities were assigned to events: a component fails, a test returns a positive result, or a system experiences a particular combination of outcomes. In many applications, however, the outcome of interest is numerical. We may want to record the number of failed requests in an hour, the number of defective components in a batch, or the number of attempts required before a transmission succeeds. To analyse such quantities systematically, we introduce the concept of a **random variable**.

A random variable connects the outcomes of a random experiment with numbers. This allows an uncertain process to be described using familiar mathematical ideas such as functions, sums, averages, and variation. The possible values of the variable do not all need to be equally likely. Their probabilities form a distribution, which provides a complete probabilistic description of the variable and makes it possible to answer questions about both individual outcomes and long-run behaviour.

This chapter focuses on **discrete random variables**, whose possible values form a finite or countable set. A packet is either lost or delivered; a test suite contains an integer number of failures; and a queue receives a countable number of requests during a specified interval. These examples lead naturally to three important models. The Bernoulli distribution describes a single trial with two possible outcomes. The binomial distribution describes the number of successes in a fixed number of independent Bernoulli trials. The Poisson distribution describes counts occurring within a fixed interval of time, space, or another exposure measure.

Before studying these models, we develop the general language used to describe discrete random variables. Probability mass functions assign probabilities to individual values, while cumulative distribution functions describe probabilities up to a specified threshold. Expected value and variance then summarise the centre and spread of a distribution. Together, these concepts provide the foundation for selecting, interpreting, and applying discrete probability models throughout statistics and data analysis.

3.1 Random variables, PMFs, and CDFs

An event describes a collection of outcomes, but it does not necessarily describe them numerically. For example, suppose that two components are inspected and that each component either passes or fails. The sample space may be written as

$$S = \{PP, PF, FP, FF\},$$

where PF means that the first component passes and the second fails. If the quantity of interest is the *number* of failed components, the four outcomes are represented by the numbers 0, 1, and 2. A random variable formalises this assignment.

Definition 3.1 (Random Variable)

Let S be the sample space of a random experiment. A **random variable** is a function

$$X : S \longrightarrow \mathbb{R}$$

that assigns a real number $X(\omega)$ to each outcome $\omega \in S$.

♣

The word *variable* can be slightly misleading. A random variable is a function with a fixed rule; it is the

outcome of the underlying experiment, and therefore the resulting numerical value, that is uncertain. Before the experiment is observed, the random variable is denoted by an uppercase letter such as X. A particular realised value is denoted by the corresponding lowercase letter x.

Example 3.1 Number of Failed Components

Let X denote the number of failed components when two components are inspected. The mapping from the sample space to the real numbers is

$$X(PP) = 0, \qquad X(PF) = 1, \qquad X(FP) = 1, \qquad X(FF) = 2.$$

Notice that different outcomes can produce the same value of X. In particular, the event that exactly one component fails is

$$\{X = 1\} = \{PF, FP\}.$$

Thus, a statement involving a random variable defines an event in the original sample space.

Random variables may be discrete or continuous. A discrete random variable takes values in a finite or countably infinite set. Counts are therefore discrete, even when there is no fixed upper bound. A continuous random variable can take values throughout an interval and will be studied in chapter 4.

Definition 3.2 (Discrete Random Variable and Support)

A random variable X is **discrete** if its possible values form a finite or countably infinite set. The set of values with positive probability is called the **support** of X:

$$\operatorname{supp}(X) = \{x \in \mathbb{R} : P(X = x) > 0\}.$$

♣

For the component example, $\operatorname{supp}(X) = \{0, 1, 2\}$. The support records the possible numerical values, while the probability distribution specifies how probability is allocated among them.

The probability mass function

The distribution of a discrete random variable is described by its probability mass function. It gives the probability of every individual value in the support.

Definition 3.3 (Probability Mass Function)

The **probability mass function** (PMF) of a discrete random variable X is the function

$$p_X(x) = P(X = x).$$

For values outside the support of X, $p_X(x) = 0$.

♣

Theorem 3.1 (Properties of a Probability Mass Function)

A function p_X is a valid PMF if and only if

1. $p_X(x) \geq 0$ for every x, and
2. the probabilities over all values in the support sum to one:
$$\sum_{x \in \text{supp}(X)} p_X(x) = 1.$$

For any set $A \subseteq \mathbb{R}$,
$$P(X \in A) = \sum_{x \in A \cap \text{supp}(X)} p_X(x).$$

♡

The final formula is particularly important: probabilities involving a discrete random variable are found by adding the probability masses of the values that satisfy the event.

Example 3.2 PMF for Failed Components

Suppose the two component inspections are independent and that each component fails with probability 0.1. The probabilities associated with the four outcomes are

$$P(PP) = 0.9 \cdot 0.9 = 0.81,$$
$$P(PF) = 0.9 \cdot 0.1 = 0.09,$$
$$P(FP) = 0.1 \cdot 0.9 = 0.09,$$
$$P(FF) = 0.1 \cdot 0.1 = 0.01.$$

Since X counts the number of failures, its PMF is

x	0	1	2
$p_X(x)$	0.81	0.18	0.01

where $p_X(1) = P(PF) + P(FP) = 0.18$. The values are non-negative and

$$0.81 + 0.18 + 0.01 = 1,$$

so the PMF is valid. The probability that at least one component fails is

$$P(X \geq 1) = p_X(1) + p_X(2) = 0.18 + 0.01 = 0.19.$$

Equivalently, the complement rule gives $P(X \geq 1) = 1 - p_X(0) = 1 - 0.81 = 0.19$.

The notation $p_X(x)$ identifies both the random variable and the value being considered. When several random variables occur in the same analysis, the subscript prevents ambiguity. The symbol p without a subscript is often used for a model parameter, so keeping the subscript on a PMF is useful.

Example 3.3 Determining an Unknown PMF Constant

Suppose a discrete random variable X has support $\{0, 1, 2, 3\}$ and PMF

$$p_X(x) = c(x + 1), \qquad x \in \{0, 1, 2, 3\},$$

where c is an unknown constant. Since the probability masses must sum to one,

$$\sum_{x=0}^{3} p_X(x) = c(1+2+3+4) = 10c = 1.$$

Therefore, $c = 0.1$, and the PMF values are

x	0	1	2	3
$p_X(x)$	0.1	0.2	0.3	0.4.

For example,

$$P(X \geq 2) = p_X(2) + p_X(3) = 0.3 + 0.4 = 0.7.$$

The cumulative distribution function

A PMF assigns probability to individual values. The cumulative distribution function instead accumulates all probability at or below a threshold. Unlike a PMF, a CDF is defined for every real number and can therefore be used for discrete as well as continuous random variables.

Definition 3.4 (Cumulative Distribution Function)

The **cumulative distribution function** (CDF) of a random variable X is

$$F_X(x) = P(X \leq x), \qquad x \in \mathbb{R}.$$

For a discrete random variable, the CDF is obtained by summing the probability masses up to x:

$$F_X(x) = \sum_{t \leq x} p_X(t).$$

The summation includes only values t in the support of X. Consequently, the CDF of a discrete random variable is a step function. It remains constant between successive support values and jumps by $p_X(x)$ at each value x in the support.

Theorem 3.2 (Properties of a Cumulative Distribution Function)

Every CDF F_X has the following properties:

1. $0 \leq F_X(x) \leq 1$ for every $x \in \mathbb{R}$;
2. F_X is non-decreasing: if $a < b$, then $F_X(a) \leq F_X(b)$;
3. F_X is right-continuous;
4. $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$.

These properties follow from the interpretation of $F_X(x)$ as accumulated probability. Raising the threshold cannot remove outcomes, and a sufficiently small or sufficiently large threshold eventually includes none or all of the possible values.

Example 3.4 CDF for Failed Components

For the number of failed components, the PMF values are 0.81, 0.18, and 0.01 at 0, 1, and 2, respectively.

Therefore,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.81, & 0 \le x < 1, \\ 0.99, & 1 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

At $x = 1$, for example,

$$F_X(1) = P(X \le 1) = p_X(0) + p_X(1) = 0.81 + 0.18 = 0.99.$$

The CDF jumps by 0.81 at 0, by 0.18 at 1, and by 0.01 at 2. Each jump is exactly the corresponding PMF value.

The CDF is especially convenient for threshold and interval probabilities. For any real numbers $a < b$,

$$P(a < X \le b) = F_X(b) - F_X(a).$$

Tail probabilities follow from the complement rule:

$$P(X > x) = 1 - F_X(x).$$

Care is required with strict and non-strict inequalities for a discrete random variable because an individual value can have positive probability. For an integer-valued random variable,

$$P(X \ge k) = 1 - F_X(k-1),$$

not $1 - F_X(k)$.

Example 3.5 Probabilities from a CDF

Return to the random variable with PMF

$$p_X(0) = 0.1, \qquad p_X(1) = 0.2, \qquad p_X(2) = 0.3, \qquad p_X(3) = 0.4.$$

Its CDF at the support values is

$$F_X(0) = 0.1, \qquad F_X(1) = 0.3, \qquad F_X(2) = 0.6, \qquad F_X(3) = 1.$$

The probability that X is strictly greater than 1 is

$$P(X > 1) = 1 - F_X(1) = 1 - 0.3 = 0.7.$$

The probability that X lies between 1 and 3, excluding 1 but including 3, is

$$P(1 < X \le 3) = F_X(3) - F_X(1) = 1 - 0.3 = 0.7.$$

If the lower endpoint is included instead, then

$$P(1 \le X \le 3) = 1 - F_X(0) = 1 - 0.1 = 0.9.$$

The difference occurs because $P(X = 1) = 0.2$.

The PMF and CDF contain the same information for a discrete random variable. The CDF is obtained by accumulating the PMF, and the PMF is recovered from the jumps of the CDF. Writing

$$F_X(x^-) = \lim_{t \uparrow x} F_X(t)$$

for the value immediately to the left of x, we have

$$p_X(x) = F_X(x) - F_X(x^-).$$

For an integer-valued random variable, this becomes

$$p_X(k) = F_X(k) - F_X(k-1).$$

The choice between PMF and CDF therefore depends on the question. The PMF is natural for exact values such as $P(X = 2)$, whereas the CDF is often more convenient for thresholds, ranges, and tail probabilities.

3.2 Expectation and variance

A probability distribution describes all possible values of a random variable and the probability attached to each value. For many purposes, however, it is useful to summarise the distribution using a small number of quantities. The **expected value** describes its centre, while the **variance** and **standard deviation** describe how widely its probability is spread around that centre.

These quantities are properties of a probability model. They should therefore be distinguished from the descriptive measures introduced in chapter 1. The expected value $E[X]$ is the theoretical mean of the random variable X, whereas a sample mean $\bar{x}$ is calculated from observed data. Similarly, $\mathrm{Var}(X)$ describes variation in the probability distribution, whereas a sample variance s^2 is calculated from a particular sample.

Expected value

Suppose that a discrete random variable X has support $\{x_1, x_2, \ldots\}$. Its expected value is a probability-weighted average of these possible values. Values with high probability contribute more than values with low probability.

Definition 3.5 (Expected Value of a Discrete Random Variable)

Let X be a discrete random variable with PMF p_X. The **expected value**, or **mean**, of X is

$$E[X] = \sum_{x \in \mathrm{supp}(X)} x\, p_X(x),$$

provided that the sum exists. The expected value is also denoted by μ_X or simply μ when the random variable is clear from context. ♣

The expected value is not necessarily a possible value of the random variable. It describes the long-run average obtained by repeating the random experiment under the same conditions, not the result expected in any one trial.

Example 3.6 Expected Number of Failed Components

For the number X of failed components introduced in section 3.1, the PMF is

x	0	1	2
$p_X(x)$	0.81	0.18	0.01.

The expected number of failed components is therefore

$$\begin{aligned} E[X] &= 0 \cdot 0.81 + 1 \cdot 0.18 + 2 \cdot 0.01 \\ &= 0.20. \end{aligned}$$

This does not mean that 0.20 of a component will fail in a particular inspection. It means that over many inspections of two components, the average number of failures per inspection will approach 0.20 if the model is appropriate. For example, across 1000 such inspections, the model predicts approximately $1000 \cdot 0.20 = 200$ failed components in total.

Example 3.7 Expected Value of a Fair Die

Let X be the number shown on a fair six-sided die. Since every value has probability $1/6$,

$$\begin{aligned} E[X] &= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \cdots + 6 \cdot \frac{1}{6} \\ &= \frac{1+2+3+4+5+6}{6} \\ &= 3.5. \end{aligned}$$

No single roll can produce 3.5. The value is nevertheless the centre of the probability distribution and the long-run average of repeated rolls.

Expected value is linear. Multiplying every possible value by a constant multiplies the expected value by the same constant, and adding a constant shifts the expected value by that amount.

Theorem 3.3 (Linearity of Expected Value)

If X is a random variable with a finite expected value and a and b are constants, then

$$E[aX + b] = aE[X] + b.$$

♡

More generally, an engineering quantity of interest may be a function of a random variable. If $Y = g(X)$, it is not necessary to determine the complete PMF of Y before calculating its expectation. Each possible value $g(x)$ can be weighted directly by the probability $p_X(x)$.

Theorem 3.4 (Expectation of a Function of a Random Variable)

Let X be a discrete random variable with PMF p_X, and let g be a real-valued function. Then

$$E[g(X)] = \sum_{x \in \text{supp}(X)} g(x) p_X(x),$$

provided that the sum exists.

♡

The linearity result follows by choosing $g(x) = ax + b$. The more general form is useful when costs, energy consumption, response penalties, or other consequences depend nonlinearly on the outcome.

Example 3.8 Expected Inspection Cost

Suppose that inspecting a pair of components has a fixed setup cost of EUR 50, and every failed component produces an additional cost of EUR 600. If X is the number of failed components, the total cost is

$$C = 50 + 600X.$$

Using linearity and the result $E[X] = 0.20$,

$$E[C] = 50 + 600E[X] = 50 + 600 \cdot 0.20 = 170.$$

Thus, the expected cost is EUR 170 per inspection. This is a long-run average cost; an individual inspection costs EUR 50, EUR 650, or EUR 1250 depending on whether zero, one, or two components fail.

Example 3.9 Expected Loss under a Nonlinear Penalty

Suppose instead that a simultaneous failure of both components is especially disruptive. Let $L = g(X)$ be the loss, where $g(0) = 0$, $g(1) = 500$, and $g(2) = 5000$, measured in euros. The expected loss is

$$\begin{aligned} E[L] &= g(0)p_X(0) + g(1)p_X(1) + g(2)p_X(2) \\ &= 0 \cdot 0.81 + 500 \cdot 0.18 + 5000 \cdot 0.01 \\ &= 140. \end{aligned}$$

Using $g(E[X])$ would not give the correct result: the penalty is nonlinear, and $g(0.20)$ is not even defined by the stated loss table. The probabilities must be applied to the possible losses themselves.

Variance and standard deviation

Two distributions can have the same expected value while assigning probability very differently around it. Expected value alone therefore does not describe the uncertainty in a random variable. Variance measures the expected squared distance from the mean.

Definition 3.6 (Variance and Standard Deviation)

Let X be a random variable with mean $\mu_X = E[X]$. The **variance** of X is

$$\operatorname{Var}(X) = E\big[(X - \mu_X)^2\big].$$

For a discrete random variable,

$$\operatorname{Var}(X) = \sum_{x \in \operatorname{supp}(X)} (x - \mu_X)^2 p_X(x).$$

The **standard deviation** of X is

$$\sigma_X = \sqrt{\operatorname{Var}(X)}.$$

♣

The distances from the mean are squared for two reasons. First, positive and negative deviations cannot cancel. Second, large deviations receive greater weight than small deviations. Variance is always non-negative and equals zero only when X is constant with probability one.

Because the deviations are squared, variance has squared units. If X is measured in milliseconds, then $\operatorname{Var}(X)$ is measured in square milliseconds. Standard deviation has the same unit as X and is therefore usually easier to interpret directly.

The definition of variance is conceptually important, but the following equivalent formula is often more convenient for calculation.

Theorem 3.5 (Computational Formula for Variance)

If $E[X^2]$ is finite, then

$$\operatorname{Var}(X) = E[X^2] - \big(E[X]\big)^2.$$

♡

Example 3.10 Variance of the Number of Failed Components

For the component model, $E[X] = 0.20$. To use the computational formula, first calculate

$$\begin{aligned} E[X^2] &= 0^2 \cdot 0.81 + 1^2 \cdot 0.18 + 2^2 \cdot 0.01 \\ &= 0.22. \end{aligned}$$

Therefore,

$$\operatorname{Var}(X) = E[X^2] - \big(E[X]\big)^2 = 0.22 - 0.20^2 = 0.18,$$

and

$$\sigma_X = \sqrt{0.18} \approx 0.424.$$

The expected number of failures is 0.20 per inspection, and the standard deviation is approximately 0.424 failures.

Variance responds differently from expectation to a linear transformation. Adding a constant shifts every possible value and the mean by the same amount, so it does not change the distances from the mean. Multiplication by a multiplies every distance by a and every squared distance by a^2.

Theorem 3.6 (Variance under a Linear Transformation)

If X has finite variance and a and b are constants, then

$$\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

Consequently,

$$\sigma_{aX+b} = |a|\sigma_X.$$

♡

Example 3.11 Variation in Inspection Cost

For the inspection cost $C = 50 + 600X$, the fixed setup cost changes the mean but contributes no variation. Hence,

$$\begin{aligned} \operatorname{Var}(C) &= 600^2 \operatorname{Var}(X) \\ &= 600^2 \cdot 0.18 \\ &= 64\,800, \end{aligned}$$

measured in square euros. The standard deviation is

$$\sigma_C = 600\sqrt{0.18} \approx 254.56 \text{ EUR}.$$

Thus, the model gives an expected cost of EUR 170 with a standard deviation of approximately EUR 255. The standard deviation is large relative to the expected cost because the usually low cost is accompanied by a small probability of a much larger failure cost.

The mean and variance do not determine the distribution

Expected value and variance provide useful summaries, but they do not replace the PMF or CDF. Even distributions with the same mean and variance can differ in other ways, and distributions with the same mean can have very different levels of uncertainty.

Example 3.12 Same Mean, Different Variability

Consider two random variables X and Y with PMFs

x	0	1	2
$p_X(x)$	0.25	0.50	0.25
$p_Y(x)$	0.50	0	0.50.

Both variables have expected value one:

$$E[X] = 0 \cdot 0.25 + 1 \cdot 0.50 + 2 \cdot 0.25 = 1$$

and

$$E[Y] = 0 \cdot 0.50 + 1 \cdot 0 + 2 \cdot 0.50 = 1.$$

For X,

$$E[X^2] = 0^2 \cdot 0.25 + 1^2 \cdot 0.50 + 2^2 \cdot 0.25 = 1.5,$$

so

$$\mathrm{Var}(X) = 1.5 - 1^2 = 0.5.$$

For Y,

$$E[Y^2] = 0^2 \cdot 0.50 + 1^2 \cdot 0 + 2^2 \cdot 0.50 = 2,$$

so

$$\mathrm{Var}(Y) = 2 - 1^2 = 1.$$

Although the two models have the same expected value, Y never takes the central value 1 and places all its probability at the two extremes. Its variance is therefore twice as large. A decision based only on the expected value would overlook this difference in uncertainty.

Expectation and variance will be calculated for each of the named distributions in the following sections. Their values help describe and compare the models, but the choice of model must still be based on the underlying random experiment and the complete form of the distribution.

3.3 Bernoulli and binomial models

Many engineering measurements begin with a binary question. Did a request complete successfully? Was an intrusion attempt blocked? Did the temperature exceed a specified threshold? Was a delivery late? Did a bearing fail, a casting pass inspection, or a structural element satisfy its load requirement? Each question has two possible answers. A **Bernoulli model** describes one such trial, while a **binomial model** describes the number of successes in a fixed collection of repeated Bernoulli trials.

The words *success* and *failure* are conventional labels only. A success is the outcome coded as 1 and may represent an undesirable event such as a defect, packet loss, late delivery, or security alert. What matters is that the event of interest is defined clearly before the probability model is constructed.

Bernoulli trials

A single trial with two possible outcomes can be represented numerically by assigning the value 1 to the event of interest and 0 to its complement. This numerical representation is also called an indicator random variable.

Definition 3.7 (Bernoulli Random Variable)

A random variable X has a **Bernoulli distribution** with parameter p if

$$P(X = 1) = p \qquad \text{and} \qquad P(X = 0) = 1 - p,$$

where $0 \le p \le 1$. We write

$$X \sim \text{Bernoulli}(p).$$

The value 1 denotes success and the value 0 denotes failure. ♣

The PMF can be written compactly as

$$p_X(x) = p^x (1-p)^{1-x}, \qquad x \in \{0, 1\}.$$

When $x = 1$, this expression equals p; when $x = 0$, it equals $1 - p$. The CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

Theorem 3.7 (Mean and Variance of a Bernoulli Random Variable)

If $X \sim \text{Bernoulli}(p)$, then

$$E[X] = p, \qquad \text{Var}(X) = p(1-p).$$

♡

The variance is zero when $p = 0$ or $p = 1$ because the outcome is then certain. It is largest when $p = 0.5$, where the two outcomes are equally likely and the uncertainty about a single trial is greatest.

Example 3.13 Cybersecurity Alert for One Authentication Attempt

Suppose a monitoring rule flags an authentication attempt with probability 0.03 under a specified operating condition. Let

$$X = \begin{cases} 1, & \text{the attempt is flagged,} \\ 0, & \text{the attempt is not flagged.} \end{cases}$$

Then $X \sim \text{Bernoulli}(0.03)$, and

$$P(X = 1) = 0.03, \qquad P(X = 0) = 0.97.$$

Its mean and variance are

$$E[X] = 0.03$$

and

$$\text{Var}(X) = 0.03(1 - 0.03) = 0.0291.$$

The expected value 0.03 is also the long-run proportion of attempts that the model predicts will be flagged. It is not, by itself, the probability that a flagged attempt is genuinely malicious; that would require information about the accuracy of the rule and the prevalence of malicious attempts, as discussed in section 2.7.

From Bernoulli trials to a binomial count

Suppose the same type of Bernoulli trial is repeated n times and that X counts the number of successes. A binomial model is appropriate only when the structure of the repeated experiment satisfies specific assumptions.

Definition 3.8 (Binomial Experiment)

A **binomial experiment** consists of trials satisfying all four conditions:

1. the number of trials n is fixed in advance;
2. each trial has exactly two outcomes, labelled success and failure;
3. the probability of success is the same value p on every trial; and
4. the trials are independent.

The random variable of interest is the number of successes among the n trials. ♣

The assumptions describe the data-generating process, not merely the observed data. Seeing 7 failures in 100 tests does not establish a binomial model. We must also have a reason to regard the 100 tests as a fixed number of independent trials with a common failure probability.

The binomial coefficient

To derive the probability of exactly k successes, first consider one particular ordering. For example, the sequence

$$SSFSS$$

contains four successes and one failure. If the trials are independent, the probability of this particular sequence is

$$p^4(1 - p).$$

Every other sequence with four successes and one failure has the same probability. We therefore need to count how many different sequences contain exactly k successes among n positions.

Definition 3.9 (Binomial Coefficient)

For integers $0 \le k \le n$, the **binomial coefficient**

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

is the number of ways to choose the k positions occupied by successes among n positions. By convention, $0! = 1$.

♣

Example 3.14 Placing Two Defects among Five Units

Suppose five units are inspected and exactly two are defective. Once the two positions containing defects have been chosen, the entire defect pattern is determined. The number of possible patterns is

$$\binom{5}{2} = \frac{5!}{2!3!} = 10.$$

If each unit is independently defective with probability p, every one of these ten patterns has probability

$$p^2(1-p)^3.$$

Consequently, the total probability of exactly two defective units is

$$10p^2(1-p)^3.$$

This is the only counting principle needed for the binomial model: $\binom{n}{k}$ counts the possible placements of k successes, while independence gives the probability of each placement.

The binomial distribution

Combining the number of possible placements with the probability of each placement gives the binomial PMF.

Definition 3.10 (Binomial Distribution)

Let X be the number of successes in n independent Bernoulli trials, each with success probability p. Then X has a **binomial distribution**, written

$$X \sim \mathrm{Bin}(n, p),$$

with PMF

$$p_X(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

♣

The four factors in the PMF have distinct meanings:

$$\underbrace{\binom{n}{k}}_{\text{placements}} \underbrace{p^k}_{\text{successes}} \underbrace{(1-p)^{n-k}}_{\text{failures}}.$$

The support is $\{0, 1, \ldots, n\}$. For an integer k in this support, the CDF is

$$F_X(k) = P(X \le k) = \sum_{j=0}^{k} \binom{n}{j} p^j (1-p)^{n-j}.$$

Theorem 3.8 (Mean and Variance of a Binomial Random Variable)

If $X \sim \text{Bin}(n, p)$, then

$$E[X] = np, \qquad \text{Var}(X) = np(1-p),$$

and

$$\sigma_X = \sqrt{np(1-p)}.$$

♡

The mean has a direct interpretation. Each of the n trials contributes a success with probability p, so the expected number of successes is n times p. The variance also adds across the independent Bernoulli trials, giving n times the Bernoulli variance $p(1-p)$. Independence is essential for this variance result.

Example 3.15 Defective Units in a Production Batch

Suppose a production process has a defect probability of 0.03 for each unit. Twenty units are selected under conditions that make their defect indicators reasonably independent, and X denotes the number of defective units. Under the stated assumptions,

$$X \sim \text{Bin}(20, 0.03).$$

The probability of exactly one defective unit is

$$\begin{aligned} P(X = 1) &= \binom{20}{1}(0.03)^1(0.97)^{19} \\ &\approx 0.3364. \end{aligned}$$

The expected number and standard deviation are

$$E[X] = 20 \cdot 0.03 = 0.60$$

and

$$\sigma_X = \sqrt{20 \cdot 0.03 \cdot 0.97} \approx 0.763.$$

Thus, the model predicts an average of 0.60 defective units per group of twenty. This does not imply that most inspected groups contain exactly 0.60 defects, which is not a possible count.

Exact, cumulative, and tail probabilities

The wording of a problem determines whether the PMF, CDF, or complement rule is most convenient. For $X \sim \text{Bin}(n, p)$,

$$P(X = k) = p_X(k)$$

is an exact probability,

$$P(X \le k) = F_X(k)$$

is cumulative, and

$$P(X \ge k) = 1 - F_X(k-1)$$

is an upper-tail probability. For integers $a \le b$,

$$P(a \le X \le b) = F_X(b) - F_X(a-1).$$

Careful attention to the endpoints prevents the off-by-one errors discussed for general CDFs in section 3.1.

For realistic values of n, p, and λ, exact, cumulative, and tail probabilities are normally evaluated with statistical software such as Python. The formulas remain the foundation of the calculation: they identify the model, translate the engineering event into probability notation, and determine which tail or interval is required. Software evaluates the resulting sums accurately, but it does not select the model or correct an event that has been formulated incorrectly.

Example 3.16 Packet Loss in a Software System

Suppose a message contains 100 packets. Under a simplified model, each packet is lost independently with probability 0.01, and X is the number of packets that must be retransmitted. Then

$$X \sim \text{Bin}(100, 0.01).$$

The probability that at least one packet must be retransmitted is most easily found using the complement:

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - \binom{100}{0}(0.01)^0(0.99)^{100} \\ &= 1 - (0.99)^{100} \\ &\approx 0.6340. \end{aligned}$$

For at least two retransmissions, subtract the probabilities of zero and one:

$$\begin{aligned} P(X \geq 2) &= 1 - P(X = 0) - P(X = 1) \\ &= 1 - (0.99)^{100} - \binom{100}{1}(0.01)(0.99)^{99} \\ &\approx 0.2642. \end{aligned}$$

The mean and standard deviation are

$$E[X] = 100 \cdot 0.01 = 1$$

and

$$\sigma_X = \sqrt{100 \cdot 0.01 \cdot 0.99} \approx 0.995.$$

Although only one packet is lost on average, the probability of at least one retransmission is approximately 63.4%. Expected counts and event probabilities answer different questions.

Example 3.17 Late Deliveries in a Supply Chain

A supplier schedules 12 deliveries during a planning period. Suppose each delivery is late with probability 0.08, independently of the others, and let X be the number of late deliveries. Then

$$X \sim \text{Bin}(12, 0.08).$$

The probability of at most one late delivery is

$$\begin{aligned} P(X \le 1) &= P(X = 0) + P(X = 1) \\ &= (0.92)^{12} + \binom{12}{1}(0.08)(0.92)^{11} \\ &\approx 0.7513. \end{aligned}$$

The probability of exactly two late deliveries is

$$\begin{aligned} P(X = 2) &= \binom{12}{2}(0.08)^2(0.92)^{10} \\ &\approx 0.1835. \end{aligned}$$

The expected number of late deliveries is

$$E[X] = 12 \cdot 0.08 = 0.96.$$

These calculations are meaningful only if a common delay mechanism, such as the same port closure or severe weather system, does not create substantial dependence among deliveries.

Example 3.18 Structural Fastener Tests

Twelve structural fasteners are subjected to identical acceptance tests. Suppose each fastener passes with probability 0.98, independently of the others, and let X be the number that pass. Then

$$X \sim \mathrm{Bin}(12, 0.98).$$

The probability that all fasteners pass is

$$P(X = 12) = (0.98)^{12} \approx 0.7847.$$

The probability that at least eleven pass is

$$\begin{aligned} P(X \ge 11) &= P(X = 11) + P(X = 12) \\ &= \binom{12}{11}(0.98)^{11}(0.02) + (0.98)^{12} \\ &\approx 0.9769. \end{aligned}$$

Requiring every fastener to pass is therefore much more stringent than allowing one failure: the corresponding probabilities are approximately 78.5% and 97.7% under this model.

Assessing the binomial assumptions

A binomial calculation can be numerically correct and still be an inappropriate analysis if its assumptions do not match the engineering process. Each assumption should be connected to a feature of the application.

Example 3.19 Can Exceedance Days Be Modelled as Binomial?

An environmental engineer records whether a river exceeds a warning level on each of 30 consecutive days and proposes the model

$$X \sim \text{Bin}(30, p),$$

where X is the number of exceedance days. The number of trials is fixed and every day has two coded outcomes. However, the remaining assumptions require scrutiny.

Exceedances on consecutive days are likely to be dependent because river levels persist and several days may be influenced by the same storm. The probability of exceedance may also vary with season, upstream releases, and soil saturation. Consequently, a binomial model is not justified merely because the response is binary on each day. It may be a rough approximation under stable conditions and weak dependence, but those conditions must be argued rather than assumed.

Dependence also arises through common-cause mechanisms. Fasteners installed in the same concrete pour may share material and installation conditions. Software requests handled by the same failing service instance may fail together. Deliveries routed through the same port may be delayed together. In each case, treating the outcomes as independent usually understates the chance of clusters of failures.

The constant-probability assumption can fail when equipment wears, operators learn, weather changes, or suppliers differ. Sampling without replacement from a small finite lot creates another problem: after a defective item is selected, the composition of the remaining lot changes. For example, drawing ten bearings without replacement from a lot of fifty containing four defective bearings does not produce independent Bernoulli trials with a constant defect probability. A binomial model may sometimes be an adequate approximation when the sampled fraction is small, but it is not the exact model.

The Bernoulli and binomial distributions therefore provide more than formulas. They express a particular structure: one binary trial, or a fixed number of comparable and independent binary trials. In the next section, the focus shifts from a fixed number of trials to events counted over an interval of time, space, area, volume, or another exposure measure.

3.4 Poisson models

The binomial distribution counts successes among a fixed number of trials. Many engineering counts arise differently. A server may experience an uncertain number of failures during a day, a sheet of material may contain an uncertain number of surface flaws, or a warehouse may receive an uncertain number of urgent orders during an hour. In these situations, the number of opportunities is not naturally fixed in advance. Instead, events are counted within a specified **exposure**, such as time, distance, area, volume, number of transactions, or operating hours.

The **Poisson distribution** is a fundamental model for such counts. It is governed by one parameter, λ, which is the expected number of events in the stated exposure. The exposure is part of the parameter definition: an expected count of two events per hour is not the same model as an expected count of two events per day.

When a Poisson model is plausible

A Poisson model is often considered when the following features are reasonable approximations:

1. X counts identifiable events in a clearly specified exposure;
2. the average event rate is approximately stable throughout that exposure;
3. occurrences do not substantially increase or decrease the chance of further occurrences, so clustering and inhibition are limited; and
4. there is no small fixed upper bound on the count that is central to the problem.

These are modelling conditions, not facts implied by observing count data. A column of non-negative integers is not automatically Poisson distributed. The mechanism producing the counts and the definition of the exposure must support the model.

Definition 3.11 (Poisson Distribution)

A discrete random variable X has a **Poisson distribution** with parameter $\lambda > 0$ if

$$P(X = k) = p_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

We write

$$X \sim \text{Pois}(\lambda).$$

♣

Here, $e \approx 2.71828$ is the base of the natural logarithm, and $0! = 1$. The support is the set of all non-negative integers. Although the mathematical model has no upper bound, very large counts may have extremely small probabilities when λ is modest.

The parameter λ is both the centre of the distribution and an expected count for one specific exposure. If a system has a stable average rate r events per unit of exposure and the exposure has size t, then

$$\lambda = rt.$$

The units cancel in the multiplication, but they remain essential to the interpretation. For example,

$$\left(0.8\ \frac{\text{failures}}{\text{hour}}\right)(4\ \text{hours}) = 3.2\ \text{failures}.$$

Thus, the appropriate Poisson parameter for four hours is 3.2, not 0.8.

PMF, CDF, and tail probabilities

The PMF gives the probability of an exact count. For $X \sim \text{Pois}(\lambda)$,

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}.$$

The CDF is obtained by summing the PMF:

$$F_X(k) = P(X \le k) = \sum_{j=0}^{k} \frac{e^{-\lambda}\lambda^j}{j!}.$$

As with the binomial distribution, complements are often the most efficient way to calculate upper-tail probabilities:

$$P(X \ge k) = 1 - F_X(k-1).$$

In particular,

$$P(X = 0) = e^{-\lambda}$$

and

$$P(X \ge 1) = 1 - e^{-\lambda}.$$

Example 3.20 Software Failures during an Operating Window

A service experiences failures at an average rate of 0.8 per hour under stable operating conditions. Let X be the number of failures during a four-hour monitoring window. The parameter for this exposure is

$$\lambda = \left(0.8 \frac{\text{failures}}{\text{hour}}\right)(4 \text{ hours}) = 3.2 \text{ failures},$$

so

$$X \sim \text{Pois}(3.2).$$

The probability of no failures is

$$P(X = 0) = e^{-3.2} \approx 0.0408.$$

The probability of at most two failures is

$$\begin{aligned} P(X \le 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= e^{-3.2}\left(1 + 3.2 + \frac{3.2^2}{2!}\right) \\ &\approx 0.3799. \end{aligned}$$

Therefore,

$$P(X \ge 3) = 1 - P(X \le 2) \approx 0.6201.$$

Under this model, three or more failures during the four-hour window are more likely than fewer than three. Whether that rate is operationally acceptable is an engineering decision, not a conclusion supplied by the probability model alone.

Mean and variance

The Poisson distribution has the distinctive property that its mean and variance are equal.

Theorem 3.9 (Mean and Variance of a Poisson Random Variable)

If $X \sim \text{Pois}(\lambda)$, then

$$E[X] = \lambda, \qquad \text{Var}(X) = \lambda,$$

and

$$\sigma_X = \sqrt{\lambda}.$$

♡

The equality of mean and variance is a property of the theoretical model. It does not mean that the sample mean and sample variance calculated from a finite dataset must be exactly equal. Their relationship can nevertheless be useful when assessing whether a Poisson model is plausible.

Example 3.21 Surface Flaws in Manufactured Panels

Suppose surface flaws occur at an average rate of 0.4 per square metre, and a panel has an area of 5 square metres. If the rate is stable across the surface and flaws occur without substantial clustering, then

$$\lambda = \left(0.4 \, \frac{\text{flaws}}{\text{m}^2}\right)(5 \text{ m}^2) = 2 \text{ flaws}.$$

Let X be the number of flaws on one panel. Then

$$X \sim \text{Pois}(2).$$

The probability of exactly three flaws is

$$\begin{aligned} P(X = 3) &= \frac{e^{-2}2^3}{3!} \\ &\approx 0.1804. \end{aligned}$$

The probability of at least one flaw is

$$P(X \geq 1) = 1 - e^{-2} \approx 0.8647.$$

Finally,

$$E[X] = 2, \qquad \sigma_X = \sqrt{2} \approx 1.414.$$

The model therefore predicts two flaws per panel on average, with a standard deviation of approximately 1.41 flaws.

Changing the exposure

When the rate is constant, changing the exposure changes λ proportionally. If the event rate is r per unit and the exposure changes from t_1 to t_2, then

$$\lambda_1 = rt_1 \qquad \text{and} \qquad \lambda_2 = rt_2.$$

It is incorrect to reuse a parameter defined for one exposure without rescaling it to the exposure in the probability question.

Example 3.22 Urgent Orders during Twenty Minutes

A distribution centre receives urgent orders at an average rate of six per hour. Let X be the number received during the next twenty minutes. Since twenty minutes is one third of an hour,

$$\lambda = \left(6 \ \frac{\text{orders}}{\text{hour}}\right)\left(\frac{1}{3}\ \text{hour}\right) = 2 \text{ orders}.$$

Thus, $X \sim \text{Pois}(2)$. The probability of more than four urgent orders is

$$\begin{aligned} P(X > 4) &= 1 - P(X \leq 4) \\ &= 1 - \sum_{k=0}^{4} \frac{e^{-2}2^k}{k!} \\ &\approx 0.0527. \end{aligned}$$

The probability of more than four arrivals in twenty minutes is therefore approximately 5.3%. Using $\lambda = 6$ would answer a different question: the number of arrivals in one hour.

Poisson approximation to the binomial distribution

The Poisson distribution can also approximate a binomial distribution when the number of trials is large, the success probability is small, and

$$\lambda = np$$

is of moderate size. The approximation replaces

$$X \sim \text{Bin}(n, p)$$

with

$$X \approx \text{Pois}(np).$$

It is especially useful for rare-event counts when evaluating the binomial PMF directly is inconvenient. The approximation does not remove the binomial assumptions: the underlying trials must still be independent with a common success probability.

Example 3.23 Rare Security Incidents

Suppose $10\,000$ independent connection attempts each have probability 0.0002 of producing a particular security incident. If X is the number of incidents, the exact model is

$$X \sim \text{Bin}(10\,000, 0.0002).$$

Since n is large and p is small, use the Poisson parameter

$$\lambda = np = 10\,000 \cdot 0.0002 = 2.$$

The probability of at least three incidents is approximately

$$\begin{aligned} P(X \geq 3) &\approx 1 - P(Y \leq 2), \qquad Y \sim \text{Pois}(2), \\ &= 1 - e^{-2}\left(1 + 2 + \frac{2^2}{2!}\right) \\ &\approx 0.3233. \end{aligned}$$

The exact binomial probability is also 0.3233 to four decimal places in this case. Such close agreement is not guaranteed when p is larger or when n is small.

Assessing the Poisson assumptions

The mean–variance relationship provides a useful diagnostic. If repeated counts from comparable exposures have a sample variance substantially larger than their sample mean, the data are said to show **overdispersion** relative to a Poisson model. Possible causes include clustering, unobserved differences between exposures, a changing event rate, or common-cause failures. A sample variance substantially smaller than the sample mean is called **underdispersion** and may arise from spacing, inhibition, capacity limits, or unusually regular processes.

Neither pattern proves that a Poisson model is wrong, because sample summaries vary from one dataset to another. They identify features that should be investigated using process knowledge, visualisation, and later statistical methods.

Example 3.24 Annual Flood-Warning Events

Suppose annual counts of flood-warning events over comparable years have a sample mean of 3.5 and a sample variance of 9.1. A Poisson model with $\lambda = 3.5$ would have theoretical variance 3.5, considerably smaller than the observed sample variance.

The difference suggests overdispersion. A plausible explanation is that some years are much wetter than others and that several warnings may arise from the same persistent weather system. The annual rate is then not stable across years, and events within a year may be clustered. The comparison is a diagnostic rather than a formal rejection of the model, but it gives a clear reason to examine the mechanism before using Poisson tail probabilities for planning.

A Poisson model is also inappropriate when the count has a small fixed maximum that matters. The number of failed components among twenty tested components cannot exceed twenty and naturally begins with a binomial model. By contrast, the number of failures during a period of operation may have no comparable fixed trial count and can be a plausible Poisson quantity. The model choice follows from the event-generating structure, not simply from whether the observed variable is a count.

Together, the Bernoulli, binomial, and Poisson distributions cover three common engineering structures: one binary outcome, successes among a fixed number of trials, and events within a specified exposure. Selecting among them requires both a correctly calculated probability and a credible account of how the data are generated.

The models in this chapter describe binary outcomes and event counts. Many engineering quantities, however, are measured on a continuous scale, including response times, dimensions, temperatures, and component lifetimes. The next chapter develops probability models for such continuous measurements.

Chapter 4 Continuous Distributions in Practice

Many engineering quantities are measured on a continuous scale. Response times are recorded in milliseconds, component dimensions in millimetres, temperatures in degrees, and lifetimes in hours or cycles. Although an instrument reports a finite number of digits, the underlying quantity is usually modelled as capable of taking any value in an interval. Continuous probability models provide a mathematical description of the uncertainty in such measurements.

This setting differs fundamentally from the discrete models developed in the previous chapter. A discrete random variable assigns probability to individual values through a probability mass function. For a continuous random variable, probabilities are assigned to intervals instead. The probability of observing any one exact value is zero, even though intervals containing that value can have positive probability. A probability density describes how probability is distributed locally, while the cumulative distribution function records the probability of falling below a threshold.

The chapter develops three continuous models that occur frequently in engineering. The uniform distribution represents uncertainty spread evenly across a bounded interval. The normal distribution describes symmetric variation around a central value and is widely used for measurement error and stable process variation. The exponential distribution describes waiting times and lifetimes under a constant event-rate model, providing a direct connection with the Poisson counts introduced previously. Each distribution is an idealised model whose usefulness depends on the mechanism that generated the data and on whether its assumptions are credible.

We first introduce densities, distribution functions, survival probabilities, and quantiles. We then study the uniform, normal, and exponential distributions, including standardisation and interval calculations. Finally, we consider how graphical evidence, especially quantile–quantile plots, can be used to assess whether a proposed model is a reasonable approximation. The emphasis throughout is on interpretation and model choice rather than on manual integration.

4.1 Continuous Random Variables, Densities, and Distribution Functions

A count changes in separate steps: a system can receive 12 or 13 requests, but not 12.4 requests. A measurement behaves differently. Between two possible response times there is, at least conceptually, another possible response time. This distinction motivates a different way of assigning probability.

Continuous random variables and support

Definition 4.1 (Continuous Random Variable)

A random variable X is **continuous** if its probabilities can be described by a probability density function over intervals of real numbers. The set of values on which the density is positive is called the **support** of the distribution. ♣

The support may be bounded, such as the interval from 0 to 10 volts, or unbounded, such as all positive waiting times. It may also be the complete real line, as in the normal distribution. The support is part of the model and should agree with the possible values of the engineering quantity.

Recorded data can look discrete even when the underlying model is continuous. A temperature sensor that reports to the nearest $0.1\,^{\circ}\mathrm{C}$ produces values such as 18.3 and 18.4, but the physical temperature is not assumed to jump between those values. The rounding belongs to the measurement system; the continuous model describes the underlying quantity.

Example 4.1 Measurement Resolution and a Continuous Model

A laser scanner reports the thickness of a steel plate to the nearest 0.01 mm. A displayed value of 12.37 mm represents a range of underlying thicknesses that round to that number. It does not mean that the true thickness can take only multiples of 0.01 mm. Modelling the underlying thickness as continuous is therefore compatible with the rounded observations.

Probability density functions

For a discrete random variable, the height of the PMF at x is the probability $P(X = x)$. For a continuous random variable, the height of a density is not a probability. Probability is represented by area under the density over an interval.

Definition 4.2 (Probability Density Function)

A function f_X is a **probability density function** (PDF) for a continuous random variable X if

$$f_X(x) \geq 0$$

for all x, the total area under the density is 1,

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1,$$

and

$$P(a < X \leq b) = \int_a^b f_X(x)\,dx$$

for any $a < b$.

♣

The integral notation expresses an area. In this book, standard distributions supply the relevant areas through their distribution functions; manual integration is not required. The notation remains important because it explains what a density means.

The units of a density are the reciprocal of the units of the random variable. If X is measured in hours, then $f_X(x)$ has units of probability per hour. This is another reason why a density height cannot itself be interpreted as a probability.

Theorem 4.1 (Basic Properties of a Continuous Distribution)

If X is continuous, then

$$P(X = x) = 0$$

for every exact value x. Consequently, for $a < b$,

$$\begin{aligned} P(a < X < b) &= P(a \leq X < b) \\ &= P(a < X \leq b) \\ &= P(a \leq X \leq b). \end{aligned}$$

The choice of strict or inclusive endpoints does not change an interval probability.

♡

The statement $P(X = x) = 0$ does not mean that observing x is impossible. It means that a single point has

no area. Every realised measurement is an individual value, but probability calculations concern intervals of possible values.

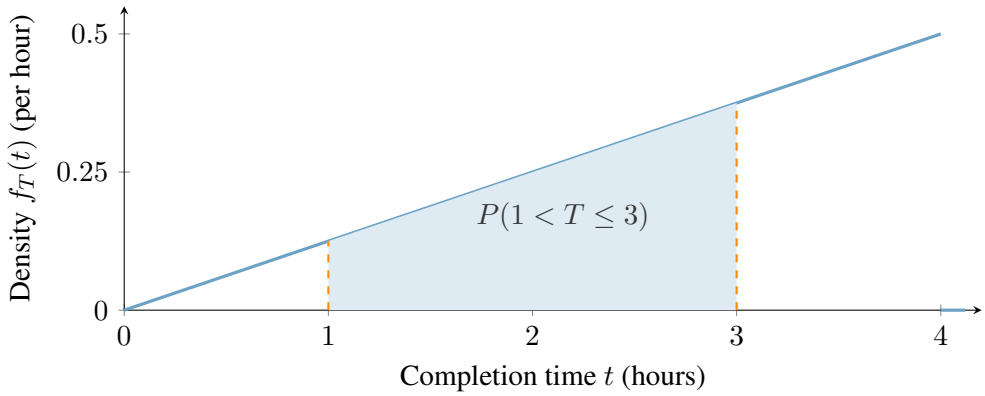


Figure 4.1: For a continuous variable, the shaded area over an interval represents probability. The height of the density at one point is not a probability.

The cumulative distribution function

The cumulative distribution function was introduced in Section 3.1 for random variables in general. For a continuous variable it accumulates area under the density from the far left up to a threshold.

Definition 4.3 (Cumulative Distribution Function for a Continuous Variable)

The **cumulative distribution function** (CDF) of a continuous random variable X is

$$F_X(x) = P(X \leq x).$$

It is related to the density by

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt.$$

♣

The letter t inside the integral is a temporary variable. The argument x remains the threshold at which the CDF is evaluated. A continuous CDF is non-decreasing, takes values between 0 and 1, approaches 0 far to the left, and approaches 1 far to the right.

Theorem 4.2 (Probabilities from a Continuous CDF)

For a continuous random variable X with CDF F_X,

$$P(X \leq x) = F_X(x),$$

$$P(X > x) = 1 - F_X(x),$$

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

♡

These identities are the main computational tools for continuous models. They translate the position of an interval into accumulated probability without requiring direct evaluation of an area.

Example 4.2 Completion Time from a CDF

Let T be the time, in hours, required to complete a maintenance task. Suppose

$$F_T(t) = \begin{cases} 0, & t < 0, \\ t^2/16, & 0 \le t \le 4, \\ 1, & t > 4. \end{cases}$$

The probability of completing the task within one hour is

$$P(T \le 1) = F_T(1) = \frac{1}{16} = 0.0625.$$

The probability that the task takes more than two hours is

$$P(T > 2) = 1 - F_T(2) = 1 - \frac{4}{16} = 0.75.$$

Finally,

$$\begin{aligned} P(1 < T \le 3) &= F_T(3) - F_T(1) \\ &= \frac{9}{16} - \frac{1}{16} \\ &= 0.50. \end{aligned}$$

Thus half of all such tasks are expected to take between one and three hours under this model. The corresponding density and interval area are illustrated in Figure 4.1.

Survival functions and quantiles

Many engineering questions concern exceeding a threshold: a component surviving beyond 500 hours, a response taking longer than 100 ms, or a river level exceeding a warning limit. Such questions can be written compactly using a survival function.

Definition 4.4 (Survival Function)

For a continuous random variable X, the **survival function** is

$$S_X(x) = P(X > x) = 1 - F_X(x).$$

♣

The word survival is especially natural for lifetime data, but the function applies to any upper-tail probability. It is often preferable to writing $1 - F_X(x)$ repeatedly.

A probability calculation starts from a value and finds an area. A quantile calculation reverses the direction: it starts from an accumulated probability and finds the corresponding value.

Definition 4.5 (Quantile)

For $0 < p < 1$, a **p-quantile** of a continuous distribution is a value q_p satisfying

$$F_X(q_p) = p.$$

When the CDF is strictly increasing, the quantile is unique and can be written

$$q_p = F_X^{-1}(p).$$

The median is the 0.5-quantile, and the $100p$th percentile is q_p.

♣

Example 4.3 A Service-Time Guarantee

For the maintenance-time model in Example 4.2, suppose a planner wants a time limit that is sufficient for 90% of tasks. The required value is the 0.9-quantile. Within the support,

$$F_T(t) = \frac{t^2}{16}.$$

Setting this equal to 0.9 gives

$$\frac{q_{0.9}^2}{16} = 0.9, \qquad q_{0.9} = 4\sqrt{0.9} \approx 3.79 \text{ hours}.$$

A planning allowance of approximately 3.79 hours is therefore exceeded by 10% of tasks under the model.

Expected value and variance retain the interpretations developed for discrete random variables: they describe the centre and spread of a distribution. For the named continuous models in this chapter, their values are stated as model properties and interpreted in context. Their calculus-based derivations are not needed here.

4.2 The Uniform Distribution

The uniform distribution is the simplest continuous model. It describes a quantity restricted to a finite interval when equal-length subintervals are assigned equal probability. The density is constant, so probability depends only on interval length.

Definition 4.6 (Continuous Uniform Distribution)

A random variable X has a **continuous uniform distribution** on the interval $[a, b]$, where $a < b$, if its density is

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

We write

$$X \sim U(a, b).$$

Its CDF is

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}$$

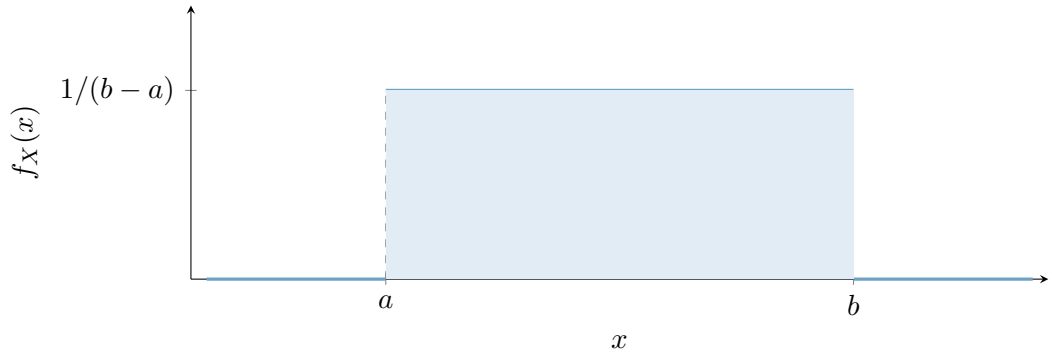


Figure 4.2: The uniform density is constant on $[a, b]$. Its rectangular area is height $1/(b-a)$ times width $b-a$, giving total probability 1.

For any subinterval $[c, d] \subseteq [a, b]$,

$$P(c \le X \le d) = \frac{d-c}{b-a}.$$

The formula compares the length of the favourable interval with the length of the complete support. This geometric interpretation is visible in Figure 4.2.

Theorem 4.3 (Mean, Variance, and Quantiles of a Uniform Variable)

If $X \sim U(a, b)$, then

$$E[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12},$$

and

$$q_p = a + p(b-a), \qquad 0 < p < 1.$$

♡

The mean is the midpoint of the support. The standard deviation depends only on the width $b-a$: shifting both endpoints by the same amount shifts the mean but leaves the variability unchanged.

Example 4.4 Delivery within a Scheduled Window

A supplier commits to deliver a specialised component between 18 and 30 days after an order. Suppose that, within this interval, no delivery time is considered more likely than another. Let

$$X \sim U(18, 30).$$

The probability of delivery between day 22 and day 25 is

$$P(22 \le X \le 25) = \frac{25-22}{30-18} = \frac{3}{12} = 0.25.$$

Similarly,

$$P(X > 27) = \frac{30-27}{12} = 0.25.$$

The mean and standard deviation are

$$E[X] = \frac{18+30}{2} = 24 \text{ days}$$

and

$$\sqrt{\mathrm{Var}(X)} = \sqrt{\frac{(30-18)^2}{12}} = \sqrt{12} \approx 3.46 \text{ days}.$$

The 90th percentile is

$$q_{0.9} = 18 + 0.9(30-18) = 28.8 \text{ days}.$$

Thus 90% of deliveries occur by day 28.8 under the uniform model.

Example 4.5 Bounded Calibration Error

Suppose the rounding error E of a digital measurement is modelled as uniform between -0.05 and 0.05 units. Then

$$E \sim U(-0.05, 0.05).$$

The probability that the absolute error is no more than 0.02 is

$$\begin{aligned} P(|E| \le 0.02) &= P(-0.02 \le E \le 0.02) \\ &= \frac{0.02 - (-0.02)}{0.05 - (-0.05)} \\ &= 0.40. \end{aligned}$$

The model is centred at zero, but only 40% of its probability lies within 0.02 units of zero.

When is a uniform model reasonable?

A uniform model requires more than known minimum and maximum values. It asserts that intervals of equal length are equally likely throughout the support. This may be plausible for a deliberately randomised start time, an unknown phase in a repeating cycle, or rounding error under suitable conditions. It is usually not plausible for manufacturing measurements that cluster around a target or for waiting times concentrated near zero.

The bounded support is also a firm assumption: the model assigns zero probability outside $[a, b]$. If values beyond the stated limits can occur, even rarely, the uniform distribution is not an adequate model for the complete process.

4.3 The Normal Distribution

The normal distribution is central to statistics because many stable measurement processes produce approximately symmetric, bell-shaped variation. It also arises as an approximation to sums and averages of many small contributions, a connection developed formally in Chapter 5. Normality is nevertheless a modelling assumption, not a default property of every continuous dataset.

Definition 4.7 (Normal Distribution)

A random variable X has a **normal distribution** with mean μ and variance σ^2, where $\sigma > 0$, if its density is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \qquad -\infty < x < \infty.$$

We write

$$X \sim N(\mu, \sigma^2).$$

♣

The first parameter is the mean and the second is the variance. The standard deviation is therefore σ, not σ^2. The parameter μ determines the centre of the curve, while σ determines its horizontal spread.

Theorem 4.4 (Properties of a Normal Distribution)

If $X \sim N(\mu, \sigma^2)$, then

$$E[X] = \mu, \qquad \operatorname{Var}(X) = \sigma^2.$$

The density is symmetric about μ, so

$$P(X \le \mu) = P(X \ge \mu) = 0.5.$$

Its mean, median, and mode are all equal to μ.

♡

Although the normal support is the complete real line, values far from the mean have very small probability. A normal model may therefore be useful for a positive quantity when the mean is many standard deviations above

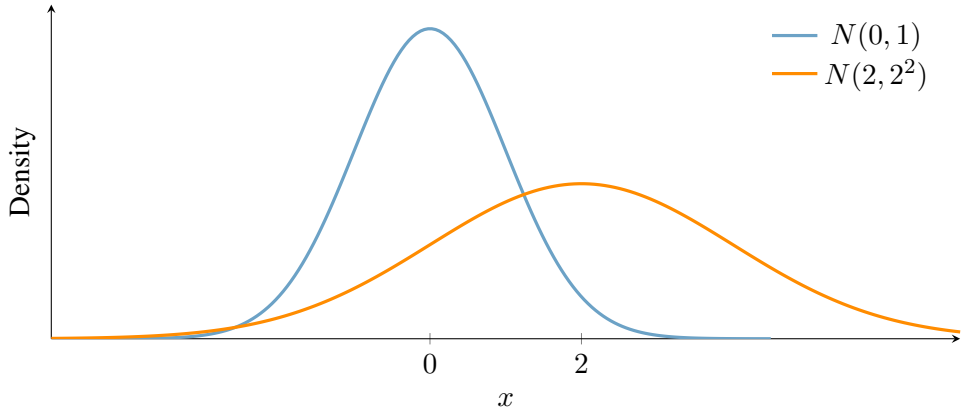


Figure 4.3: The mean determines the centre of a normal curve and the standard deviation determines its spread. Every normal density is symmetric and has total area 1.

zero. It is inappropriate when the model assigns non-negligible probability to impossible values or when the data are strongly skewed.

The standard normal distribution

All normal distributions can be related to one reference distribution.

Definition 4.8 (Standard Normal Distribution)

A normal random variable with mean 0 and variance 1 is called a **standard normal random variable**. It is denoted by

$$Z \sim N(0,1).$$

Its CDF is written

$$\Phi(z) = P(Z \le z).$$

There is no elementary closed formula for $\Phi(z)$. Its values are obtained from numerical evaluation or a standard normal table. Symmetry gives the useful identity

$$\Phi(-z) = 1 - \Phi(z).$$

Using a standard normal table

A standard normal table gives cumulative left-tail probabilities $\Phi(z) = P(Z \le z)$. In the common layout illustrated by Table 4.1, the row supplies the units and first decimal place of z, while the column supplies the second decimal place. For example, $z = 1.25$ is found in row 1.2 and column 0.05, giving

$$\Phi(1.25) = 0.8944.$$

Table 4.1: Excerpt from a cumulative standard normal table. Each entry is $\Phi(z) = P(Z \le z)$.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

Tables often list only nonnegative z-scores. Negative values are then handled by symmetry. For example,

$$\Phi(-1.50) = 1 - \Phi(1.50) = 1 - 0.9332 = 0.0668.$$

Right-tail probabilities are found by complementation, and interval probabilities are found by subtracting two cumulative probabilities. The same table can be read in reverse to locate quantiles. An entry of 0.9750, for instance, occurs at row 1.9 and column 0.06, so $z_{0.975} \approx 1.96$.

In practical analysis, statistical software such as Python is normally used to evaluate normal probabilities and quantiles. The table remains useful because it makes the direction of a probability calculation visible: the entry is a left-tail probability, while right tails and intervals still require the complement or subtraction rules. Software performs the numerical evaluation; the analyst must still standardise correctly and identify the intended event.

Theorem 4.5 (Standardisation of a Normal Variable)

If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

♡

Consequently,

$$P(X \le x) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

The standardised value

$$z = \frac{x - \mu}{\sigma}$$

is called a z**-score**. It describes how many standard deviations x lies above or below the mean. A positive z-score is above the mean, and a negative z-score is below it.

Example 4.6 Shaft Diameter within an Interval

The diameter X of a machined shaft is modelled as

$$X \sim N(20.00, 0.04^2) \text{ mm}.$$

The probability that a shaft diameter is between 19.94 mm and 20.05 mm is found by standardising both limits:

$$z_{\text{lower}} = \frac{19.94 - 20.00}{0.04} = -1.50, \qquad z_{\text{upper}} = \frac{20.05 - 20.00}{0.04} = 1.25.$$

Therefore,

$$\begin{aligned} P(19.94 \le X \le 20.05) &= \Phi(1.25) - \Phi(-1.50) \\ &\approx 0.8944 - 0.0668 \\ &= 0.8276. \end{aligned}$$

Approximately 82.8% of shafts lie in this interval if the normal model and its parameters are appropriate.

Normal quantiles and percentiles

Let z_p denote the p-quantile of the standard normal distribution, so that

$$\Phi(z_p) = p.$$

If $X \sim N(\mu, \sigma^2)$, its p-quantile is

$$q_p = \mu + z_p \sigma.$$

This formula converts a standard normal percentile to the original measurement scale.

Example 4.7 A High Measurement-Error Threshold

The output of a voltage reference is modelled as

$$X \sim N(5.00, 0.02^2) \text{ V}.$$

The 99th standard normal percentile is $z_{0.99} \approx 2.326$. Hence

$$\begin{aligned} q_{0.99} &= 5.00 + 2.326(0.02) \\ &\approx 5.0465 \text{ V}. \end{aligned}$$

Under the model, 99% of outputs are at or below approximately 5.0465 V, and 1% exceed that value.

The empirical rule

The standard normal CDF gives three useful reference probabilities:

$$\begin{aligned} P(\mu - \sigma \le X \le \mu + \sigma) &\approx 0.6827, \\ P(\mu - 2\sigma \le X \le \mu + 2\sigma) &\approx 0.9545, \\ P(\mu - 3\sigma \le X \le \mu + 3\sigma) &\approx 0.9973. \end{aligned}$$

These are often remembered as the **68–95–99.7 rule**. The rule applies to a normal distribution, not to an arbitrary dataset merely because its mean and standard deviation are known.

Example 4.8 Process Yield within Symmetric Tolerances

Return to the shaft model $X \sim N(20.00, 0.04^2)$ mm. Suppose the acceptable diameter is from 19.92 mm to 20.08 mm. These limits are two standard deviations from the mean:

$$\frac{19.92 - 20.00}{0.04} = -2, \qquad \frac{20.08 - 20.00}{0.04} = 2.$$

Thus

$$P(19.92 \le X \le 20.08) \approx 0.9545.$$

The model predicts a yield of approximately 95.45% before considering any other defect mechanism. The remaining 4.55% is split equally between the two tails because the process is centred at the midpoint of the tolerance interval.

When is a normal model reasonable?

A normal model is most plausible when variation is approximately symmetric around a stable centre and results from many small contributions rather than one dominant mechanism. Measurement error, controlled process dimensions, and aggregated environmental variation can sometimes have this form.

The model should be questioned when observations are strongly skewed, bounded close to the centre, contain several distinct clusters, or include more extreme values than a normal distribution would predict. Waiting times

and response times are often right-skewed and nonnegative; a raw normal model can then be misleading even if it is convenient. Graphical diagnostics are discussed in section 4.5.

4.4 The Exponential Distribution

The exponential distribution models the time until an event when events occur at a constant rate and independently of how long one has already waited. Examples include the time until the next request under a stable arrival mechanism or the operating time until a sudden failure under a constant failure-rate model.

Definition 4.9 (Exponential Distribution)

A nonnegative random variable T has an **exponential distribution** with rate $\lambda > 0$ if its density is

$$f_T(t) = \begin{cases} \lambda e^{-\lambda t}, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

We write

$$T \sim \operatorname{Exp}(\lambda).$$

Its CDF and survival function are

$$F_T(t) = \begin{cases} 0, & t < 0, \\ 1 - e^{-\lambda t}, & t \geq 0, \end{cases}$$

and

$$S_T(t) = P(T > t) = e^{-\lambda t}, \qquad t \geq 0.$$

The parameter λ is a rate and therefore has reciprocal time units. If t is measured in hours, then λ is measured in events per hour. A larger rate produces shorter waiting times and a density that decreases more rapidly, as shown in Figure 4.4.

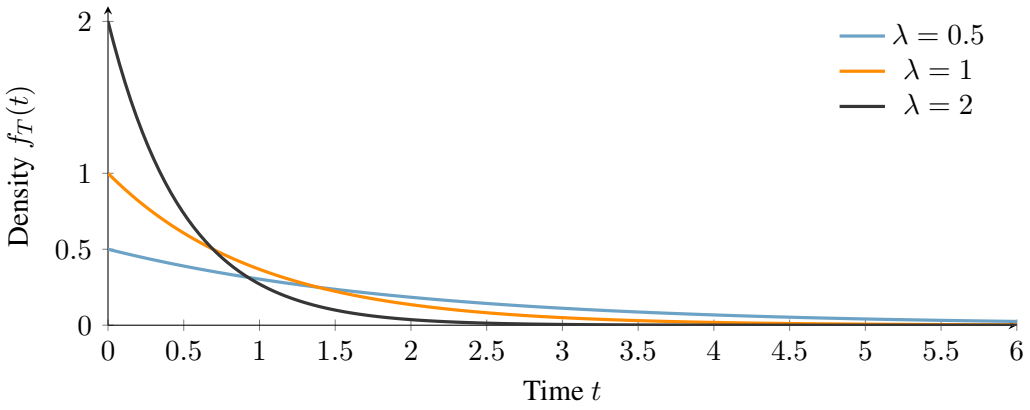


Figure 4.4: Exponential densities for three rates. Increasing λ concentrates more probability near zero and decreases the mean waiting time.

Theorem 4.6 (Mean, Variance, and Quantiles of an Exponential Variable)

If $T \sim \operatorname{Exp}(\lambda)$, then

$$E[T] = \frac{1}{\lambda}, \qquad \operatorname{Var}(T) = \frac{1}{\lambda^2}.$$

Its p-quantile is

$$q_p = -\frac{\ln(1-p)}{\lambda}, \qquad 0 < p < 1.$$

The mean and standard deviation are both $1/\lambda$. The distribution is therefore strongly right-skewed: most waiting times are relatively short, but occasional long waits remain possible.

Example 4.9 Time until the Next Server Request

A server receives requests at a stable rate of 25 requests per second. Suppose the time T until the next request is modelled as

$$T \sim \text{Exp}(25),$$

with time measured in seconds. The mean waiting time is

$$E[T] = \frac{1}{25} = 0.04 \text{ s} = 40 \text{ ms}.$$

The probability of waiting more than 10 ms is

$$\begin{aligned} P(T > 0.01) &= e^{-25(0.01)} \\ &= e^{-0.25} \\ &\approx 0.7788. \end{aligned}$$

The 95th percentile is

$$\begin{aligned} q_{0.95} &= -\frac{\ln(0.05)}{25} \\ &\approx 0.1198 \text{ s} \\ &= 119.8 \text{ ms}. \end{aligned}$$

Thus 95% of interarrival times are no more than approximately 119.8 ms under the constant-rate model.

Memorylessness

The exponential distribution has a distinctive property: after a waiting time has already elapsed, the distribution of the additional waiting time is unchanged.

Theorem 4.7 (Memoryless Property)

If $T \sim \text{Exp}(\lambda)$, then for $s, t \geq 0$,

$$P(T > s + t \mid T > s) = P(T > t).$$

The exponential distribution is said to be **memoryless**. ♡

Memorylessness does not say that the same physical component becomes new again. It states that the model assigns the same distribution to its remaining life after survival to age s. This is appropriate only when the event rate does not change with age.

Example 4.10 Remaining Component Life

Suppose the operating life T of an electronic component is exponentially distributed with mean 500 hours. The rate is

$$\lambda = \frac{1}{500} = 0.002 \text{ per hour}.$$

The unconditional probability of surviving beyond 600 hours is

$$P(T > 600) = e^{-0.002(600)} = e^{-1.2} \approx 0.3010.$$

Now suppose the component has already survived 400 hours. The probability that its total life exceeds 600 hours is the probability of surviving an additional 200 hours:

$$\begin{aligned} P(T > 600 \mid T > 400) &= P(T > 200) \\ &= e^{-0.002(200)} \\ &= e^{-0.4} \\ &\approx 0.6703. \end{aligned}$$

The conditional probability is larger than the unconditional probability because the component is known to have survived the first 400 hours. Under the exponential model, however, those 400 hours do not change the distribution of its additional lifetime.

The connection with Poisson counts

The Poisson and exponential distributions describe two views of the same constant-rate mechanism. A Poisson variable counts events in a fixed interval; an exponential variable measures the waiting time until the next event.

Suppose events occur independently at a constant rate λ per unit time. If $N(t)$ denotes the number of events in an interval of length t, then

$$N(t) \sim \text{Pois}(\lambda t).$$

The waiting time T until the first event exceeds t exactly when no event occurs during the interval. Therefore,

$$\begin{aligned} P(T > t) &= P(N(t) = 0) \\ &= e^{-\lambda t}, \end{aligned}$$

which is the exponential survival function.

Example 4.11 Counts and Waiting Times for Network Traffic

Return to a request rate of 25 per second and consider a 10 ms interval. The expected number of requests in the interval is

$$\lambda t = 25(0.01) = 0.25.$$

Using the Poisson count model,

$$P(N(0.01) = 0) = e^{-0.25} \approx 0.7788.$$

Using the exponential waiting-time model,

$$P(T > 0.01) = e^{-25(0.01)} = e^{-0.25} \approx 0.7788.$$

The two calculations agree because “no request in the next 10 ms” and “the next request arrives after more than 10 ms” are the same event.

Limitations of the exponential model

An exponential lifetime model assumes a constant failure rate. It can describe sudden failures caused by independent random shocks, but it does not describe gradual wear, early manufacturing defects, or changing operating conditions unless those effects are negligible over the period studied.

Mechanical components often experience a changing failure rate: early failures may be caused by defects, a relatively stable middle period may follow, and wear may increase the failure rate later. A single exponential distribution cannot represent all three phases. The memoryless property is therefore both the model's defining feature and its strongest practical limitation.

4.5 Choosing and Assessing a Continuous Model

A probability distribution is not selected only because its formula is familiar. Model choice begins with the measurement scale and the mechanism that could generate the observations. Graphical evidence then helps assess whether the proposed shape is credible.

Comparing the three models

Table 4.2: Interpretive comparison of the three continuous models in this chapter.

Model	Support and shape	Typical mechanism	Important limitation
Uniform $U(a, b)$	Bounded; constant density	Equal-length intervals are equally likely	Known bounds alone do not justify equal density
Normal $N(\mu, \sigma^2)$	Entire real line; symmetric and bell-shaped	Stable centre with many small sources of variation	Can assign probability to impossible values and misses skewness or heavy tails
Exponential $\mathrm{Exp}(\lambda)$	Nonnegative; right-skewed and decreasing	Waiting time under a constant event rate	Requires memorylessness and a constant rate

The supports provide an immediate first check. A uniform model needs meaningful lower and upper limits. A normal model permits negative and positive values. An exponential model permits only nonnegative values and begins with its highest density at zero. The shape and mechanism must then be considered together.

Example 4.12 Selecting a Model from the Engineering Mechanism

Consider the following quantities:

1. a random activation time selected anywhere within a 20-minute testing window;
2. the dimensional error from a centred and stable machining process;
3. the time until the next alarm under a constant alarm rate.

A uniform model is plausible for the deliberately randomised activation time because equal-length parts of the window are equally likely. A normal model may be plausible for the machining error because small independent sources of variation combine around a target. An exponential model may be plausible for the waiting time because the question concerns time to an event under a constant rate.

These choices are conditional, not automatic. Preferential scheduling would challenge the uniform model, tool wear could shift or skew the machining distribution, and a varying alarm rate would invalidate the exponential model.

Graphical assessment

A histogram can reveal support, skewness, multiple clusters, and extreme observations. A fitted density can be compared with the histogram, but agreement depends on the selected bin widths and may hide tail discrepancies. The empirical CDF gives another comparison because it shows accumulated proportions without bins.

No plot proves that a population follows a distribution. A sample contains limited information, and several models can look similar over the observed range. Graphs are diagnostics: they can expose important disagreements and indicate whether a model is a useful approximation for the intended decision.

Quantile–quantile plots

A quantile–quantile plot compares the ordered observations with quantiles expected under a proposed distribution. It is commonly abbreviated as a **QQ-plot**.

Definition 4.10 (Quantile–Quantile Plot)

Let

$$x_{(1)} \le x_{(2)} \le \cdots \le x_{(n)}$$

be the ordered observations. For a proposed CDF F, define theoretical quantiles

$$q_i = F^{-1}\left(\frac{i-0.5}{n}\right), \qquad i = 1, \ldots, n.$$

A **QQ-plot** displays the pairs

$$(q_i, x_{(i)}).$$

If the data follow the proposed location–scale family reasonably well, the points should lie approximately on a straight line.

♣

The plotting probabilities $(i-0.5)/n$ avoid using 0 and 1, whose quantiles may be infinite. Other closely related plotting conventions exist and lead to nearly the same diagnostic for moderate or large samples.

In a normal QQ-plot, an intercept reflects location and a slope reflects scale. Systematic curvature matters more than small random deviations. A few common patterns are:

- approximately linear points: the normal shape may be a reasonable approximation;
- an upper end bending strongly above the line: the data have a longer right tail than the fitted normal model;
- both ends departing outward: the data may have heavier tails than the normal model;
- separated groups or abrupt changes in slope: the data may combine different populations.

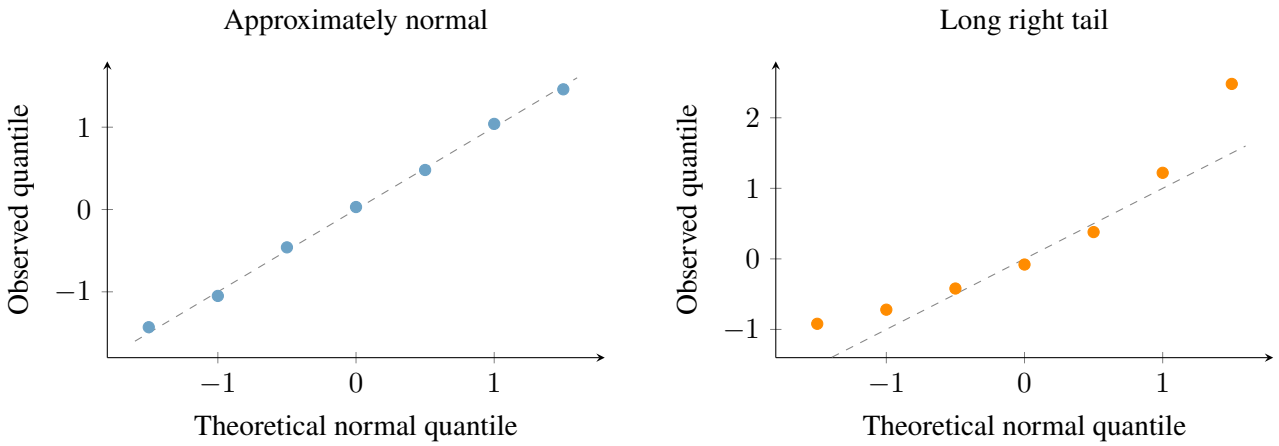


Figure 4.5: Schematic normal QQ-plots. Random scatter around a line is expected, whereas systematic curvature indicates a shape discrepancy.

Example 4.13 A Normal QQ-Plot for Response Times

Seven ordered response times, in milliseconds, are

$$85,\ 91,\ 94,\ 98,\ 103,\ 111,\ 139.$$

For $n = 7$, the plotting probabilities are approximately

$$0.071,\ 0.214,\ 0.357,\ 0.500,\ 0.643,\ 0.786,\ 0.929.$$

The corresponding standard normal quantiles are approximately

$$-1.47,\ -0.79,\ -0.37,\ 0,\ 0.37,\ 0.79,\ 1.47.$$

Plotting the ordered response times against these theoretical quantiles would place the largest response, 139 ms, noticeably above the trend formed by the central observations. This suggests a longer right tail than a normal model would predict.

With only seven observations the evidence is weak, so the plot does not establish a different distribution. It does identify the upper tail as the part of the normal approximation that deserves further investigation, especially if high response-time percentiles are important.

The purpose of a model

Model adequacy depends on the question. A normal approximation that describes the centre well may still be unsuitable for estimating a very rare failure probability. An exponential model may describe short-term waiting times adequately while failing over a component's complete lifetime. The relevant range and consequence of model error should therefore guide how demanding the assessment must be.

A useful model combines three forms of evidence:

1. the support agrees with possible values;
2. the generating mechanism makes the distribution plausible;
3. the observed data do not show a practically important disagreement with the proposed shape.

Together, these models connect engineering thresholds with probabilities for practical decisions.

Part II

Sampling and Statistical Inference

Chapter 5 Sampling Distributions and the Central Limit Theorem

A dataset gives one realised view of a population. If a different random sample had been collected, its observations and numerical summaries would have been different. This sample-to-sample variation is not a defect in statistical analysis; it is the uncertainty that statistical inference must describe. A reported sample mean, defect proportion, or sample variance is therefore not only a number calculated from data. Before the data are observed, it is a random variable with its own probability distribution.

That distribution is called a **sampling distribution**. Sampling distributions connect the probability models developed in earlier chapters with confidence intervals and hypothesis tests. They explain why an average based on 100 measurements is generally more stable than an average based on four, why uncertainty decreases at a square-root rate, and why normal distributions appear repeatedly in statistical analysis even when the individual observations are not normally distributed.

The chapter begins by distinguishing a population distribution, a realised sample, and the sampling distribution of a statistic. We then derive the centre and spread of the sample mean and introduce its standard error. The law of large numbers explains why averages stabilise, while the central limit theorem explains the approximate shape of their sampling distribution. Finally, we study the sample variance under normal sampling and establish the results that lead to Student's t-distribution in the next chapter.

The theory concerns repeated random sampling. In practice, statistical software such as Python is used to simulate that repetition, display sampling distributions, and evaluate probabilities. The definitions, assumptions, and formulas in this chapter determine what such computations mean; code and simulation workflows belong in the accompanying practical material.

5.1 Random Samples, Statistics, and Sampling Distributions

Statistical reasoning begins with a population model and a mechanism for obtaining observations. The mathematical model of a random sample represents repeated observations made under comparable conditions.

Definition 5.1 (Study Design)

A **study design** is the structure by which observations are obtained for a statistical analysis. It specifies the target population and observational units, how units are selected or assigned to conditions, what is measured, and how the observations may be related.

♣

The study design determines which repeated-sampling model is credible and therefore which sampling distribution and standard error should be used. A calculation cannot repair a design that selects the wrong population, treats dependent measurements as independent, or changes the measurement process between observations. Later chapters develop these consequences for estimation and for comparisons between groups.

Definition 5.2 (Random Sample)

Let a population be described by a random variable X with distribution function F. A collection

$$X_1, X_2, \ldots, X_n$$

is a **random sample of size** n from that population if the variables are independent and all have the same distribution F. The abbreviation **i.i.d.** means independent and identically distributed.

♣

Identical distribution represents a stable population: the observations are generated by the same probability model. Independence means that knowing one observation does not change the distribution of another. Both assumptions are approximations whose credibility depends on the sampling mechanism.

Consecutive measurements from a drifting production process may not be identically distributed. Daily energy demands can be dependent because neighbouring days experience similar weather. Measurements from several sensors can share a calibration error. Increasing the number of observations does not remove such structural problems.

Example 5.1 Random Sampling from a Production Process

A filling machine is stable during a morning shift. An engineer selects one container every ten minutes, and the selected times are sufficiently separated that short-term machine effects are negligible. Treating the fill weights as approximately i.i.d. may be reasonable.

If the machine warms throughout the shift, later containers can have a different mean fill weight from earlier containers. The identical-distribution assumption is then questionable. If several consecutive containers are selected after each machine adjustment, observations within a group may also be dependent. A large dataset from this design is not automatically a random sample from one stable population.

The random variables $X_1, \ldots, X_n$ describe observations before they are collected. After collection, they take realised values $x_1, \ldots, x_n$. The same distinction applies to every numerical summary calculated from them.

Definition 5.3 (Statistic and Observed Statistic)

A **statistic** is a function of the random sample that does not contain an unknown population parameter. It is therefore a random variable before the sample is observed. If

$$T = g(X_1, \ldots, X_n),$$

then its observed value is

$$t = g(x_1, \ldots, x_n).$$

Examples include the sample mean $\bar{X}$, the sample variance S^2, and a sample proportion $\widehat{p}$.

♣

Uppercase notation refers to the random statistic, while lowercase notation refers to its realised value. Thus $\bar{X}$ varies across possible samples, whereas $\bar{x} = 248.6$ is the mean of one observed sample. This distinction becomes essential when probability statements are made about a statistic.

Definition 5.4 (Sampling Distribution)

The **sampling distribution** of a statistic T is the probability distribution of T over all random samples of a fixed size drawn under the stated population model and sampling mechanism.

♣

Figure 5.1 shows the repeated-sampling idea. One realised sample produces one statistic. Conceptually repeating the entire sampling process produces many possible values of that statistic, and their distribution

describes its sampling uncertainty.

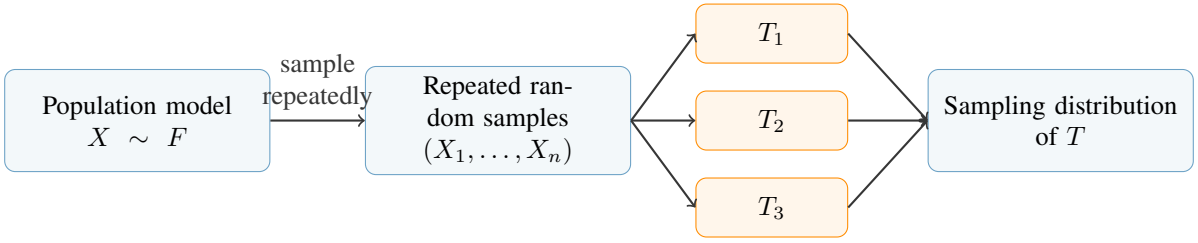


Figure 5.1: A sampling distribution is generated conceptually by repeating the complete sampling process and recalculating the statistic each time.

Three different distributions must therefore be kept separate, as summarised in Table 5.1.

Table 5.1: Population, sample, and sampling distributions answer different questions.

Object	What varies?	Typical question
Population distribution	Individual values of X	How do component lifetimes vary?
Observed sample distribution	The realised values $x_1, \ldots, x_n$	What shape and spread were observed in this dataset?
Sampling distribution	Values of a statistic such as $\bar{X}$ across repeated samples	How much would the reported mean vary if the study were repeated?

An exact sampling distribution

A small discrete example makes the construction explicit. Suppose an observation X takes the values 2, 4, and 6 with equal probability. Its population mean and variance are

$$\mu = 4, \qquad \sigma^2 = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3} = \frac{8}{3}.$$

Draw two observations independently and define $\bar{X} = (X_1 + X_2)/2$. The nine ordered pairs are equally likely. Grouping pairs that produce the same mean gives the sampling distribution in Table 5.2.

Table 5.2: Exact sampling distribution of $\bar{X}$ for two independent draws from $\{2, 4, 6\}$.

$\bar{x}$	2	3	4	5	6
$P(\bar{X} = \bar{x})$	$1/9$	$2/9$	$3/9$	$2/9$	$1/9$

The statistic is centred correctly:

$$\begin{aligned} E[\bar{X}] &= 2\left(\frac{1}{9}\right) + 3\left(\frac{2}{9}\right) + 4\left(\frac{3}{9}\right) + 5\left(\frac{2}{9}\right) + 6\left(\frac{1}{9}\right) \\ &= 4. \end{aligned}$$

Its variance is

$$\operatorname{Var}(\bar{X}) = \frac{4}{3},$$

which is half the population variance $8/3$. The sample mean varies, but it varies less than an individual observation.

5.2 The Sample Mean and Standard Error

The sample mean is the most important sampling statistic in this book. It estimates a population centre, enters engineering decisions about average performance, and forms the basis of many confidence intervals and hypothesis tests.

Definition 5.5 (Sample Mean)

For a random sample $X_1, \ldots, X_n$, the **sample mean** is the statistic

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

For observed values $x_1, \ldots, x_n$, its realised value is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

♣

The expectation and variance rules developed for random variables immediately determine the centre and spread of its sampling distribution.

Theorem 5.1 (Mean and Variance of the Sample Mean)

Let $X_1, \ldots, X_n$ be independent random variables with common mean μ and common variance $\sigma^2 < \infty$. Then

$$E[\bar{X}] = \mu$$

and

$$\operatorname{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Consequently, the standard deviation of $\bar{X}$ is

$$\operatorname{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

♡

The result $E[\bar{X}] = \mu$ says that repeated sample means balance around the population mean. The variance decreases because independent deviations partly cancel when they are averaged. The square-root relationship is fundamental: multiplying the sample size by four divides the standard deviation of the sample mean by two.

Definition 5.6 (Standard Error)

The **standard error** of a statistic is the standard deviation of its sampling distribution. For the sample mean,

$$\operatorname{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

If σ is unknown, it is commonly estimated by

$$\widehat{\operatorname{SE}}(\bar{X}) = \frac{S}{\sqrt{n}}.$$

♣

Standard deviation and standard error describe different kinds of variation. Confusing them makes a dataset appear more precise than it is or a reported mean appear more variable than it is.

Exact sampling from a normal population

The mean and variance formulas do not specify the complete shape of a sampling distribution. When the population itself is normal, however, the sample mean has an exact normal distribution.

Table 5.3: Standard deviation and standard error refer to different distributions.

Quantity	Describes	Effect of increasing n
σ or s	Variation among individual observations	Does not become smaller merely because more observations are collected
$\sigma/\sqrt{n}$ or $s/\sqrt{n}$	Variation among means from repeated samples of size n	Decreases at the rate $1/\sqrt{n}$

Theorem 5.2 (Sample Mean from a Normal Population)

If

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2),$$

then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

This result is exact for every positive sample size n. ♡

The second parameter in the normal notation is the variance σ^2/n. The corresponding standard deviation and standard error are $\sigma/\sqrt{n}$.

Example 5.2 Mean Thickness of a Production Sample

The thickness X of a manufactured plate is normally distributed with mean $250\,\mu\text{m}$ and standard deviation $4.2\,\mu\text{m}$. For a random sample of 36 plates,

$$\bar{X} \sim N\left(250, \frac{4.2^2}{36}\right) = N(250, 0.7^2).$$

The probability that the sample mean is within $1.4\,\mu\text{m}$ of the population mean is

$$\begin{aligned} P(248.6 \le \bar{X} \le 251.4) &= P\left(\frac{248.6 - 250}{0.7} \le Z \le \frac{251.4 - 250}{0.7}\right) \\ &= P(-2 \le Z \le 2) \\ &\approx 0.9545. \end{aligned}$$

The individual plate thicknesses have standard deviation $4.2\,\mu\text{m}$, while the mean of 36 plates has standard error $0.7\,\mu\text{m}$. These describe different distributions.

Example 5.3 Designing for a Target Standard Error

A vibration measurement has population standard deviation 8 units. An engineer wants the standard error of the reported mean to be at most 1 unit. The requirement is

$$\frac{8}{\sqrt{n}} \le 1.$$

Therefore

$$\sqrt{n} \ge 8, \qquad n \ge 64.$$

If the target standard error were reduced from 1 to 0.5, the required sample size would become

$$n \ge \left(\frac{8}{0.5}\right)^2 = 256.$$

Halving the standard error requires four times as many independent observations.

Sums and sample proportions

Some engineering questions concern totals rather than averages. If

$$T_n = \sum_{i=1}^{n} X_i,$$

then independence gives

$$E[T_n] = n\mu, \qquad \mathrm{Var}(T_n) = n\sigma^2.$$

For a normal population, T_n is exactly normal. For other populations, the central limit theorem will often provide a normal approximation when n is sufficiently large.

A sample proportion is also a sample mean. Let $Y_i = 1$ when observation i has a specified property and $Y_i = 0$ otherwise. If the probability of the property is p, then Y_i is Bernoulli with mean p and variance $p(1-p)$. The sample proportion is

$$\widehat{p} = \frac{1}{n}\sum_{i=1}^{n} Y_i,$$

so

$$E[\widehat{p}] = p, \qquad \mathrm{Var}(\widehat{p}) = \frac{p(1-p)}{n}, \qquad \mathrm{SE}(\widehat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$

Example 5.4 Variation in an Observed Alert Proportion

Suppose a stable monitoring rule flags an authentication attempt with probability $p = 0.08$, independently across attempts. For 200 attempts, the sample proportion of flagged attempts has mean

$$E[\widehat{p}] = 0.08$$

and standard error

$$\mathrm{SE}(\widehat{p}) = \sqrt{\frac{0.08(0.92)}{200}} \approx 0.0192.$$

Thus repeated samples of 200 attempts would produce flag proportions with a standard deviation of about 1.92 percentage points. This sampling variation is present even though the underlying flag probability remains fixed at 8%.

5.3 The Law of Large Numbers and the Central Limit Theorem

The previous section established the centre and spread of $\bar{X}$ without determining its shape for a general population. Two limit results explain what happens as the sample size grows. They answer different questions and should not be treated as interchangeable.

Why averages stabilise

Theorem 5.3 (Law of Large Numbers)

Let $X_1, X_2, \ldots$ be i.i.d. random variables with mean μ and finite variance. Then, for every $\varepsilon > 0$,

$$P\big(|\bar{X} - \mu| \geq \varepsilon\big) \longrightarrow 0 \qquad \text{as } n \longrightarrow \infty.$$

Equivalently, the sample mean converges in probability to μ. ♡

The law of large numbers formalises the idea that an average based on many independent observations is likely to be close to the population mean. It does not say that a finite-sample mean equals μ, and it does not describe the shape of the sampling distribution. Its conclusion is about concentration around the correct value.

The variance formula makes that concentration visible:

$$\operatorname{Var}(\bar{X}) = \frac{\sigma^2}{n} \longrightarrow 0.$$

The assumptions still matter. An average of dependent measurements, or of observations from a process whose mean is drifting, need not stabilise around the intended population mean.

The central limit theorem

The central limit theorem gives the missing information about shape. It is a statement about a standardised mean, not about the raw observations.

Theorem 5.4 (Central Limit Theorem)

Let $X_1, X_2, \ldots$ be i.i.d. random variables with mean μ and finite positive variance σ^2. Then

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{\mathrm{d}} N(0,1) \qquad \text{as } n \longrightarrow \infty.$$

Consequently, for sufficiently large n,

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

♡

The symbol $\xrightarrow{\mathrm{d}}$ denotes convergence in distribution. Operationally, the theorem says that probabilities concerning a suitably standardised sample mean can be approximated by standard normal probabilities when the sampling distribution is sufficiently close to normal.

The approximation improves as the sample size grows, but the required size depends on the population shape. A symmetric, light-tailed population can produce an approximately normal mean for a modest n. A strongly skewed or heavy-tailed population may require far more observations. There is no universal rule that $n = 30$ is always adequate.

Figure 5.2 illustrates the effect for a population with an exponential distribution of mean 1. The individual observations are strongly right-skewed. Means of four observations remain skewed, while means of 25 observations are much more concentrated and nearly symmetric.

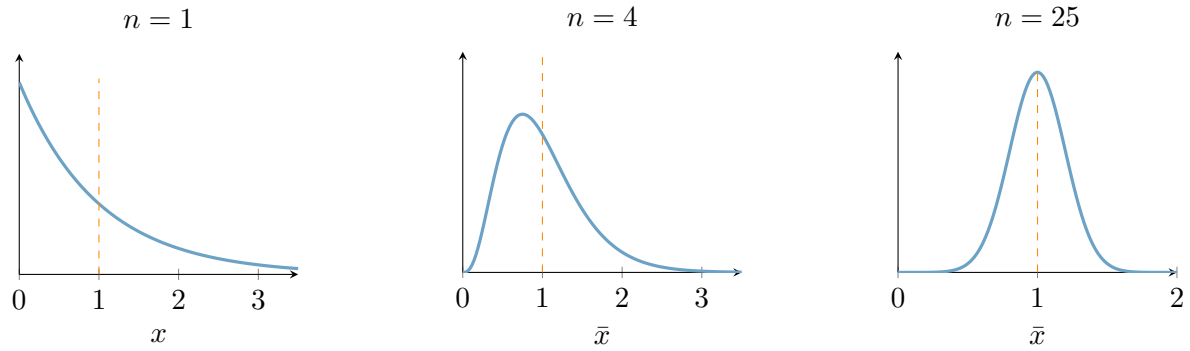


Figure 5.2: The population distribution ($n = 1$), the exact distribution of the mean for $n = 4$, and a normal approximation for the mean when $n = 25$. Each distribution is centred at 1, while the standard error decreases as $1/\sqrt{n}$.

Table 5.4: Exact normal sampling and the central limit approximation must be distinguished.

Situation	Distribution of $\bar{X}$	Status
Normal population	$N(\mu, \sigma^2/n)$ for every n	Exact
Non-normal i.i.d. population with finite variance	Approximately $N(\mu, \sigma^2/n)$ when n is sufficiently large	CLT approximation whose quality depends on population shape and n
Dependent, drifting, or non-representative observations	Not justified by the standard CLT	Sampling mechanism must be reconsidered

Example 5.5 Mean Server Response Time

Individual response times are right-skewed with population mean 120 ms and standard deviation 120 ms. Suppose 64 response times can reasonably be treated as i.i.d. The central limit theorem gives

$$\bar{X} \approx N\left(120, \frac{120^2}{64}\right) = N(120, 15^2).$$

Therefore

$$\begin{aligned} P(\bar{X} > 150) &\approx P\left(Z > \frac{150 - 120}{15}\right) \\ &= P(Z > 2) \\ &\approx 0.0228. \end{aligned}$$

The approximation concerns the mean of 64 response times. It does not imply that individual response times are approximately normal; their population remains right-skewed.

The CLT for sums

Because $T_n = n\bar{X}$, the central limit theorem also gives

$$T_n = \sum_{i=1}^{n} X_i \approx N(n\mu, n\sigma^2)$$

when its assumptions are credible and n is sufficiently large. This form is useful for total demand, aggregate load, accumulated error, and resource planning.

Example 5.6 Snowfall and Road-Salt Capacity

A highway department has enough salt for 80 inches of snowfall. Suppose daily snowfall has mean 1.5 inches and standard deviation 0.3 inch. If the next 50 daily amounts were independent and generated by a stable process, their total T_{50} would have

$$E[T_{50}] = 50(1.5) = 75$$

and

$$\mathrm{SD}(T_{50}) = \sqrt{50}(0.3) \approx 2.121.$$

Using the CLT,

$$\begin{aligned} P(T_{50} \le 80) &\approx \Phi\left(\frac{80 - 75}{2.121}\right) \\ &= \Phi(2.36) \\ &\approx 0.9909. \end{aligned}$$

The numerical result is high, but the independence assumption is doubtful because snowfall on neighbouring days is affected by common weather systems. The calculation is only as credible as that assumption.

Normal approximation for a sample proportion

Applying the CLT to Bernoulli indicators gives

$$\widehat{p} \approx N\left(p, \frac{p(1-p)}{n}\right).$$

The approximation is most credible when both expected counts np and $n(1-p)$ are sufficiently large. It can be poor for rare events or small samples. Exact binomial probabilities remain preferable when they are readily available and the question concerns a discrete count. The normal approximation becomes especially useful in the confidence intervals and tests developed later.

What the theorem does not repair

The CLT is powerful but specific. It does not make raw data normal, guarantee that every statistic is normally distributed, or correct a biased sampling design. Its usual form requires independence, a stable common distribution, and finite variance. Strong dependence, changing process conditions, very heavy tails, or influential extreme observations can make the approximation unreliable.

Simulation is valuable for investigating a proposed sample size and population shape, but simulation does not replace the assumptions. It shows the behaviour of the model that was simulated; it cannot establish that the real data-generating mechanism follows that model.

5.4 The Sample Variance and Normal-Population Results

The sample mean is not the only random statistic. The sample variance also changes from sample to sample. Its sampling distribution explains why replacing an unknown population standard deviation by a sample standard deviation introduces additional uncertainty.

Definition 5.7 (Sample Variance)

For a random sample $X_1, \ldots, X_n$, the **sample variance** is

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Its realised value for observed data is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The sample standard deviation is $S = \sqrt{S^2}$, with observed value s.

♣

The use of $n-1$ is connected to the fact that the deviations from the sample mean satisfy

$$\sum_{i=1}^{n}(X_i - \bar{X}) = 0.$$

Once $n-1$ deviations are known, the final deviation is determined. The sample therefore has $n-1$ independent pieces of information about variation, commonly described as $n-1$ **degrees of freedom**.

Theorem 5.5 (Unbiasedness of the Sample Variance)

If $X_1, \ldots, X_n$ are i.i.d. with finite population variance σ^2, then

$$E[S^2] = \sigma^2.$$

Thus S^2 is centred at the population variance across repeated random samples.

♡

Unbiasedness does not mean that an observed s^2 equals σ^2. It means that over repeated sampling, positive and negative estimation errors balance in expectation. The distribution of S^2 can still be wide for a small sample.

The chi-square connection

For a normal population, the sampling distribution of S^2 can be expressed through a chi-square distribution.

Definition 5.8 (Chi-Square Distribution)

If $Z_1, \ldots, Z_\nu$ are independent standard normal random variables, then

$$Q = \sum_{i=1}^{\nu} Z_i^2$$

has a **chi-square distribution with ν degrees of freedom**, written

$$Q \sim \chi^2_\nu.$$

Its mean is ν and its variance is 2ν.

♣

A chi-square variable is nonnegative and right-skewed, especially for small degrees of freedom. It measures an accumulated squared standardised distance.

Theorem 5.6 (Sampling from a Normal Population)

If

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2),$$

then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Moreover, the sample mean $\bar{X}$ and the sample variance S^2 are independent.

♡

The independence is a special property of normal sampling. It is not generally true for arbitrary population

distributions. Together with the exact normal distribution of $\bar{X}$, it provides the theoretical basis for inference about a normal mean when σ is unknown.

Example 5.7 Sampling Variation in a Process Variance

Suppose fill weights are normally distributed with population standard deviation $\sigma = 2$ g, and a random sample of $n = 10$ containers is collected. Then

$$\frac{9S^2}{4} \sim \chi^2_9.$$

The expected sample variance remains

$$E[S^2] = \sigma^2 = 4 \text{ g}^2.$$

If one sample produces $s = 2.4$ g, then $s^2 = 5.76$ g^2, and its corresponding chi-square value is

$$q = \frac{9(5.76)}{4} = 12.96.$$

This transformation locates the observed variance within its exact normal-sampling distribution. Statistical software can evaluate the associated tail probability when inference about a variance is required.

From normal standardisation to Student's t

If σ were known, the standardised sample mean

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

would be standard normal under normal sampling. In practice, σ is usually unknown and is replaced by the random quantity S. The resulting statistic has a different distribution.

Theorem 5.7 (Student's t Statistic)

If $X_1, \ldots, X_n$ are i.i.d. normal random variables, then

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1},$$

where t_{n-1} denotes Student's t-distribution with $n - 1$ degrees of freedom. ♡

The t-distribution is symmetric about zero but has heavier tails than the standard normal distribution. Those heavier tails reflect the extra uncertainty created by estimating σ. As the degrees of freedom increase, the t-distribution approaches the standard normal distribution.

Example 5.8 Standardising a Mean when Variability Is Unknown

A random sample of 16 normally distributed battery capacities has mean $\bar{x} = 102$ units and standard deviation $s = 12$ units. Relative to a candidate population mean of 100 units, the observed t statistic is

$$t = \frac{102 - 100}{12/\sqrt{16}} = \frac{2}{3} \approx 0.667.$$

The relevant reference distribution has $16 - 1 = 15$ degrees of freedom. The statistic alone is not a conclusion; the next chapters use its t_{15} distribution to construct confidence intervals and evaluate hypotheses.

Sampling distributions describe the uncertainty of statistics before a particular dataset is observed. The sample mean is centred at μ with standard error $\sigma/\sqrt{n}$; the law of large numbers explains its concentration, and the central limit theorem supplies an approximate normal shape under appropriate conditions. Under normal sampling, the exact distributions of $\bar{X}$ and S^2 lead to Student's t. These results provide the probability foundation for estimation and statistical testing.

Chapter 6 Estimation and Confidence Intervals

A sample contains incomplete information about a population. Its mean, proportion, and variance provide useful summaries, but each would change if a different random sample were collected. Statistical estimation turns those sample statistics into disciplined statements about unknown population parameters.

Point estimation reports a single value, such as a sample mean of 48.6 milliseconds. A single value, however, does not communicate how strongly the data determine the answer. An estimate based on 10 observations is normally less precise than one based on 1,000 comparable observations, and an estimate from a highly variable process is less precise than one from a stable process. Interval estimation adds this missing information by reporting a range of parameter values that are compatible with the sampling uncertainty.

Confidence intervals are built from the sampling distributions developed in the previous chapter. The standard error describes how an estimator varies across repeated samples, while a critical value determines how much of that sampling distribution is covered. The resulting interval must be interpreted as a property of a repeated procedure: the parameter is fixed, but the interval would change from sample to sample.

This chapter begins by distinguishing estimators from their observed estimates and by examining bias and precision. It then develops the general logic of confidence intervals and repeated-sampling coverage. We construct intervals for a population mean when the population standard deviation is known or unknown, followed by intervals for a population proportion. The final section connects interval width to engineering decisions, study design, and computational resampling.

6.1 Point Estimation and Precision

A population parameter is a fixed but usually unknown characteristic of the population model. Examples include a population mean μ, variance σ^2, and proportion p. A statistic calculated from a random sample can be used to estimate such a parameter.

Definition 6.1 (Estimator and Estimate)

An **estimator** of a parameter θ is a statistic

$$\widehat{\Theta} = g(X_1, \ldots, X_n)$$

used to learn about θ. Because it is a function of a random sample, the estimator is a random variable.

After data $x_1, \ldots, x_n$ have been observed, the realised value

$$\widehat{\theta} = g(x_1, \ldots, x_n)$$

is called an **estimate** of θ. ♣

The distinction is the same as that between $\bar{X}$ and $\bar{x}$. The sample mean $\bar{X}$ is an estimator because it varies across possible samples. The number $\bar{x} = 48.6$ is an estimate obtained from one particular sample.

Example 6.1 Estimates from a Monitoring Sample

An engineer randomly selects 40 transactions from a stable service and records their response times and whether

each transaction violates a latency requirement. The observed sample has

$$\bar{x} = 48.6 \text{ ms}, \qquad s = 7.4 \text{ ms}, \qquad 5 \text{ violations}.$$

The statistic $\bar{X}$ is an estimator of the population mean response time μ, and its observed estimate is 48.6 ms. The sample proportion $\widehat{p}$ is an estimator of the population violation probability p, and its observed estimate is

$$\widehat{p} = \frac{5}{40} = 0.125.$$

Neither estimate equals its parameter by definition. Each is the result produced by one random sample.

Bias

An estimator should be centred appropriately across repeated samples. Bias measures a systematic displacement from the target parameter.

Definition 6.2 (Bias and Unbiasedness)

The **bias** of an estimator $\widehat{\Theta}$ for a parameter θ is

$$\text{Bias}(\widehat{\Theta}) = E[\widehat{\Theta}] - \theta.$$

The estimator is **unbiased** if

$$E[\widehat{\Theta}] = \theta.$$

♣

Unbiasedness concerns the centre of a sampling distribution, not the accuracy of every observed estimate. An unbiased estimator can be far from the parameter in one sample, especially when its sampling distribution is wide.

For an i.i.d. sample with mean μ, the sample mean satisfies

$$E[\bar{X}] = \mu,$$

so $\bar{X}$ is unbiased for μ. For independent Bernoulli observations with success probability p,

$$E[\widehat{p}] = p,$$

so the sample proportion is unbiased for p. The sample variance S^2 was constructed with divisor $n-1$ so that $E[S^2] = \sigma^2$.

Example 6.2 Calibration Bias Does Not Vanish with Sample Size

Suppose a temperature sensor has a constant calibration offset of $0.6°\text{C}$. A measurement can be modelled as

$$X_i = \mu + 0.6 + \varepsilon_i, \qquad E[\varepsilon_i] = 0,$$

where μ is the true mean temperature. The sample mean has expectation

$$E[\bar{X}] = \mu + 0.6,$$

and therefore

$$\text{Bias}(\bar{X}) = 0.6°\text{C}.$$

Collecting more observations reduces random variation in $\bar{X}$, but it does not remove the calibration offset. If the offset is known, the corrected estimator

$$\bar{X} - 0.6$$

is unbiased for μ. This is a design and calibration correction, not a consequence of a larger sample.

Precision and standard error

Bias describes where a sampling distribution is centred. Precision describes how tightly it is concentrated. The standard error provides the relevant scale.

Definition 6.3 (Precision of an Estimator)

The **precision** of an estimator refers to the amount of sampling variation it has. An estimator with a smaller variance or standard error is more precise than an estimator of the same parameter with a larger variance or standard error. ♣

For the sample mean and sample proportion,

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

and

$$\mathrm{SE}(\widehat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$

When the population quantities are unknown, they are replaced by suitable sample estimates. Precision improves with additional independent information, but only at a square-root rate.

Example 6.3 Averaging Independent Sensor Readings

Two independent calibrated sensors measure the same physical quantity. Each reading is unbiased for μ and has variance 9 units^2. Using only the first sensor gives the estimator X_1 with standard error

$$\mathrm{SE}(X_1) = 3.$$

The average

$$\widehat{\Theta} = \frac{X_1 + X_2}{2}$$

is also unbiased, and independence gives

$$\begin{aligned}\mathrm{Var}(\widehat{\Theta}) &= \frac{1}{4}\big(\mathrm{Var}(X_1) + \mathrm{Var}(X_2)\big) \\ &= \frac{1}{4}(9+9) = 4.5.\end{aligned}$$

Thus

$$\mathrm{SE}(\widehat{\Theta}) = \sqrt{4.5} \approx 2.12.$$

Averaging the independent readings improves precision while preserving unbiasedness. If the sensors share a calibration error, independence is no longer a credible description of all uncertainty.

Bias and precision are separate qualities. An estimator can be unbiased but imprecise, biased but highly concentrated, or poor in both respects. A narrow confidence interval describes sampling precision under the model; it does not demonstrate that the estimator is free from systematic error.

6.2 Confidence Intervals and Coverage

A point estimate answers the question "What value did this sample produce?" A confidence interval answers the additional question "How much would that estimate vary under repeated sampling?" The interval is calculated from random endpoints before the data are observed.

Definition 6.4 (Confidence Interval)

Let $L = L(X_1, \ldots, X_n)$ and $U = U(X_1, \ldots, X_n)$ be statistics with $L \leq U$. The random interval

$$[L, U]$$

is a $100(1-\alpha)\%$ **confidence interval** for a parameter θ if its coverage probability is

$$P(L \leq \theta \leq U) = 1 - \alpha$$

under the stated population model and sampling procedure.

♣

The number $1-\alpha$ is the confidence level. Common choices are 0.90, 0.95, and 0.99. After a sample is observed, L and U become numerical endpoints l and u, while the parameter remains fixed.

The estimate–standard error–critical value pattern

Many confidence intervals have the form

$$\text{estimate} \quad \pm \quad \text{critical value} \times \text{standard error}.$$

Suppose the standardised estimator is exactly or approximately standard normal:

$$Z = \frac{\widehat{\Theta} - \theta}{\mathrm{SE}(\widehat{\Theta})} \approx N(0,1).$$

Let $z_{1-\alpha/2}$ denote the standard normal quantile satisfying

$$P\big(Z \leq z_{1-\alpha/2}\big) = 1 - \frac{\alpha}{2}.$$

Then

$$P\left(-z_{1-\alpha/2} \leq \frac{\widehat{\Theta} - \theta}{\mathrm{SE}(\widehat{\Theta})} \leq z_{1-\alpha/2}\right) \approx 1 - \alpha.$$

Rearranging gives the general interval

$$\widehat{\Theta} \pm z_{1-\alpha/2}\,\mathrm{SE}(\widehat{\Theta}).$$

Definition 6.5 (Margin of Error)

For an interval of the form

$$\widehat{\theta} \pm E,$$

the quantity

$$E = \text{critical value} \times \text{standard error}$$

is called the **margin of error**. The total interval width is $2E$.

♣

The critical value controls confidence, while the standard error controls sampling precision. Statistical software is normally used to evaluate critical values, but their meaning is determined by the probability distribution and confidence level.

Repeated-sampling coverage

Figure 6.1 illustrates repeated confidence intervals. Each horizontal line is produced from a different random sample using the same procedure. The vertical line marks the fixed population parameter. Nineteen of the twenty illustrated intervals cover the parameter; one does not.

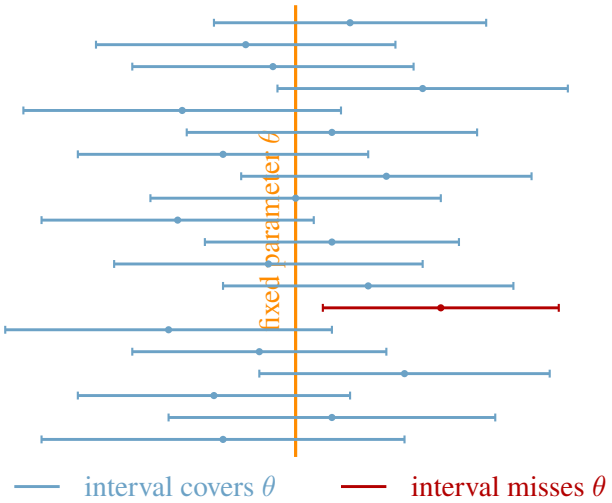


Figure 6.1: Repeated confidence intervals from the same sampling procedure. The parameter is fixed; the interval endpoints vary from sample to sample.

The figure is an illustration rather than a guarantee that every group of 20 intervals contains exactly 19 successes. Coverage is a long-run probability. Over many repetitions, approximately $100(1-\alpha)\%$ of the intervals produced by a valid procedure cover the parameter.

Example 6.4 Interpreting Repeated Coverage

Suppose 1,000 independent laboratories each collect a random sample using the same design and construct a 95% confidence interval for the mean concentration of a material. Under the model assumptions, the procedure is expected to produce approximately

$$0.95(1000) = 950$$

intervals that cover the fixed population mean and approximately 50 that miss it.

For one realised interval, say $[18.2, 19.7]$ mg/L, the frequentist statement is not that the fixed mean has probability 0.95 of lying between the observed endpoints. The defensible statement is that the interval was produced by a procedure with 95% coverage under repeated sampling.

What determines interval width?

For an estimate–critical value–standard error interval, width is controlled by three general features, summarised in Table 6.1.

Table 6.1: Factors affecting the width of a confidence interval.

Change	Mathematical effect	Consequence for width
Higher confidence level	Larger critical value	Wider interval
Larger independent sample	Smaller standard error, commonly at rate $1/\sqrt{n}$	Narrower interval
Greater population variability	Larger standard error	Wider interval
More efficient estimator	Smaller sampling variance	Narrower interval for the same confidence level

A narrow interval is desirable only when the sampling design and model are credible. Systematic calibration error, selection bias, dependence, or process drift can produce a narrow interval around the wrong target.

6.3 Confidence Intervals for a Population Mean

The sample mean is the natural point estimator of a population mean. Its standard error is $\sigma/\sqrt{n}$, and the relevant critical distribution depends on whether the population standard deviation is known.

Known population standard deviation: the z interval

If a random sample comes from a normal population with known standard deviation σ, then

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

Solving the central probability statement for μ gives an exact interval.

Theorem 6.1 (Confidence Interval for a Mean with Known Standard Deviation)

Let $X_1, \ldots, X_n$ be an i.i.d. sample from $N(\mu, \sigma^2)$, where σ is known. A $100(1-\alpha)\%$ confidence interval for μ is

$$\bar{X} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

For observed data, the endpoints are

$$\bar{x} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

♡

When the population is not normal, the same form can be an approximate interval if the central limit theorem gives an adequate approximation to the sampling distribution of $\bar{X}$. Known population standard deviations are uncommon in new engineering studies, but they arise when a measurement process has been characterised reliably from extensive historical data.

Example 6.5 Mean Compressive Strength with Known Variability

A civil engineer studies the compressive strength of a concrete formulation. The strength distribution is modelled as normal, and historical process validation supports a population variance of 1000 psi^2, so

$$\sigma = \sqrt{1000} \approx 31.62 \text{ psi}.$$

A random sample of 12 specimens has mean strength $\bar{x} = 3250$ psi. The standard error is

$$\frac{\sigma}{\sqrt{n}} = \frac{31.62}{\sqrt{12}} \approx 9.129 \text{ psi}.$$

For 95% confidence, $z_{0.975} \approx 1.960$, giving

$$\begin{aligned} 3250 \pm 1.960(9.129) &= 3250 \pm 17.892 \\ &= [3232.11, 3267.89] \text{ psi}. \end{aligned}$$

For 99% confidence, $z_{0.995} \approx 2.576$, so

$$3250 \pm 2.576(9.129) = [3226.49, 3273.51] \text{ psi}.$$

The 99% interval is wider because the procedure must cover a larger central part of the sampling distribution.

Unknown population standard deviation: the t interval

In most applications, σ is unknown and the standard error is estimated by $S/\sqrt{n}$. Under normal sampling,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

The t critical value is larger than the corresponding standard normal critical value, reflecting the additional uncertainty from estimating σ.

Theorem 6.2 (Confidence Interval for a Mean with Unknown Standard Deviation)

Let $X_1, \ldots, X_n$ be an i.i.d. sample from $N(\mu, \sigma^2)$, where σ is unknown. A $100(1-\alpha)\%$ confidence interval for μ is

$$\bar{X} \pm t_{n-1,\,1-\alpha/2} \frac{S}{\sqrt{n}}.$$

For observed data, the endpoints are

$$\bar{x} \pm t_{n-1,\,1-\alpha/2} \frac{s}{\sqrt{n}}.$$

♡

The notation $t_{\nu,q}$ denotes the q quantile of a Student t distribution with ν degrees of freedom. As ν increases, the t distribution approaches the standard normal distribution and its critical values approach the corresponding z values.

Example 6.6 Mean Parallel-Processing Speedup

Thirteen measurements of the speedup achieved by a parallel computing architecture have

$$n = 13, \qquad \bar{x} = 4.313, \qquad s = 0.4328.$$

Treating the measurements as an independent sample from an approximately normal population, the estimated standard error is

$$\frac{s}{\sqrt{n}} = \frac{0.4328}{\sqrt{13}} \approx 0.1200.$$

For a 95% interval with 12 degrees of freedom,

$$t_{12,0.975} \approx 2.179.$$

Therefore

$$\begin{aligned}\bar{x} \pm t_{12,0.975}\frac{s}{\sqrt{n}} &= 4.313 \pm 2.179(0.1200)\\ &= 4.313 \pm 0.262\\ &= [4.052, 4.575].\end{aligned}$$

The interval estimates the population mean speedup. It does not state that 95% of individual speedup measurements lie between these endpoints.

Assumptions and robustness

The one-sample t interval is exact under i.i.d. normal sampling. For a non-normal population, it is commonly used as an approximation when the sample mean is approximately normal and the sample standard deviation estimates the population spread adequately.

The quality of that approximation depends on more than sample size. Strong skew, heavy tails, or influential outliers can be important for small and moderate samples. Dependence and process drift affect the sampling distribution directly and are not repaired by a t critical value. A graphical examination of the data and knowledge of the sampling mechanism should accompany the interval calculation.

Table 6.2 summarises the two mean intervals.

Table 6.2: Choosing between z and t intervals for one population mean.

Situation	Standard error	Critical distribution
σ known	$\sigma/\sqrt{n}$	Standard normal; exact for a normal population
σ unknown	$s/\sqrt{n}$	Student t with $n-1$ degrees of freedom; exact for a normal population
Non-normal population	Depends on whether σ is known	Either interval may be approximate; adequacy depends on the sampling distribution of $\bar{X}$

Planning for a target margin of error

If σ is known or a credible planning value is available, the margin of error of a two-sided z interval is

$$E = z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

Solving for n gives

$$n \geq \left(\frac{z_{1-\alpha/2}\sigma}{E}\right)^2.$$

The calculated value must always be rounded up because rounding down would exceed the target margin of error.

Example 6.7 Sample Size for a Vibration Study

A vibration measurement has a planning standard deviation of 8 units. An engineer wants a 95% confidence interval for the mean with margin of error no greater than 1 unit. Using $z_{0.975} = 1.960$,

$$\begin{aligned} n &\geq \left(\frac{1.960(8)}{1}\right)^2 \\ &= 245.86. \end{aligned}$$

The study therefore requires at least

$$n = 246$$

independent observations under the planning assumptions. The calculation addresses random precision; it does not determine whether the measurements are representative or free from calibration bias.

When σ is unknown, a previous study, pilot sample, engineering specification, or conservative planning value can be used. The final t interval still uses the standard deviation observed in the completed study.

6.4 Confidence Intervals for a Population Proportion

Binary outcomes occur in defect detection, reliability testing, cybersecurity monitoring, and survey data. If X is the number of successes in n independent Bernoulli trials with probability p, then

$$\widehat{p} = \frac{X}{n}$$

is an unbiased estimator of the population proportion p.

For sufficiently large n, the sample proportion has the approximate sampling distribution

$$\widehat{p} \approx N\left(p, \frac{p(1-p)}{n}\right).$$

Replacing the unknown p in the standard error by $\widehat{p}$ gives

$$\widehat{\mathrm{SE}}(\widehat{p}) = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

The normal or Wald interval

Applying the general estimate–critical value–standard error pattern gives the traditional normal interval.

Definition 6.6 (Wald Confidence Interval for a Proportion)

The **Wald interval** with confidence level $100(1-\alpha)\%$ is

$$\widehat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

It is an approximate interval based on the normal approximation to the sampling distribution of $\widehat{p}$. ♣

The approximation is most credible when the observations are independent, the probability p is stable, and both the observed success and failure counts are reasonably large. Conditions such as $n\widehat{p} \geq 10$ and $n(1-\widehat{p}) \geq 10$ are useful diagnostics, not universal guarantees.

Example 6.8 Interval for an Authentication-Alert Rate

In a random sample of 200 authentication attempts, 24 trigger an alert. The estimated alert probability is

$$\widehat{p} = \frac{24}{200} = 0.12.$$

The estimated standard error is

$$\sqrt{\frac{0.12(0.88)}{200}} \approx 0.02298.$$

Using $z_{0.975} = 1.960$, the 95% Wald interval is

$$\begin{aligned} 0.12 \pm 1.960(0.02298) &= 0.12 \pm 0.0450 \\ &= [0.075, 0.165]. \end{aligned}$$

The procedure estimates the stable population alert probability under the assumed sampling mechanism. It does not estimate the probability that an alert represents a genuine attack; that is a different conditional probability.

Why the Wald interval can fail

The Wald interval is easy to derive but can have poor coverage for small samples or proportions near 0 or 1. Its endpoints can even fall outside the parameter space $[0, 1]$. A widely used improvement is the Wilson score interval.

For $z = z_{1-\alpha/2}$, the Wilson interval is

$$\frac{\widehat{p} + \dfrac{z^2}{2n} \quad \pm \quad z\sqrt{\dfrac{\widehat{p}(1-\widehat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}.$$

The expression is less simple than the Wald formula, but statistical software evaluates it directly. It generally provides more reliable coverage and respects the range of a proportion more effectively.

Example 6.9 Rare Defects and the Wilson Interval

Two non-conforming components are observed among 20 inspected components, so

$$\widehat{p} = \frac{2}{20} = 0.10.$$

The 95% Wald interval is

$$\begin{aligned} 0.10 \pm 1.960\sqrt{\frac{0.10(0.90)}{20}} &\approx 0.10 \pm 0.131 \\ &= [-0.031, 0.231]. \end{aligned}$$

The negative lower endpoint is impossible for a probability and signals that the approximation is unsuitable.

Using the Wilson formula with the same confidence level gives approximately

$$[0.028, 0.301].$$

The Wilson interval is asymmetric and remains within $[0, 1]$. It is also wider on the upper side, reflecting the substantial uncertainty created by only two observed defects.

Planning a sample size for a proportion

Using the approximate margin of error

$$E = z_{1-\alpha/2}\sqrt{\frac{p(1-p)}{n}},$$

the planning requirement is

$$n \geq \frac{z_{1-\alpha/2}^2 p(1-p)}{E^2}.$$

If no credible planning value for p is available, the product $p(1-p)$ is largest at $p = 0.5$. Using 0.25 therefore gives a conservative approximate sample size.

For example, a 95% interval with a target margin of error of 0.03 requires

$$\begin{aligned} n &\geq \frac{1.960^2(0.25)}{0.03^2} \\ &\approx 1067.11. \end{aligned}$$

At least 1,068 independent observations are required under this conservative calculation.

6.5 Interpretation, Limitations, and Computational Extensions

A confidence interval combines an estimate with a model-based description of sampling uncertainty. It is valuable only when the parameter, target population, sampling mechanism, and assumptions have been stated clearly.

Table 6.3 summarises the core intervals in this chapter.

Table 6.3: Core confidence intervals for one population.

Target parameter	Core interval	Principal conditions
Mean μ, σ known	z interval using $\sigma/\sqrt{n}$	Independent representative sample; exact normal model or adequate large-sample approximation
Mean μ, σ unknown	t interval using $s/\sqrt{n}$	Independent representative sample; exact normal model or adequate approximation for the sample mean
Proportion p	Wilson interval preferred in practical work; Wald interval explains the normal approximation	Independent Bernoulli observations with stable p; approximation quality checked

A mean is not an individual observation

A confidence interval for μ describes uncertainty about a population mean. It does not describe the spread of individual observations and is not a prediction interval for a new observation. As n grows, an interval for μ can become very narrow even when individual measurements remain highly variable. Prediction intervals require an additional allowance for the variability of a new observation and are introduced with regression modelling later in the book.

Precision is not practical importance

A narrow interval can show that an effect has been estimated precisely without showing that the effect matters in practice.

Example 6.10 A Precise but Small Energy Reduction

An energy-management intervention is estimated to reduce daily consumption by 0.8 kWh, with a 95% confidence interval from 0.5 to 1.1 kWh. The interval excludes zero, so the data support a positive average reduction under the study design.

Suppose, however, that implementation costs are justified only by a reduction of at least 5 kWh per day. The entire interval lies far below that engineering requirement. The estimated effect is statistically distinguishable from zero but not large enough to satisfy the practical decision criterion.

What additional data cannot repair

Increasing n reduces standard error when the additional observations contribute independent information from the same target population. It does not correct a biased sensor, a convenience sample, missing parts of the population, changing operating conditions, or duplicated measurements treated as independent. Confidence intervals quantify uncertainty under a model; they do not validate the data-generating process automatically.

Bootstrap intervals as a computational perspective

Classical intervals use a theoretical sampling distribution such as the normal or Student t distribution. A bootstrap approximates sampling variation by repeatedly resampling the observed data.

Definition 6.7 (Bootstrap Sample)

Given observed values $x_1, \ldots, x_n$, a **bootstrap sample** is a sample of size n drawn with replacement from those observed values. Recalculating an estimator across many bootstrap samples produces a **bootstrap distribution**.

Percentiles of the bootstrap distribution can be used to form a computational interval. This approach is useful when the sampling distribution is difficult to derive, and it is explored through software in the practical material. It does not create information beyond the observed sample and cannot repair a biased or unrepresentative sampling design. Its quality still depends on the observed data representing the target population adequately.

Confidence intervals are procedures for converting sampling variation into a quantitative statement about precision. Their critical values, standard errors, and coverage all depend on the underlying probability model. A careful analysis therefore reports the estimate and interval together, states the target parameter and assumptions, and interprets the result in the engineering context rather than treating the interval as an automatic decision.

Chapter 7 Hypothesis Testing I: Principles and One-Sample Tests

Estimation and hypothesis testing answer related but different questions. A point estimate describes the value suggested by a sample, and a confidence interval describes the precision of that estimate. A hypothesis test begins with a specified value or claim and asks whether the observed data are reasonably compatible with it.

Engineering decisions often have this form. Is the mean thickness of a manufactured plate still equal to its target? Has average server response time exceeded a service threshold? Is the proportion of defective components greater than the stated limit? The sample rarely agrees exactly with the claimed value, even when the claim is correct, because statistics vary from sample to sample. A test therefore measures the observed discrepancy relative to the sampling variation expected under the claim.

The result is not a proof that a hypothesis is true or false. It is a quantified assessment of evidence under a probability model. The reasoning depends on the sampling distribution developed in Chapter 5 and the standard errors and confidence intervals developed in Chapter 6. This chapter develops the general logic of testing, explains p-values and decision errors, and then applies the framework to one population mean and one population proportion.

7.1 Claims, Hypotheses, and Statistical Evidence

A statistical analysis should begin with the population quantity of interest. Without a clearly stated parameter, the hypotheses and the conclusion can easily refer to different questions.

Definition 7.1 (Statistical Hypothesis)

A **statistical hypothesis** is a statement about a population parameter or probability model. A **hypothesis test** is a procedure that uses sample data to assess the compatibility of that statement with the observations. ♣

Typical one-sample parameters include a population mean μ and a population proportion p. The parameter is fixed, although unknown. The sample statistic is random before the data are collected and provides the evidence used by the test.

Table 7.1: Examples of one-sample engineering questions and their parameters.

Engineering question	Population quantity	Parameter
Is mean API response time equal to its service target?	Mean response time for requests under the stated operating conditions	μ
Does mean concrete strength exceed the minimum design requirement?	Mean strength for material produced by the process	μ
Is the fraction of compromised login attempts above the alert threshold?	Probability that a login attempt is compromised	p
Is the proportion of late deliveries greater than the contractual limit?	Probability that a delivery from the process is late	p

The null and alternative hypotheses

Two competing statements are used to organise the test.

Definition 7.2 (Null and Alternative Hypotheses)

The **null hypothesis**, denoted H_0, is the claim used to construct the reference distribution of the test statistic. The **alternative hypothesis**, denoted H_1 or H_A, describes the departures from H_0 that the study is designed to detect.

For a population mean, the most common null hypothesis is

$$H_0 : \mu = \mu_0,$$

where μ_0 is a specified target, standard, or historical value. Three common alternatives are shown in Table 7.2.

Table 7.2: The alternative determines which discrepancies count as evidence against the null hypothesis.

Test	Null hypothesis	Alternative hypothesis	Question
Two-sided	$H_0 : \mu = \mu_0$	$H_1 : \mu \neq \mu_0$	Is the mean different in either direction?
Upper-tailed	$H_0 : \mu = \mu_0$	$H_1 : \mu > \mu_0$	Is the mean greater than the specified value?
Lower-tailed	$H_0 : \mu = \mu_0$	$H_1 : \mu < \mu_0$	Is the mean less than the specified value?

The equality is conventionally placed in H_0 because the null value supplies the sampling distribution used in the calculation. For a one-sided test, the broader null may be written $\mu \leq \mu_0$ or $\mu \geq \mu_0$. The test is calculated at the boundary $\mu = \mu_0$, which is the null value most favourable to the alternative.

Example 7.1 Choosing the Direction for a Structural Test

The specified mean breaking load of a fastener is 42 kN. If departures in either direction indicate that the manufacturing process has changed, the appropriate hypotheses are

$$H_0 : \mu = 42 \qquad \text{and} \qquad H_1 : \mu \neq 42.$$

If the engineering concern is specifically inadequate strength, the question is instead whether the mean has fallen below the requirement:

$$H_0 : \mu = 42 \qquad \text{and} \qquad H_1 : \mu < 42.$$

An observed sample mean above 42 kN is not evidence for this lower-tailed alternative, however far above the target it lies.

Direction must follow the question

A one-sided test concentrates all of the stated type I error probability in one tail and can therefore detect a change in that direction more readily. It is justified only when a discrepancy in the opposite direction would not lead to the same scientific claim. The direction must be chosen from the engineering question before inspecting the data. Selecting the favourable tail after observing the sample invalidates the stated significance level.

For example, a production manager should not choose $H_1 : \mu > \mu_0$ merely because the observed mean happens to exceed μ_0. If both increases and decreases would matter, the test must be two-sided even when the realised discrepancy points in only one direction.

The claim is only as credible as the design

A test describes sampling variation under its statistical model. It does not establish that observations are representative, independent, correctly measured, or generated under stable conditions. A very small p-value from a biased sensor or a convenience sample may precisely quantify the wrong comparison. The population, sampling unit, measurement procedure, and operating conditions must therefore be specified before the hypotheses are interpreted.

7.2 Test Statistics, p-Values, and Decision Errors

The raw difference between an estimate and a null value is not enough to assess evidence. A deviation of 0.5 units may be remarkable in a highly stable process and ordinary in a variable one. A test statistic compares the discrepancy with its standard error under H_0.

Definition 7.3 (Test Statistic and Null Distribution)

A **test statistic** is a statistic used to measure disagreement between the sample and H_0. Its **null distribution** is its sampling distribution calculated under the assumption that H_0 is true. ♣

Many one-sample test statistics have the standardised form

$$\text{test statistic} = \frac{\text{estimate} - \text{null value}}{\text{standard error under } H_0}.$$

A value near zero means that the estimate is close to the null value relative to expected sampling variation. A value far into a tail of the null distribution indicates greater disagreement.

The p-value

Definition 7.4 (p-Value)

Assume that the null hypothesis H_0 is true and that the statistical model assumptions hold. The **p-value** is the probability, over hypothetical repetitions of the sampling process, of obtaining a test statistic at least as incompatible with H_0 as the statistic observed. ♣

For a symmetric standardised statistic T with observed value t_{obs}, the relevant tail probability depends on H_1:

$$\begin{aligned} H_1 : \theta > \theta_0 \quad &\Longrightarrow \quad p = P_{H_0}(T \geq t_{\text{obs}}), \\ H_1 : \theta < \theta_0 \quad &\Longrightarrow \quad p = P_{H_0}(T \leq t_{\text{obs}}), \\ H_1 : \theta \neq \theta_0 \quad &\Longrightarrow \quad p = P_{H_0}(|T| \geq |t_{\text{obs}}|). \end{aligned}$$

The last expression counts discrepancies at least as far from zero in either direction. Figure 7.1 shows the two tail areas for an observed standardised statistic of magnitude 2.1.

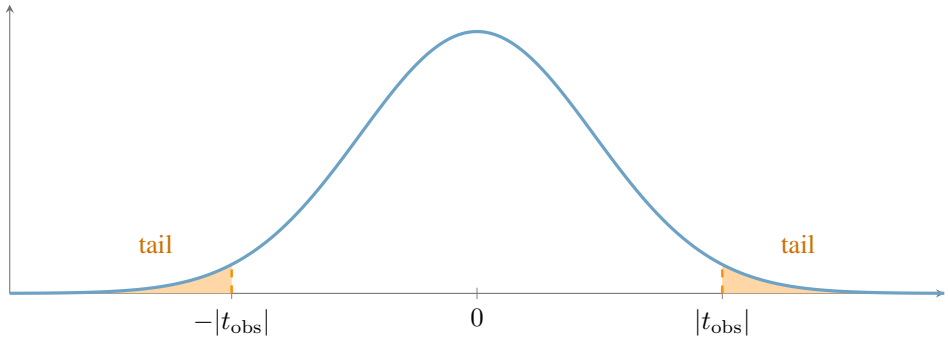


Figure 7.1: For a two-sided test, the p-value is the combined null probability in both tails beyond the observed magnitude.

The direction of the conditioning is essential. A p-value is calculated *given that the null hypothesis is true*:

$$p\text{-value} = P\big(\text{test statistic at least as extreme as the observed statistic} \;\big|\; H_0 \text{ is true}\big)\,.$$

In words, we assume H_0 and its model assumptions, and then ask how unusual the observed test statistic would be under that assumption. This is not the reversed conditional probability

$$p\text{-value} \neq P(H_0 \text{ is true} \mid \text{observed data})\,.$$

The expression on the right is a posterior probability for H_0. It requires a Bayesian model, including prior probabilities, and is not supplied by a frequentist hypothesis test. Nor is the p-value the probability that the result occurred "by chance." Random variation is already part of the null model; the p-value measures how extreme the observation is within that model.

Example 7.2 Interpreting a Small p-Value

Suppose a test of a sensor's mean calibration error produces $p = 0.018$. If the null value and model assumptions are correct, results at least as incompatible with the null hypothesis as the observed result would occur with probability 0.018 under repeated sampling.

The result does not mean that H_0 has probability 0.018, that there is a 98.2% probability of a calibration shift, or that the shift is large enough to matter operationally.

Significance level and decision language

Definition 7.5 (Significance Level)

The **significance level**, denoted α, is the chosen upper bound on the probability of rejecting H_0 when the null hypothesis is true, for a test procedure whose assumptions hold. For many continuous test statistics the type I error probability equals α; for a discrete exact test it may be smaller. ♣

The significance level is chosen before the data are examined. Common choices such as 0.05 and 0.01 are conventions rather than universal laws. Safety-critical decisions may justify a smaller type I error probability, while screening procedures may use a different balance between false alarms and missed changes.

The standard decision rule is

$$p \le \alpha \quad \Longrightarrow \quad \text{reject } H_0,$$
$$p > \alpha \quad \Longrightarrow \quad \text{fail to reject } H_0.$$

Failing to reject H_0 does not establish that H_0 is true. The data may be compatible with the null value, or the study may simply have insufficient information to detect the departure that is present.

Type I error, type II error, and power

Every decision rule can make errors because samples vary.

Definition 7.6 (Type I and Type II Errors)

A **type I error** occurs when a test rejects H_0 although H_0 is true. A **type II error** occurs when a test fails to reject H_0 although a specific alternative is true. ♣

Definition 7.7 (Power)

The **power** of a test at a particular alternative parameter value is the probability of rejecting H_0 when that alternative is true. If the type II error probability at that value is β, then

$$\text{power} = 1 - \beta.$$

♣

Table 7.3: Possible outcomes of a hypothesis-testing procedure.

Decision	H_0 is true	A specified alternative is true
Reject H_0	Type I error, with probability at most α	Correct detection, with probability equal to the power
Fail to reject H_0	Correct non-rejection	Type II error, with probability β

Example 7.3 False Alarms and Missed Intrusions

A cybersecurity team tests whether the proportion of malicious requests has risen above a baseline value. Rejecting H_0 triggers an expensive incident response.

A type I error is a false alarm: the baseline rate has not increased, but the test triggers the response. A type II error is a missed increase: the malicious-request rate has risen, but the test does not detect it. The relative consequences of these errors should influence the design, sample size, and choice of α before monitoring begins.

Power is not a single fixed property of a test. A large departure from H_0 is easier to detect than a small departure. Power generally increases with a larger sample size, a larger true effect, and lower observation variability. Reducing α makes false rejection less likely but, with all else fixed, also makes rejection harder and reduces power. These competing consequences are part of study design rather than something a p-value can resolve after the data have been collected.

7.3 One-Sample Tests for a Population Mean

Suppose $X_1, \dots, X_n$ are independent observations from a population with mean μ. We wish to test a specified mean μ_0. The sample mean $\bar{X}$ estimates μ, and its standard error determines how large a discrepancy $\bar{X} - \mu_0$ must be to count as unusual.

Known population standard deviation

When the population standard deviation σ is genuinely known, the standardised statistic is

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

Theorem 7.1 (One-Sample z Statistic under the Null Hypothesis)

Let $X_1, \ldots, X_n$ be an i.i.d. sample with mean μ and known standard deviation σ. Under

$$H_0 : \mu = \mu_0,$$

the statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

has a standard normal distribution when the population is normal. It is approximately standard normal when the sampling distribution of $\bar{X}$ is adequately approximated by the central limit theorem. ♡

Known σ is uncommon outside a well-characterised and stable process. A historical standard deviation should not automatically be treated as known when operating conditions, equipment, or suppliers may have changed.

Example 7.4 Mean Concentration with Known Process Variability

A laboratory process has a long-established standard deviation of 0.9 mg/L. For $n = 36$ independent specimens, the observed mean concentration is 14.7 mg/L. Test

$$H_0 : \mu = 15 \qquad \text{against} \qquad H_1 : \mu \neq 15.$$

The standard error and observed statistic are

$$\frac{\sigma}{\sqrt{n}} = \frac{0.9}{\sqrt{36}} = 0.15,$$

and

$$z_{\text{obs}} = \frac{14.7 - 15}{0.15} = -2.00.$$

The two-sided p-value is

$$p = 2P(Z \geq 2.00) \approx 0.0455.$$

At $\alpha = 0.05$, the null hypothesis is rejected. The data provide evidence that the mean differs from 15 mg/L under the stated process model. The estimated difference is -0.3 mg/L; whether that difference matters must be judged in concentration units, not from the p-value alone.

Unknown population standard deviation

In most applications, σ is unknown and is estimated by the sample standard deviation S. This additional uncertainty changes the reference distribution from standard normal to Student's t distribution.

Theorem 7.2 (One-Sample t Statistic under the Null Hypothesis)

Let $X_1, \ldots, X_n$ be an i.i.d. sample from a normal population with mean μ and unknown variance. Under

$$H_0 : \mu = \mu_0,$$

the statistic

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

has a Student t distribution with $n - 1$ degrees of freedom. ♡

The observed statistic is therefore

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

The same value is used for all three alternatives; only the tail area changes. For $\nu = n - 1$ degrees of freedom,

$$\begin{aligned} H_1 : \mu > \mu_0 &\implies p = P(T_\nu \geq t_{\text{obs}}), \\ H_1 : \mu < \mu_0 &\implies p = P(T_\nu \leq t_{\text{obs}}), \\ H_1 : \mu \neq \mu_0 &\implies p = P(|T_\nu| \geq |t_{\text{obs}}|). \end{aligned}$$

Conditions for the one-sample t procedure

The exact t result assumes independent sampling from a normal population. In practice, the procedure is often reasonably robust to moderate non-normality, especially for larger samples, because the sample mean becomes more nearly normal. Robustness is not a licence to ignore the data-generating process.

Table 7.4: Conditions to examine before interpreting a one-sample t test.

Condition	Practical question
Quantitative response	Is a population mean a meaningful summary of the measured quantity?
Representative observations	Does the sample address the stated target population and operating conditions?
Independence	Are repeated measurements, clusters, time dependence, or duplicated units being treated as independent?
Distributional adequacy	For a small sample, is a normal population model credible? For a larger sample, are extreme skewness and influential outliers limited enough for the mean and its standard error to be useful?
Measurement validity	Are calibration, resolution, censoring, and missingness capable of producing systematic error?

No universal sample-size threshold guarantees validity. A sample of 30 highly dependent observations is not repaired by its size, while a smaller sample from a well-controlled normal process may support exact t inference.

Assessing a normality assumption

A normal quantile plot, also called a Q–Q plot or normal probability plot, compares the ordered observations with the quantiles expected from a normal distribution. A nearly straight pattern supports the shape of a normal model, while systematic curvature, asymmetric tails, or isolated points reveal how the data depart from that model. The plot is valuable because it displays the form and practical size of a discrepancy rather than reducing the assessment to one number.

A formal normality test answers a related but narrower question. For independent observations $X_1, \ldots, X_n$, the hypotheses may be written

$$H_0 : X_1, \ldots, X_n \text{ are sampled from } N(\mu, \sigma^2) \text{ for some } \mu \text{ and } \sigma > 0$$

against

$$H_1 : \text{the population distribution is not normal.}$$

The null hypothesis specifies the normal family rather than fixed numerical values of μ and σ. Any calibration of the test must therefore account for estimating these parameters from the data.

Definition 7.8 (Shapiro–Wilk Normality Statistic)

Let $X_{(1)} \le \cdots \le X_{(n)}$ denote the ordered observations. The **Shapiro–Wilk statistic** has the form

$$W = \frac{\left(\sum_{i=1}^{n} a_i X_{(i)}\right)^2}{\sum_{i=1}^{n} (X_i - \bar{X})^2},$$

where the constants a_i are determined from the expected order statistics and covariance structure of a standard normal sample of size n. Values of W close to 1 indicate that the ordered observations align closely with a normal pattern; unusually small values indicate departure from normality.

The Shapiro–Wilk p-value is a lower-tail probability,

$$p = P(W \le W_{\text{obs}} \mid H_0 \text{ and independent sampling}).$$

The null distribution of W is not a chi-square distribution. Statistical software uses a specialised approximation to its null distribution; the reference distribution can also be approximated by repeatedly simulating normal samples of the same size and recalculating W.

The Anderson–Darling statistic is another formal goodness-of-fit measure. It compares the empirical distribution with a fitted normal cumulative distribution and gives relatively strong weight to tail discrepancies. When μ and σ are estimated, a Monte Carlo implementation should refit them within each simulated sample before comparing the simulated statistic with the observed statistic. Shapiro–Wilk is a convenient primary normality test, while Anderson–Darling is a useful alternative when tail behaviour is especially important.

Example 7.5 Formal Assessment of Recovery Times

Eleven independently selected recovery operations have durations, in seconds,

$$48,\ 54,\ 58,\ 60,\ 61,\ 62,\ 66,\ 70,\ 82,\ 95,\ 136.$$

The upper observations bend away from the line in a normal probability plot. A Shapiro–Wilk analysis reports

$$W = 0.789, \qquad p = 0.0067.$$

If recovery times were sampled independently from a normal population, a Shapiro–Wilk statistic no greater than 0.789 would occur with probability about 0.0067. The data therefore provide evidence against a normal population model. The test does not identify the engineering cause: the long recovery may represent a valid heavy tail, a separate failure mode, a recording error, or changing operating conditions. Those possibilities require examination before selecting a replacement model or procedure.

The quantity assessed must match the statistical model. A one-sample t procedure concerns the distribution

of the observations around one population mean. A paired t procedure concerns the paired differences, not the two measurements separately. One-way ANOVA concerns within-group errors, and regression concerns residuals around the fitted line; pooling distinct groups or testing the marginal response can create apparent non-normality that is irrelevant to the assumed model.

Formal tests should not be used as automatic gatekeepers. In a small sample, a large p-value may reflect low power rather than convincing normality. In a large sample, a small p-value may detect a mild departure that has little effect on a robust mean comparison. A defensible assessment combines the Q–Q plot, the formal test, sample size, outliers, study design, and the sensitivity of the intended inference. Independence and representative sampling remain more fundamental and cannot be established by a normality test.

Example 7.6 Two-Sided Test of Mean Shaft Diameter

A machining process targets a mean shaft diameter of 40.00 mm. A random sample of $n = 16$ shafts gives

$$\bar{x} = 40.18 \text{ mm}, \qquad s = 0.24 \text{ mm}.$$

To detect a shift in either direction, test

$$H_0 : \mu = 40.00 \qquad \text{against} \qquad H_1 : \mu \neq 40.00.$$

The standard error is

$$\text{SE}(\bar{X}) = \frac{0.24}{\sqrt{16}} = 0.06 \text{ mm},$$

so

$$t_{\text{obs}} = \frac{40.18 - 40.00}{0.06} = 3.00.$$

With 15 degrees of freedom, the two-sided p-value is approximately

$$p = 0.0090.$$

At $\alpha = 0.05$, reject H_0. The sample provides evidence that the process mean differs from the target.

Using $t_{15,0.975} = 2.131$, the corresponding 95% confidence interval is

$$40.18 \pm 2.131(0.06) = [40.052, 40.308] \text{ mm}.$$

The null value 40.00 mm lies outside the interval, giving the same decision as the two-sided test. The estimated shift is 0.18 mm, which must also be compared with engineering tolerances.

Example 7.7 One-Sided Test of Mean Server Response Time

A service-level investigation asks whether mean response time has exceeded 200 ms. A representative sample of $n = 25$ requests gives $\bar{x} = 205.4$ ms and $s = 12.0$ ms. The hypotheses are

$$H_0 : \mu = 200 \qquad \text{and} \qquad H_1 : \mu > 200.$$

The observed statistic is

$$t_{\text{obs}} = \frac{205.4 - 200}{12/\sqrt{25}} = \frac{5.4}{2.4} = 2.25.$$

With 24 degrees of freedom, the upper-tail probability is

$$p = P(T_{24} \geq 2.25) \approx 0.0169.$$

At the 5% level, reject H_0. The data support the conclusion that mean response time exceeds 200 ms under the sampled operating conditions. The relevant estimated exceedance is 5.4 ms; the test alone does not decide whether that increase has operational importance.

The one-sided alternative in this example was determined by the service question. A two-sided analysis of the same data would have a p-value approximately twice as large because it would count extreme discrepancies in both directions.

7.4 One-Sample Tests for a Population Proportion

Binary outcomes arise when a unit passes or fails, an order is late or on time, an alarm is true or false, or a component survives or fails. Let X be the number of observations with the specified outcome among n independent Bernoulli trials. Under a stable success probability p,

$$X \sim \text{Binomial}(n, p), \qquad \hat{p} = \frac{X}{n}.$$

The test compares $\hat{p}$ with a null value p_0.

The large-sample score test

Under $H_0 : p = p_0$, the null standard error of the sample proportion is

$$\text{SE}_0(\hat{p}) = \sqrt{\frac{p_0(1 - p_0)}{n}}.$$

This gives the score statistic

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

When the binomial distribution is adequately approximated by a normal distribution under H_0, this statistic is approximately standard normal.

Notice that the denominator uses p_0, not $\hat{p}$. The test asks how variable $\hat{p}$ would be if the null hypothesis were true. This differs from the Wald standard error used in the simplest confidence interval.

A common diagnostic is to check whether both expected null counts

$$np_0 \qquad \text{and} \qquad n(1 - p_0)$$

are reasonably large. Values near a boundary or small expected counts make the discrete binomial distribution poorly represented by a continuous normal curve. The check is a guideline for approximation quality, not a substitute for examining independence and a stable success probability.

Example 7.8 Has an Automated Inspection Flag Rate Increased?

An automated inspection system is expected to flag 5% of units for manual review. During a stable production period, it flags 74 of $1{,}000$ independently sampled units. Test

$$H_0 : p = 0.05 \qquad \text{against} \qquad H_1 : p > 0.05.$$

The observed proportion is $\hat{p} = 0.074$. Under H_0,

$$\mathrm{SE}_0(\hat{p}) = \sqrt{\frac{0.05(0.95)}{1000}} \approx 0.006892.$$

Therefore,

$$z_{\text{obs}} = \frac{0.074 - 0.05}{0.006892} \approx 3.482.$$

The upper-tail p-value is

$$p = P(Z \geq 3.482) \approx 0.00025.$$

Reject H_0 at the 5% level. The data provide strong evidence that the flag rate exceeds 5%. The estimated increase is

$$0.074 - 0.050 = 0.024,$$

or 2.4 percentage points. Whether that increase justifies recalibration depends on review capacity, costs, and the consequences of missed defects.

The exact binomial test

When the expected counts are small, the binomial distribution can be used directly instead of relying on a normal approximation.

Definition 7.9 (Exact One-Sample Binomial Test)

Under $H_0 : p = p_0$, the count X has the exact distribution

$$X \sim \text{Binomial}(n, p_0).$$

For an observed count x, the upper- and lower-tailed exact p-values are

$$P_{p_0}(X \geq x) \qquad \text{and} \qquad P_{p_0}(X \leq x),$$

respectively.

♣

Because a binomial distribution is discrete, there is more than one convention for constructing a two-sided exact p-value. A common probability-based definition sums the null probabilities of outcomes no more likely than the observed outcome. Statistical reports should identify the method used when the distinction can affect the conclusion.

Example 7.9 Rare Failures in a Small Safety Test

A safety interlock has a claimed failure probability of 0.05. In $n = 20$ independent trials, the interlock fails $x = 3$ times. To investigate an increased failure probability, test

$$H_0 : p = 0.05 \qquad \text{against} \qquad H_1 : p > 0.05.$$

The exact upper-tail p-value is

$$\begin{aligned} p &= P(X \geq 3 \mid X \sim \text{Binomial}(20, 0.05)) \\ &= 1 - P(X \leq 2) \\ &\approx 0.0755. \end{aligned}$$

At $\alpha = 0.05$, fail to reject H_0. The data are concerning enough to motivate further investigation, but they do not meet the specified rejection rule.

The normal score statistic would be

$$z_{\text{obs}} = \frac{0.15 - 0.05}{\sqrt{0.05(0.95)/20}} \approx 2.052,$$

giving an upper-tail normal approximation near 0.020. The disagreement is substantial because the expected null count of failures is only $np_0 = 1$. The exact binomial calculation is the more credible analysis here.

Table 7.5: Choosing a method for one population proportion.

Method	Main use	Important limitation
Exact binomial test	Small samples, rare outcomes, or inadequate normal approximation	Discreteness makes attainable p-values and coverage conservative or uneven; two-sided conventions can differ
Score z test	Large-sample test of a specified proportion p_0	Requires an adequate normal approximation under H_0
Wilson interval	Practical interval estimate for p	It is an interval, not a replacement for stating the test question and alternative

Connection with the Wilson interval

The score test and Wilson confidence interval use the same null-based standardisation. A two-sided score test of $H_0 : p = p_0$ at significance level α rejects exactly when p_0 lies outside the corresponding $100(1-\alpha)\%$ Wilson score interval. This is one reason the Wilson interval developed in Chapter 6 is preferable to the Wald interval.

The method must match when applying this duality. An exact binomial test has a corresponding family of exact intervals; its decision need not agree perfectly with a Wilson interval because discrete exact methods and score methods use different constructions.

The binomial assumptions remain essential for both procedures. Dependence between repeated observations, changing risk across time, or a sample that overrepresents high-risk units can invalidate the model even when the numerical calculation is exact.

7.5 Confidence Intervals, Practical Significance, and Responsible Conclusions

A hypothesis test compresses evidence into a comparison with a specified claim. A confidence interval provides additional information by displaying the parameter values compatible with the data at the corresponding confidence level. Whenever practical, the estimate, interval, and test should be reported together.

Duality between two-sided tests and intervals

Theorem 7.3 (Confidence-Interval and Test Duality)

For a two-sided hypothesis test of

$$H_0 : \theta = \theta_0$$

at significance level α, using a test and confidence interval constructed from the same statistical method,

$$\text{reject } H_0 \iff \theta_0 \text{ lies outside the } 100(1-\alpha)\% \text{ confidence interval.}$$

♡

For a one-sample mean with unknown variance, the matching pair is the two-sided t test and the t confidence interval. For a population proportion, the matching pair is the score test and Wilson score interval. The phrase "same method" matters: combining a Wald interval with an exact test need not produce identical decisions.

An interval communicates more than a reject-or-fail-to-reject decision. It shows the direction of the estimated departure, its magnitude, and the range of parameter values that remain compatible with the data under the model.

Statistical significance is not practical importance

Statistical significance concerns the strength of evidence against a specified null value. Practical significance concerns whether the size of the estimated departure matters in the application. With a sufficiently large sample, a very small and operationally irrelevant difference can produce a small p-value. With a small or noisy sample, an important difference can remain statistically inconclusive.

Useful effect descriptions depend on the parameter and context.

Table 7.6: Effect measures that should accompany one-sample tests.

Parameter	Natural effect measure	Contextual comparison
Mean μ	$\bar{x} - \mu_0$ in the original measurement unit	Tolerance, minimum useful change, safety margin, cost, or service threshold
Proportion p	$\hat{p} - p_0$ in proportions or percentage points	Allowed defect rate, risk threshold, operational capacity, or contractual limit
Mean across incompatible scales	Standardised difference $(\bar{x} - \mu_0)/s$ when genuinely useful	Domain-specific interpretation is still required

Raw engineering units are usually easier to act upon than a standardised effect. A standardised difference can help compare effects measured on different scales, but it does not replace tolerances, costs, safety requirements, or domain knowledge.

Example 7.10 A Statistically Detectable but Small Climate-Sensor Bias

A large calibration study of $n = 400$ readings gives a mean sensor error of $\bar{x} = 0.12°\text{C}$ and a sample standard deviation of $s = 0.80°\text{C}$. Test

$$H_0 : \mu = 0 \qquad \text{against} \qquad H_1 : \mu \neq 0.$$

The test statistic is

$$t_{\text{obs}} = \frac{0.12 - 0}{0.80/\sqrt{400}} = 3.00,$$

with 399 degrees of freedom, giving

$$p \approx 0.0029.$$

The corresponding 95% confidence interval is approximately

$$[0.041, 0.199]^\circ\text{C}.$$

The data provide strong evidence of a positive mean calibration error. However, suppose the engineering tolerance is $\pm 0.5^\circ$C. The estimated bias and the entire interval are well within that tolerance. The shift is statistically detectable, but the operational response should be based on the size of the error and its consequences rather than on the p-value alone.

A complete statistical conclusion

Reporting checklist

A responsible report should enable a reader to reconstruct both the statistical result and its engineering meaning. At minimum, it should state:

1. the target population and parameter;
2. the null and alternative hypotheses, including direction;
3. the study design, sample size, and important assumptions;
4. the estimate and an appropriate confidence interval;
5. the test statistic, degrees of freedom when relevant, and p-value;
6. the decision at the pre-specified α level; and
7. the magnitude and practical consequences of the estimated departure.

For the shaft-diameter data in the earlier example, a concise report could read:

> For 16 independently sampled shafts, mean diameter was 40.18 mm ($s = 0.24$ mm). A two-sided one-sample t test against the 40.00 mm target gave $t(15) = 3.00$, $p = 0.009$, with a 95% confidence interval for the mean of $[40.052, 40.308]$ mm. The process mean is therefore statistically distinguishable from the target; the estimated 0.18 mm shift should be evaluated against the manufacturing tolerance.

Interpretations to avoid

Table 7.7: Common testing statements and their defensible replacements.

Misleading statement	Defensible interpretation
"The p-value is the probability that H_0 is true."	The p-value is a tail probability calculated under H_0.
"We accept H_0."	We fail to reject H_0; the data do not provide sufficient evidence against it at the chosen level.
"The result is significant, so the effect is important."	The data provide evidence of a departure; its estimated size and practical consequences must be assessed separately.
"The result is not significant, so there is no difference."	The study did not detect a departure; the confidence interval shows which departures remain compatible with the data.
"A larger p-value proves the model fits."	A large p-value may reflect compatibility, limited power, high variability, or an insensitive test; it does not validate every model assumption.

Finally, the nominal significance level applies to the planned procedure. Repeating many tests and reporting only the smallest p-value increases false-positive risk. Formal multiple-comparison methods appear later; the general protection begins here: define the question, parameter, direction, and primary analysis before examining the outcomes.

Connect hypothesis testing to estimation: a null model supplies the reference distribution, a test statistic measures discrepancy in standard-error units, and a p-value gives the tail area. A complete conclusion adds the interval, effect magnitude, assumptions, and context. Chapter 8 extends this to two-group and paired comparisons.

Chapter 8 Hypothesis Testing II: Comparing Two Groups

Many engineering questions concern a difference rather than a single population value. Does a new deployment strategy reduce response time compared with the current strategy? Do components from two suppliers have the same mean strength? Does an intervention change energy use? Is the defect proportion lower for one production process than for another? In each case, two samples provide evidence about a difference between population parameters.

The formulas are only part of the analysis. The relationship between the observations determines which comparison is valid. Measurements from distinct units form independent samples when the sampling mechanisms justify independence. Measurements taken twice on the same unit, or on deliberately matched units, are paired and must be analysed through their within-pair differences. Statistical software can evaluate either procedure, but it cannot determine which design produced the data.

This chapter first develops the practical meaning of study design for two-group comparisons. It then presents Welch's procedure for two independent means, the paired t procedure, and the large-sample comparison of two independent proportions. Throughout, the main object is an estimated difference with a confidence interval. A hypothesis test supplements that estimate; it does not replace the magnitude, uncertainty, or engineering interpretation of the difference.

8.1 Study Design and the Structure of a Comparison

The general concept of study design was introduced in chapter 5. For a two-group analysis, the first practical question is how observations in one condition are related to observations in the other. This relationship is fixed by the way the data were obtained, not by the appearance of the spreadsheet or by the test that produces the smallest p-value.

Definition 8.1 (Independent Samples)

Two samples are **independent** when information about the observations selected for one sample does not provide information about the observations selected for the other, and observations are independent within each sample under the stated model.

Independent samples commonly arise when different units are observed in the two groups. For example, components from supplier A and supplier B may be selected independently, or incoming service requests may be randomly routed to two server configurations. Merely placing observations in separate columns does not establish independence. Repeated measurements from the same machine, building, person, or account can remain related even when they are recorded as different rows.

Definition 8.2 (Paired Observations)

Observations are **paired** when each observation in one condition has a specific and meaningful partner in the other condition. The analysis is then based on the difference within each pair.

The most common pairing measures the same unit twice, such as response time for the same request replayed before and after a software change. Pairing can also be created by matching distinct units on important

characteristics, such as two buildings matched by size and construction type. The pairing must come from the design. Sorting two unrelated samples and matching observations by rank does not create valid pairs.

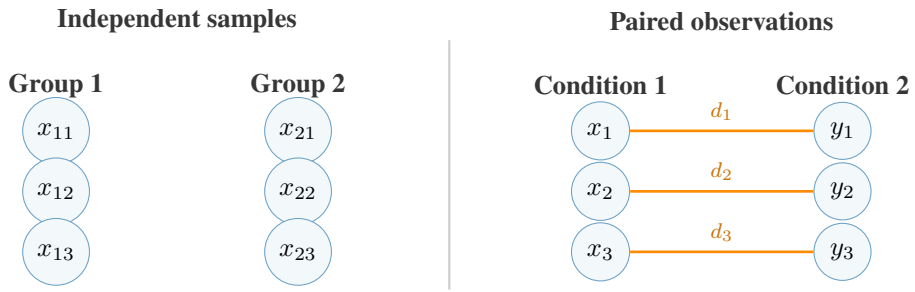


Figure 8.1: Independent groups contain different, unrelated sampling units. A paired design preserves a meaningful connection between the two measurements in each pair.

Sampling, assignment, and causal claims

Two forms of randomness answer different questions. **Random sampling** from a target population supports generalisation from the sample to that population. **Random assignment** allocates observed units to conditions by chance and helps separate a treatment effect from pre-existing differences between groups. One does not imply the other.

In an **observational study**, the researcher observes groups or exposures that already exist. A comparison can estimate an association, but other differences between the groups may explain part or all of it. In a **randomised experiment**, units are randomly assigned to conditions. When the protocol, measurement, and adherence are credible, random assignment provides stronger support for a causal interpretation.

Example 8.1 Identifying the Design Before Choosing a Test

An engineering team evaluates a database index in two ways.

First, 40 requests are sent to the current configuration and a different set of 35 requests is sent to the indexed configuration. If the requests are sampled independently from comparable traffic, this is an independent two-sample design.

Second, 30 recorded requests are replayed once on each configuration under controlled conditions. Each response time from the indexed configuration has a partner from the same request on the current configuration. This is a paired design, and the relevant observations are the 30 within-request differences.

Both datasets contain two columns of response times. The method is determined by how the rows were generated, not by the number of columns.

Table 8.1: Design questions to answer before comparing two groups.

Question	Why it matters
What are the observational units?	Repeated rows from one unit must not automatically be treated as independent units.
How were units selected?	The sampling mechanism determines the population to which results can reasonably be generalised.
Were conditions assigned or merely observed?	Random assignment can support causal interpretation; observational grouping generally supports association only.
Are observations independent or paired?	This determines whether variability is estimated between separate samples or within pairs.
Was measurement comparable across groups?	Different instruments, timing, censoring, or missing-data mechanisms can create an artificial difference.

8.2 Comparing Two Independent Means

Let population 1 have mean μ_1 and variance σ_1^2, and let population 2 have mean μ_2 and variance σ_2^2. The parameter of interest is the population mean difference

$$\Delta = \mu_1 - \mu_2.$$

The order must be stated before calculation because reversing the groups changes the sign. The natural estimator is

$$\widehat{\Delta} = \bar{X}_1 - \bar{X}_2.$$

For independent samples of sizes n_1 and n_2,

$$E[\bar{X}_1 - \bar{X}_2] = \mu_1 - \mu_2$$

and

$$\operatorname{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

Independence is what allows the two variance contributions to be added without a covariance term. Estimating the unknown population variances separately gives

$$\operatorname{SE}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

Welch's procedure

Welch's procedure allows the two populations to have different variances. It is therefore the default independent-means procedure in this book. The method uses an approximate Student t distribution with Welch–Satterthwaite degrees of freedom

$$\nu = \frac{\left(s_1^2/n_1 + s_2^2/n_2\right)^2}{\dfrac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \dfrac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}.$$

The value of ν need not be an integer. Statistical software evaluates its t probabilities and critical values directly.

Theorem 8.1 (Welch Confidence Interval and Test)

Let the two samples be independent random samples from populations for which the Welch approximation is adequate. An approximate $100(1-\alpha)\%$ confidence interval for $\Delta = \mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\nu,\,1-\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

To test

$$H_0 : \mu_1 - \mu_2 = \Delta_0,$$

use

$$t_{\text{obs}} = \frac{(\bar{x}_1 - \bar{x}_2) - \Delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}},$$

with the approximate t_ν null distribution. The common no-difference claim has $\Delta_0 = 0$. ♡

The alternative hypothesis determines whether one or both tails are used, exactly as in the one-sample setting. For the common two-sided comparison,

$$H_0 : \mu_1 - \mu_2 = 0 \qquad \text{against} \qquad H_1 : \mu_1 - \mu_2 \neq 0.$$

A matching confidence interval that excludes zero yields rejection at the corresponding significance level.

Example 8.2 Mean Strength from Two Material Suppliers

A civil engineering laboratory compares compressive strength from two suppliers. Independent specimens give

	n	$\bar{x}$ (MPa)	s (MPa)
Supplier 1	18	52.4	4.8
Supplier 2	14	48.9	6.1

Define $\Delta = \mu_1 - \mu_2$. The estimated difference is

$$\widehat{\Delta} = 52.4 - 48.9 = 3.5 \text{ MPa},$$

and its estimated standard error is

$$\text{SE}(\widehat{\Delta}) = \sqrt{\frac{4.8^2}{18} + \frac{6.1^2}{14}} \approx 1.984 \text{ MPa}.$$

The Welch–Satterthwaite calculation gives $\nu \approx 24.24$. For

$$H_0 : \Delta = 0 \qquad \text{against} \qquad H_1 : \Delta \neq 0,$$

the observed statistic is

$$t_{\text{obs}} = \frac{3.5}{1.984} \approx 1.764,$$

giving a two-sided p-value of approximately 0.090. With $t_{24.24,0.975} \approx 2.063$, the 95% confidence interval is

$$3.5 \pm 2.063(1.984) = [-0.59,\, 7.59] \text{ MPa}.$$

Supplier 1 has the larger observed mean, but the data remain compatible with population differences ranging from a small advantage for supplier 2 to a substantial advantage for supplier 1. At the 5% level, the study does not establish a difference. This is not evidence that the suppliers are equivalent; the interval is too wide to support that stronger conclusion.

Why Welch is preferred to automatic pooling

The classical pooled two-sample procedure assumes a common population variance,

$$\sigma_1^2 = \sigma_2^2 = \sigma^2.$$

It estimates that variance by

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

and uses

$$\mathrm{SE}_{\mathrm{pooled}}(\bar{X}_1 - \bar{X}_2) = s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

with $n_1 + n_2 - 2$ degrees of freedom. This model is valid when a common variance is substantively justified. Similar sample standard deviations alone do not establish equal population variances, and a preliminary variance test followed by a choice of mean test adds an unnecessary data-dependent decision. Welch's procedure performs well when variances are equal and remains appropriate when they are not.

Conditions and robustness

Table 8.2: Conditions to examine for a comparison of independent means.

Condition	Practical question
Quantitative response	Is a difference in population means meaningful for the measured outcome?
Independent groups	Are the units in one group unrelated to those in the other, and are observations independent within each group?
Representative or randomised design	Do the samples address the intended populations, or were units assigned so that a causal comparison is credible?
Distributional adequacy	For small samples, are the population distributions reasonably normal? For larger samples, are skewness and influential outliers limited enough for mean-based inference?
Comparable measurement	Were outcome definitions, instruments, timing, and missing-data handling comparable in both groups?

Welch's method does not require equal variances, but it still requires a credible independent-samples design. Large samples improve the normal approximation for the sample means; they do not repair dependence, selection bias, confounding, or incompatible measurements.

8.3 Paired Comparisons

Suppose pair i provides measurements X_i and Y_i. Choose an order with a clear interpretation and define

$$D_i = X_i - Y_i, \qquad i = 1, \ldots, n.$$

The paired problem is then a one-sample problem for the differences $D_1, \ldots, D_n$. Its population parameter is

$$\mu_D = E[D],$$

the mean within-pair difference. Under the no-change claim, $\mu_D = 0$.

Theorem 8.2 (Paired t Confidence Interval and Test)

Let $D_1, \dots, D_n$ be independent pair differences from an approximately normal population of differences.

A $100(1-\alpha)\%$ confidence interval for μ_D is

$$\bar{d} \pm t_{n-1,\,1-\alpha/2}\frac{s_D}{\sqrt{n}}.$$

To test

$$H_0 : \mu_D = \Delta_0,$$

use

$$t_{\text{obs}} = \frac{\bar{d} - \Delta_0}{s_D/\sqrt{n}},$$

with $n-1$ degrees of freedom. The common no-change claim has $\Delta_0 = 0$.

The number of observations for the analysis is the number of complete pairs, not twice the number of pairs. Normality concerns the differences, not the two marginal distributions separately. A large unit-to-unit baseline variation can be removed by pairing because each unit acts as its own comparison.

For paired variables with standard deviations σ_X and σ_Y and correlation ρ, the variance of a difference is

$$\operatorname{Var}(D) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y.$$

When the paired measurements are positively correlated, the final term reduces the variability of the differences. This can make a paired design substantially more precise than a comparison of unrelated units. Pairing is helpful only when the connection is real and preserved in the data.

Example 8.3 Response Time Before and After an Update

Twenty representative requests are replayed on the same server before and after an update. For each request define

$$D_i = (\text{response time before}) - (\text{response time after}),$$

so a positive difference represents an improvement. The 20 differences have

$$\bar{d} = 9.0 \text{ ms}, \qquad s_D = 12.0 \text{ ms}.$$

The estimated standard error is

$$\frac{s_D}{\sqrt{n}} = \frac{12.0}{\sqrt{20}} \approx 2.683 \text{ ms}.$$

Testing

$$H_0 : \mu_D = 0 \qquad \text{against} \qquad H_1 : \mu_D \neq 0$$

gives

$$t_{\text{obs}} = \frac{9.0}{2.683} \approx 3.354$$

with 19 degrees of freedom, and a two-sided p-value of approximately 0.0033. Since $t_{19,0.975} \approx 2.093$, the 95% confidence interval is

$$9.0 \pm 2.093(2.683) = [3.38, 14.62] \text{ ms}.$$

The data provide evidence that the update reduces mean response time for requests represented by this replay design. The estimated reduction is 9 ms, with plausible values from about 3.4 to 14.6 ms. Whether this improvement matters operationally depends on service requirements and on whether replay conditions represent production traffic.

Why the independent analysis is different

An independent two-sample analysis estimates variability from separate groups and ignores within-pair association. Applied to genuinely paired data, it can discard the precision gained through pairing. Conversely, applying a paired procedure to unrelated observations invents a dependence that the design does not contain. The correct choice cannot be recovered from the numerical values alone.

Incomplete pairs also require attention. Removing every pair with one missing value changes the analysed population if missingness is related to the outcome. Statistical software can omit incomplete rows automatically, but the resulting target population and possible bias still require explanation.

8.4 Comparing Two Independent Proportions

For binary outcomes, let p_1 and p_2 denote the population probabilities of the event in two independent groups. Examples include defect or no defect, compromised or secure, late or on time, and failure or success. The primary parameter is often the difference

$$\Delta_p = p_1 - p_2.$$

If group j contains x_j events among n_j independent observations, then

$$\widehat{p}_j = \frac{x_j}{n_j}, \qquad j = 1, 2,$$

and

$$\widehat{\Delta}_p = \widehat{p}_1 - \widehat{p}_2.$$

Confidence interval for a difference in proportions

The estimated standard error used for a large-sample confidence interval is

$$\mathrm{SE}(\widehat{p}_1 - \widehat{p}_2) = \sqrt{\frac{\widehat{p}_1(1-\widehat{p}_1)}{n_1} + \frac{\widehat{p}_2(1-\widehat{p}_2)}{n_2}}.$$

An approximate $100(1-\alpha)\%$ interval for $p_1 - p_2$ is therefore

$$(\widehat{p}_1 - \widehat{p}_2) \pm z_{1-\alpha/2}\,\mathrm{SE}(\widehat{p}_1 - \widehat{p}_2).$$

This interval estimates each group's variance separately because the two proportions are not assumed equal.

Testing equality of two proportions

For

$$H_0 : p_1 = p_2 \qquad \text{against} \qquad H_1 : p_1 \neq p_2,$$

the null hypothesis states that both groups share one event probability p. The pooled estimator under that assumption is

$$\widehat{p} = \frac{x_1 + x_2}{n_1 + n_2}.$$

The null standard error and observed statistic are

$$\mathrm{SE}_0 = \sqrt{\widehat{p}(1-\widehat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

and

$$z_{\mathrm{obs}} = \frac{\widehat{p}_1 - \widehat{p}_2}{\mathrm{SE}_0}.$$

Under H_0 and adequate large-sample conditions, this statistic is approximately standard normal. The pooled estimate belongs to the test because equality is assumed while constructing the null distribution. It should not replace the separate estimates in the confidence interval.

Example 8.4 Defect Proportions from Two Production Processes

Independent component samples from two production processes give

	defects x	inspected n
Process 1	18	800
Process 2	36	750

The observed defect proportions are

$$\widehat{p}_1 = \frac{18}{800} = 0.0225, \qquad \widehat{p}_2 = \frac{36}{750} = 0.0480.$$

Thus

$$\widehat{p}_1 - \widehat{p}_2 = -0.0255,$$

meaning 2.55 fewer observed defects per 100 inspected components for Process 1. The unpooled standard error is

$$\sqrt{\frac{0.0225(0.9775)}{800} + \frac{0.0480(0.9520)}{750}} \approx 0.00940.$$

The 95% confidence interval is

$$-0.0255 \pm 1.960(0.00940) = [-0.0439, -0.0071].$$

For the test of equal defect probabilities, the pooled estimate is

$$\widehat{p} = \frac{18 + 36}{800 + 750} \approx 0.03484,$$

giving

$$\mathrm{SE}_0 = \sqrt{0.03484(1 - 0.03484)\left(\frac{1}{800} + \frac{1}{750}\right)} \approx 0.00932$$

and

$$z_{\mathrm{obs}} = \frac{-0.0255}{0.00932} \approx -2.736.$$

The two-sided p-value is approximately 0.0062. The data provide evidence of different defect probabilities, with Process 1 estimated to have between 0.71 and 4.39 percentage points fewer defects. A process decision should also consider inspection validity, production conditions, cost, and whether the samples represent future production.

Risk difference, relative risk, and odds ratio

The difference $p_1 - p_2$, also called a **risk difference** for adverse events, gives an absolute effect in proportions or percentage points. Two additional measures describe relative effects:

$$\text{relative risk} = \frac{p_1}{p_2}, \qquad \text{odds ratio} = \frac{p_1/(1-p_1)}{p_2/(1-p_2)}.$$

For the production example, the estimated relative risk is

$$\frac{0.0225}{0.0480} \approx 0.469,$$

so the observed defect probability for Process 1 is about 47% of that for Process 2. The estimated odds ratio is approximately 0.457. These relative measures can be useful, but the absolute reduction of 2.55 percentage points is often easier to connect to avoided defects, cost, and production capacity. An odds ratio must not be interpreted as a relative risk, especially when the event is common.

Conditions and the boundary of the method

The normal approximation requires enough expected events and non-events in each group. Sparse data can make the approximation inaccurate; exact or specially adjusted procedures may then be more appropriate and are evaluated using statistical software. More importantly, the observations must be independent within and between groups, and each group must represent its intended population or assigned condition.

This section concerns *independent* binary samples. If the same units are classified before and after an intervention, the binary outcomes are paired. The independent two-proportion procedure is then incorrect; a paired binary method is required and lies outside the scope of this chapter.

8.5 Choosing and Reporting a Two-Group Analysis

The correct method follows from the outcome type and the study design. Table 8.3 summarises the three core cases.

Table 8.3: Core methods for comparing two groups in this chapter.

Outcome	Design	Parameter	Primary method
Quantitative	Independent groups	$\mu_1 - \mu_2$	Welch two-sample t procedure
Quantitative	Paired or matched observations	μ_D	One-sample t procedure on pair differences
Binary	Independent groups	$p_1 - p_2$	Large-sample two-proportion procedure

Six-step analysis order

A defensible two-group analysis proceeds in this order:

1. state the target populations, observational units, outcome, and order of the difference;
2. describe how units were sampled or assigned and determine whether observations are independent or paired;
3. inspect the group distributions or pair differences and examine missingness, outliers, and measurement comparability;
4. estimate the population difference and report an appropriate confidence interval;
5. if a hypothesis test addresses a relevant claim, state H_0 and H_1, report the test statistic and p-value, and use the direction selected before seeing the result; and
6. interpret the magnitude and uncertainty in engineering units, together with design limitations and the distinction between association and causation.

A difference must be tested directly

A recurring mistake is to test each group separately and then compare the two decisions. Suppose an effect is statistically significant in group 1 but not in group 2. This does not imply that the groups differ significantly. The two p-values depend on both observed effects and standard errors, and one can fall just below a threshold while the other falls just above it even when the estimates are similar. A group difference must be assessed with a confidence interval or test constructed directly for that difference.

Reporting effect and uncertainty

For quantitative outcomes, the unstandardised mean difference is usually the most useful effect measure because it retains milliseconds, megapascals, kilowatt-hours, or other engineering units. A standardised mean difference can help compare effects across incompatible scales, but it does not replace a domain threshold or tolerance. For binary outcomes, report the absolute difference in proportions and, when useful, a relative risk or odds ratio with clear labels.

Statistical software is normally used to evaluate degrees of freedom, tail probabilities, critical values, and intervals. It should reproduce the method justified by the design. It cannot decide whether requests are genuinely paired, whether production samples are representative, whether random assignment was followed, or whether a 9 ms improvement is operationally valuable.

The three procedures in this chapter share one principle: estimate the difference that answers the engineering question, quantify its sampling uncertainty, and interpret it in light of how the observations were obtained. The next chapter extends the comparison from two means to several means using analysis of variance.

Chapter 9 Analysis of Variance

Comparisons are not always limited to two groups. A production engineer may compare material from several suppliers, a software engineer may evaluate more than two caching strategies, and a climate engineer may compare energy use under several control settings. The response is quantitative in each case, but the explanatory variable identifies one of several groups.

It is tempting to compare every pair with a separate two-sample test. That approach creates two problems. First, the number of comparisons grows quickly: k groups produce $k(k-1)/2$ pairs. Second, repeatedly testing at the same significance level increases the probability of making at least one type I error across the collection of comparisons. A procedure is needed that begins with one global question and, when appropriate, follows it with multiplicity-controlled comparisons.

Analysis of variance, abbreviated ANOVA, addresses the global question by comparing two sources of variation. Differences among group means create between-group variation, while natural differences among observations in the same group create within-group variation. If the between-group variation is large relative to the within-group variation, a model with one common population mean becomes difficult to reconcile with the data.

This chapter develops one-way ANOVA for independent groups. It explains the model, the decomposition of variation, the ANOVA table, the F test, practical assumptions, and Tukey's method for identifying pairwise differences. Statistical software performs the arithmetic in realistic analyses, but the formulas reveal what the reported quantities mean and what conclusions they can support.

9.1 From Two Groups to Several Groups

A one-way analysis has one quantitative response and one categorical explanatory variable. The categorical variable determines the groups being compared.

Definition 9.1 (Factor and Levels)

A **factor** is a categorical explanatory variable used to divide observations into groups. Its possible categories are called the **levels** of the factor. ♣

In a supplier comparison, the factor may be supplier and its levels may be A, B, and C. In a software experiment, the factor may be caching strategy with levels no cache, local cache, and distributed cache. The word *one-way* means that the analysis contains one factor. It does not mean that the hypothesis test is one-sided.

Let the factor have k levels. Write μ_i for the population mean response at level i. One-way ANOVA tests

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

against

$$H_1 : \text{not all of the population means are equal.}$$

The alternative does not state that every mean differs from every other mean. It includes any configuration in which at least one pair of population means differs.

Definition 9.2 (Omnibus Test)

An **omnibus test** assesses a global hypothesis involving several parameters. The one-way ANOVA F test asks whether all group means can be represented by one common population mean; it does not by itself identify which means differ.

This distinction determines the analysis order. The F test addresses the global question. If the evidence indicates that the means are not all equal and the engineering question requires identification of particular groups, a post-hoc procedure addresses the pairwise questions while controlling error across the collection.

Why separate pairwise tests are a poor default

With three groups there are three pairwise comparisons. With five groups there are ten, and with ten groups there are 45. If every comparison is conducted as an unadjusted level-α test, the probability of at least one false rejection across the family is generally larger than α. The exact probability depends on the dependence among the tests, so it is not correctly obtained by treating all pairwise statistics as independent.

ANOVA does not solve every multiple-comparison problem automatically. It provides one protected global test of equality. A compatible post-hoc procedure, such as Tukey's method, is still needed when the goal is to determine which pairs differ.

9.2 The One-Way ANOVA Model

Suppose level i contributes n_i observations,

$$X_{i1}, X_{i2}, \ldots, X_{in_i}, \qquad i = 1, \ldots, k.$$

The total number of observations is

$$N = \sum_{i=1}^{k} n_i.$$

The group-means formulation of the one-way model is

$$X_{ij} = \mu_i + \varepsilon_{ij},$$

where μ_i is the population mean for level i and ε_{ij} is the random deviation of observation j from that mean.

For the classical one-way ANOVA F test, the errors are modelled as independent normal random variables with a common variance:

$$\varepsilon_{ij} \sim N(0, \sigma^2).$$

Equivalently,

$$X_{ij} \sim N(\mu_i, \sigma^2).$$

The means may differ across groups, but the classical model uses the same population variance in every group.

The sample mean for group i is

$$\bar{x}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} x_{ij},$$

and the grand mean of all N observations is

$$\bar{x}_{\cdot\cdot} = \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n_i} x_{ij} = \frac{1}{N}\sum_{i=1}^{k} n_i \bar{x}_i.$$

The second expression shows that the grand mean is a weighted average of the group means. Larger groups receive greater weight.

An observed within-group deviation is called a residual:

$$e_{ij} = x_{ij} - \bar{x}_i.$$

Residuals estimate the unobserved errors ε_{ij}. They are central to assessing whether the common-variance and normal-error descriptions are reasonable.

Definition 9.3 (Balanced and Unbalanced Designs)

A one-way design is **balanced** when all group sample sizes are equal. It is **unbalanced** when at least two group sample sizes differ.

Balance is not required, but it makes the group means contribute more symmetrically and often improves robustness when assumptions are only approximate. Unequal group sizes combined with unequal variances require particular care.

The design comes before the model

The observations must represent independent experimental or sampling units. Repeated measurements from the same machine, building, person, account, or component are not made independent by placing them in different groups in a data table. Clustered data and repeated-measures designs require methods beyond the one-way independent-groups model.

Random assignment of units to factor levels can support a causal interpretation when the experimental protocol is credible. In an observational comparison, ANOVA can describe evidence of an association between group membership and the mean response, but pre-existing group differences may provide alternative explanations. Random sampling and representative measurement determine how far the result can be generalised.

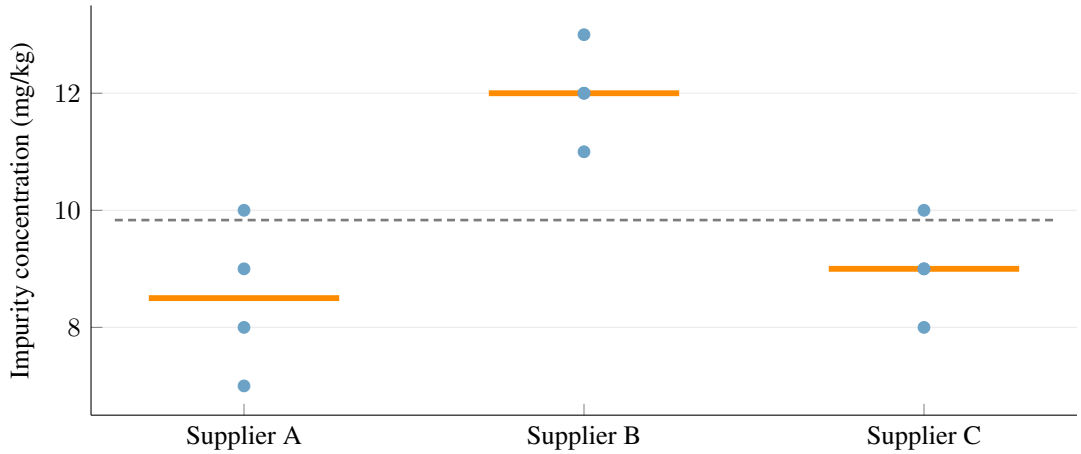


Figure 9.1: Observations from three groups. Blue points are individual measurements, orange segments are group means, and the dashed line is the grand mean. ANOVA compares the separation of the group means with the variation of observations around their own group means.

9.3 Partitioning Variation and the F Test

Every observation differs from the grand mean by an amount

$$x_{ij} - \bar{x}_{..}.$$

This total deviation can be written as

$$x_{ij} - \bar{x}_{..} = (\bar{x}_i - \bar{x}_{..}) + (x_{ij} - \bar{x}_i).$$

The first term is the deviation of the group mean from the grand mean. The second is the deviation of the observation from its own group mean. ANOVA turns this observation-level decomposition into a decomposition of squared variation.

Definition 9.4 (ANOVA Sums of Squares)

The **total sum of squares** is

$$SS_{\text{total}} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_{..})^2.$$

The **between-group sum of squares** is

$$SS_{\text{between}} = \sum_{i=1}^{k} n_i (\bar{x}_i - \bar{x}_{..})^2,$$

and the **within-group sum of squares** is

$$SS_{\text{within}} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2.$$

The group sizes appear in SS_{between} because moving a group mean away from the grand mean affects every observation represented by that group. The within-group sum of squares is the sum of the squared residuals.

Theorem 9.1 (ANOVA Sum-of-Squares Identity)

For any one-way dataset,

$$SS_{\text{total}} = SS_{\text{between}} + SS_{\text{within}}.$$

The associated degrees of freedom satisfy

$$N - 1 = (k - 1) + (N - k).$$

The identity is numerical and always holds for the stated definitions; it does not depend on normality. The distributional assumptions enter when the decomposition is used for inference.

Sums of squares grow with the number of independent pieces of information. Dividing by their degrees of freedom produces mean squares:

$$MS_{\text{between}} = \frac{SS_{\text{between}}}{k - 1}, \qquad MS_{\text{within}} = \frac{SS_{\text{within}}}{N - k}.$$

The within-group mean square is also called the error mean square. Under the classical model, it estimates the common variance σ^2 whether or not the group means are equal. When H_0 is true, the between-group mean square also estimates σ^2. If population means differ, MS_{between} tends to be enlarged by their separation.

The resulting test statistic is

$$F_{\text{obs}} = \frac{MS_{\text{between}}}{MS_{\text{within}}}.$$

Values close to 1 indicate that the two variance estimates are of similar size. Large values indicate that the group means are more separated than would be expected from the observed within-group variation under one common population mean.

The F distribution

An F distribution is a continuous probability distribution for a ratio of two independent variance estimates after appropriate scaling. It takes only nonnegative values, is generally right-skewed, and is indexed by a numerator and a denominator number of degrees of freedom.

Theorem 9.2 (One-Way ANOVA F Test)

Under the classical one-way normal model and

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k,$$

the statistic

$$F_{\text{obs}} = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

is a realised value from an F distribution with $k - 1$ numerator and $N - k$ denominator degrees of freedom. The upper-tail p-value is

$$p = P(F_{k-1,N-k} \geq F_{\text{obs}} \mid H_0 \text{ and the model assumptions}) .$$

♡

Only the upper tail is relevant because evidence against equal means appears when the between-group mean square is unusually large relative to the within-group mean square. Statistical software evaluates the F probability; an F table is not required for routine analysis.

When there are exactly two groups and the equal-variance model is used, the one-way ANOVA statistic equals the square of the classical pooled two-sample t statistic:

$$F = t^2.$$

Thus one-way ANOVA extends the equal-variance comparison of two means. It is not equivalent to Welch's unequal-variance two-sample procedure.

The ANOVA table

The calculations are conventionally organised as shown in Table 9.1.

Table 9.1: The classical one-way ANOVA table.

Source	Sum of squares	df	Mean square	Statistic
Between groups	SS_{between}	$k - 1$	$SS_{\text{between}}/(k-1)$	$F = MS_{\text{between}}/MS_{\text{within}}$
Within groups	SS_{within}	$N - k$	$SS_{\text{within}}/(N-k)$	
Total	SS_{total}	$N - 1$		

The table is more than a software format. It records how the observed variation is partitioned and makes the construction of the test statistic visible.

Example 9.1 Impurity Concentrations from Three Suppliers

A production engineer measures impurity concentration in four independently selected batches from each of three resin suppliers. The measurements, in mg/kg, are

Supplier	observations				$\bar{x}_i$
A	8	9	7	10	8.5
B	12	11	13	12	12.0
C	9	10	8	9	9.0

The grand mean is

$$\bar{x}_{\cdot\cdot} = \frac{8+9+7+10+12+11+13+12+9+10+8+9}{12} = \frac{118}{12} \approx 9.833.$$

The between-group sum of squares is

$$\begin{aligned} SS_{\text{between}} &= 4(8.5-9.833)^2 + 4(12.0-9.833)^2 + 4(9.0-9.833)^2 \\ &\approx 28.667. \end{aligned}$$

Within each supplier, the squared residuals sum to

$$\begin{aligned} SS_A &= (8-8.5)^2 + (9-8.5)^2 + (7-8.5)^2 + (10-8.5)^2 = 5, \\ SS_B &= (12-12)^2 + (11-12)^2 + (13-12)^2 + (12-12)^2 = 2, \\ SS_C &= (9-9)^2 + (10-9)^2 + (8-9)^2 + (9-9)^2 = 2. \end{aligned}$$

Therefore,

$$SS_{\text{within}} = 5 + 2 + 2 = 9,$$

and the sum-of-squares identity gives

$$SS_{\text{total}} = 28.667 + 9 = 37.667.$$

With $k = 3$ and $N = 12$, the ANOVA table is

Source	SS	df	MS	F
Between groups	28.667	2	14.333	14.333
Within groups	9.000	9	1.000	
Total	37.667	11		

For

$$H_0 : \mu_A = \mu_B = \mu_C,$$

the statistic $F = 14.333$ has 2 and 9 degrees of freedom under H_0. The upper-tail probability is approximately

$$p = 0.0016.$$

The data provide evidence that the three population mean impurity concentrations are not all equal. The result does not yet establish which suppliers differ, nor whether supplier membership caused the differences. Those conclusions require pairwise inference and consideration of the design.

9.4 Assumptions, Diagnostics, and Robustness

The exact classical F result follows from independent normal errors with a common variance. Practical validity depends on how the data were generated as well as on the observed distributions. Table 9.2 translates the model into questions that can be investigated.

Table 9.2: Conditions to examine before interpreting a one-way ANOVA.

Condition	Practical question
Quantitative response	Is a population mean meaningful for the measured outcome in every group?
Independent observations	Are different rows genuinely independent experimental or sampling units rather than repeated, clustered, or time-dependent measurements?
Representative or randomised design	Do the observations support generalisation to the target populations, or were units randomly assigned so that a causal comparison is credible?
Approximately normal errors	Within each group, are residuals reasonably compatible with a normal model, especially when group sizes are small?
Common population variance	Are the within-group spreads sufficiently similar for one pooled variance to represent all groups?
Comparable measurement	Were definitions, instruments, timing, censoring, and missing-data handling comparable across levels?

Independence and study design

Independence cannot be diagnosed reliably from a histogram, boxplot, or normality test. It follows from the sampling and experimental structure. Treating repeated observations from one unit as independent creates artificial information and usually makes standard errors too small. No transformation of the response repairs an invalid independence assumption.

The observational unit must therefore be stated explicitly. If four measurements are taken from each of ten components, the study has ten independently sampled components unless the model accounts for the repeated measurements within component. It does not automatically have 40 independent components.

Normality concerns the errors within groups

The classical model describes the distribution around each group mean. Consequently, the relevant quantities are the residuals $e_{ij} = x_{ij} - \bar{x}_i$, not the overall response distribution formed by mixing all groups. Even perfectly normal groups can produce a multimodal combined distribution when their means differ.

Group plots and residual plots can reveal strong skewness, outliers, and unequal spread. A normal quantile plot of the residuals can provide an additional diagnostic. Formal normality tests are not a mechanical gatekeeper: they may have little power in small samples and detect immaterial deviations in large samples.

Classical ANOVA is often reasonably robust to moderate non-normality when group sizes are not very small and the design is balanced or nearly balanced. Severe skewness, influential outliers, heavy tails, dependence, or censoring can still undermine a mean-based analysis. There is no universal sample-size rule that overrides these features.

Common variance and unequal group sizes

The pooled within-group mean square assumes one population variance σ^2. Similar-looking sample standard deviations do not prove equality, and a preliminary variance test should not automatically decide which analysis is reported. The judgment should combine subject-matter knowledge, study design, group sizes, and plots of the group or residual distributions.

Unequal variances are most problematic when group sizes are also unequal. In particular, a small high-variance group or a large low-variance group can distort the classical F test. A balanced design provides protection because each group contributes comparably, but balance does not make extreme variance differences irrelevant.

When the group variances are not similar

Welch's one-way ANOVA provides a practical alternative when independent groups have unequal variances. It weights the group information differently and uses adjusted degrees of freedom; statistical software performs the calculation. If pairwise follow-up is needed after Welch's ANOVA, Games–Howell comparisons provide the corresponding unequal-variance approach. Ordinary Tukey comparisons use the pooled within-group variance and should not be combined automatically with a Welch omnibus analysis.

Welch's ANOVA and Games–Howell are included here as method-selection tools, not as additional hand-calculation procedures. Transformations or rank-based methods may sometimes be suitable, but they change the modelling or interpretation and are outside the core scope of this chapter.

9.5 Post-Hoc Comparisons and Effect Magnitude

A significant omnibus result supports the statement that the population means are not all equal. It does not support statements such as "supplier B differs from supplier A" without a comparison designed for that pair. Inspecting the largest sample means after the test and then applying ordinary unadjusted t intervals ignores the selection and multiplicity that produced the question.

Family-wise error and Tukey's method

The **family-wise error rate** is the probability of making at least one type I error among a specified family of comparisons. Tukey's method constructs simultaneous inference for every pair of population means while controlling this probability at a chosen level α under the classical one-way model.

Theorem 9.3 (Tukey–Kramer Pairwise Confidence Intervals)

For all pairwise differences $\mu_i - \mu_j$ in a classical one-way ANOVA, a simultaneous $100(1-\alpha)\%$ Tukey–Kramer confidence interval is

$$(\bar{x}_i - \bar{x}_j) \pm q_{1-\alpha;k,N-k}\sqrt{\frac{MS_{\text{within}}}{2}\left(\frac{1}{n_i}+\frac{1}{n_j}\right)},$$

where $q_{1-\alpha;k,N-k}$ is a critical value from the studentized-range distribution. When all group sizes equal n, the margin of error reduces to

$$q_{1-\alpha;k,N-k}\sqrt{\frac{MS_{\text{within}}}{n}}.$$

Statistical software evaluates the studentized-range critical value and usually reports multiplicity-adjusted p-values together with simultaneous confidence intervals. A pair is distinguished at family-wise level α when its simultaneous interval excludes zero.

The usual exploratory workflow begins with the omnibus question and, when group identification is relevant, follows it with Tukey comparisons. The protection in a Tukey family comes from its simultaneous critical value, not merely from performing an F test first. Comparisons formulated in advance to answer specific engineering questions are a different design problem and lie outside this introductory treatment.

Example 9.2 Locating the Supplier Differences

Return to the impurity data from the preceding example. Since all three groups have size $n = 4$ and $MS_{\text{within}} = 1$, the 95% Tukey margin of error is

$$q_{0.95;3,9}\sqrt{\frac{1}{4}} \approx 3.948(0.5) = 1.974 \text{ mg/kg}.$$

The pairwise results are

Difference	estimate	95% simultaneous interval	adjusted p
$\mu_B - \mu_A$	3.50	[1.526, 5.474]	0.0020
$\mu_C - \mu_A$	0.50	[−1.474, 2.474]	0.7655
$\mu_C - \mu_B$	−3.00	[−4.974, −1.026]	0.0055

At the 5% family-wise level, supplier B differs from both A and C, while A and C are not distinguished. The intervals also quantify the differences: supplier B's mean impurity concentration is estimated to exceed A's by between about 1.5 and 5.5 mg/kg and C's by between about 1.0 and 5.0 mg/kg. Failure to distinguish A and C does not establish that their population means are identical.

A descriptive effect size

The omnibus p-value measures incompatibility with equal means under the model; it does not measure how important the group effect is. A simple descriptive effect size is

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}.$$

It is the proportion of the observed total sum of squares represented by differences among the observed group means. For the supplier example,

$$\eta^2 = \frac{28.667}{37.667} \approx 0.761.$$

Thus about 76.1% of the observed squared variation in this small illustrative dataset is associated with supplier-level mean differences.

This value is a sample description, not the probability that supplier determines an individual measurement and not automatically the proportion causally explained by supplier. It can overstate the corresponding population effect, especially in small samples. Group means, pairwise differences, and their confidence intervals remain essential because they retain the engineering units of the response.

9.6 Choosing and Reporting a One-Way Analysis

The method must match both the data structure and the engineering question. A classical one-way ANOVA is appropriate for a quantitative response from independent groups when a common within-group variance is a credible approximation. Welch's ANOVA is preferable when the independent groups have materially different variances. Neither method applies directly to paired, repeated, clustered, or time-dependent observations.

Seven-step analysis order for several means

A defensible one-way analysis proceeds in this order:

1. state the target population, observational unit, quantitative response, factor, and factor levels;
2. explain how units were sampled or assigned and whether the groups contain independent observations;
3. report group sample sizes, means, standard deviations, and a plot showing distributions and possible outliers;
4. assess whether the classical common-variance model is adequate or whether Welch's unequal-variance approach is more credible;
5. report the omnibus hypotheses, test statistic with degrees of freedom, p-value, and an effect magnitude;
6. when particular pairs must be identified, use a compatible multiplicity-controlled procedure and report simultaneous confidence intervals; and
7. interpret differences in engineering units, including uncertainty, practical importance, design limitations, and the distinction between association and causation.

For the illustrative supplier data, a compact report could read:

> Mean impurity concentration differed among the three sampled suppliers, $F(2, 9) = 14.33$, $p = 0.0016$, with $\eta^2 = 0.761$ in this sample. Tukey 95% simultaneous intervals indicated that supplier B had a higher population mean than A by an estimated 3.50 mg/kg, 95% simultaneous CI $[1.53, 5.47]$, and a higher mean than C by an estimated 3.00 mg/kg, 95% simultaneous CI $[1.03, 4.97]$. Suppliers A and C were not distinguished. The result concerns the sampled production conditions; a causal supplier effect requires a design that rules out systematic differences in batches, handling, and measurement.

The report combines the global question, the pairwise questions, effect magnitude, uncertainty, and design limitations. Reporting only "ANOVA was significant" omits the quantities needed for an engineering decision. Conversely, presenting many pairwise decisions without the family and adjustment obscures the error rate being controlled.

One-way ANOVA therefore has a clear division of labour. The variance decomposition explains the global F test, multiplicity-controlled intervals locate relevant group differences, and study design determines what those numerical results can mean. The next chapter replaces categorical group membership with a quantitative explanatory variable and develops simple linear regression.

Part III

Modelling and Integrated Analysis

Chapter 10 Simple Linear Regression

Fitting a straight line is computationally simple. Deciding whether the line represents a credible statistical relationship is not. A line can be calculated for almost any collection of paired numbers, including data with curvature, changing variability, influential observations, time dependence, or no scientifically meaningful connection. Regression analysis therefore begins with the question and the data-generating process, not with a fitting command.

Simple linear regression describes how the conditional mean of a quantitative response changes with one quantitative predictor. It separates systematic linear change from residual variation, estimates the magnitude of the change, and quantifies uncertainty about both mean response and individual prediction. Scatterplots and residual plots are central because the formulas are meaningful only when the model provides a reasonable description of the observations.

This chapter develops regression from a statistical perspective. Least squares explains how the fitted line is obtained; residuals reveal what the line leaves unexplained; confidence intervals and tests quantify uncertainty about the slope; and prediction intervals acknowledge variation among individual outcomes. No matrix formulation or machine-learning workflow is required. The objective is not merely to produce a line, but to understand when it can be trusted and what conclusions it supports.

10.1 The Statistical Question and Regression Model

Regression assigns different roles to the two variables.

Definition 10.1 (Response and Predictor)

The **response variable**, denoted Y, is the quantitative outcome to be explained or predicted. The **predictor variable**, denoted x, is the quantitative variable used to describe systematic change in the conditional mean of Y. ♣

For example, a controlled server experiment may use processor load as the predictor and energy use during a one-hour run as the response. A structural test may use applied load as the predictor and beam deflection as the response. Reversing the variables changes the regression question and generally changes the fitted line.

The first analysis is a scatterplot of the response against the predictor. It reveals the observed range of x, direction and strength of association, possible curvature, group structure, unusual observations, and regions with different variability. A numerical coefficient cannot replace this visual inspection.

The population regression function describes the mean response at a given predictor value:

$$E(Y \mid x) = \beta_0 + \beta_1 x.$$

The simple linear regression model for observation i is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where ε_i is the random deviation from the population line. The parameters have distinct interpretations:

- β_0 is the population mean response at $x = 0$; and
- β_1 is the change in population mean response associated with a one-unit increase in x.

The slope has units of response per predictor unit. If x is processor load in percentage points and Y is energy use in kWh, then β_1 has units kWh per percentage point. Reporting a slope without its units removes much of its engineering meaning.

The intercept may be mathematically necessary without having a useful physical interpretation. If all observed processor loads lie between 17% and 93%, the fitted intercept at 0% is outside the observed range. It should not automatically be interpreted as measured idle energy. Forcing a line through the origin is justified only when scientific knowledge supports a zero response at a zero predictor value.

Study design and causal interpretation

Suppose an experiment consists of 40 one-hour server benchmark runs. Target processor-load settings are assigned in random order, environmental conditions are held as stable as possible, and each run produces a load measurement and an energy measurement. Randomising the run order reduces the risk that warm-up, drift, or another time trend is confused with load.

This design can support a causal interpretation within the tested system if the load intervention, measurement process, and run independence are credible. In observational data, a regression slope ordinarily describes association. An omitted variable may influence both predictor and response, and a high R^2 does not remove that concern. Generalisation beyond the observed machines, workloads, and operating conditions requires additional design justification.

10.2 Least Squares, Fitted Values, and Residual Variation

The population parameters β_0 and β_1 are unknown. A sample produces the fitted line

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

Definition 10.2 (Fitted Value and Residual)

For observation (x_i, y_i), the **fitted value** is

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

The **residual** is

$$e_i = y_i - \hat{y}_i.$$

It is the observed vertical difference between the response and the fitted line. ♣

Residuals estimate the unobserved model errors, but they are not identical objects. An error ε_i is defined relative to the unknown population line; a residual e_i is calculated relative to the fitted sample line.

Ordinary least squares chooses the intercept and slope to minimise the residual sum of squares

$$SS_{\text{res}} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} e_i^2.$$

Squaring prevents positive and negative residuals from cancelling and gives relatively large discrepancies greater influence on the fit.

Define the centred sums

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2, \qquad S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}).$$

Theorem 10.1 (Least-Squares Estimates)

Provided $S_{xx} > 0$, the least-squares slope and intercept are

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

The fitted line passes through $(\bar{x}, \bar{y})$, and its residuals satisfy

$$\sum_{i=1}^{n} e_i = 0 \qquad \text{and} \qquad \sum_{i=1}^{n} x_i e_i = 0.$$

♡

The condition $S_{xx} > 0$ means that the predictor values cannot all be identical. The second residual identity expresses the least-squares balance between residuals and the predictor; it does not imply that observations are independent or that the linear model is correct.

Example 10.1 Energy Use in Controlled Server Benchmarks

For the 40 benchmark runs, let x be processor load in percentage points and let y be energy use in kWh during the one-hour run. The sample summaries are

$$n = 40, \qquad \bar{x} = 53.370, \qquad \bar{y} = 32.343,$$

$$S_{xx} = 20809.410, \qquad S_{xy} = 8659.564.$$

The fitted slope is

$$\hat{\beta}_1 = \frac{8659.564}{20809.410} \approx 0.4161 \text{ kWh per percentage point,}$$

and the intercept is

$$\hat{\beta}_0 = 32.343 - 0.4161(53.370) \approx 10.134 \text{ kWh.}$$

Thus the fitted line is

$$\hat{y} = 10.134 + 0.4161x.$$

Within the tested range, ten additional percentage points of processor load are associated with approximately

$$10(0.4161) = 4.161 \text{ kWh}$$

higher mean energy use. The intercept is an algebraic anchor for the line; it is not a measured estimate of idle energy because $x = 0$ lies outside the observed range.

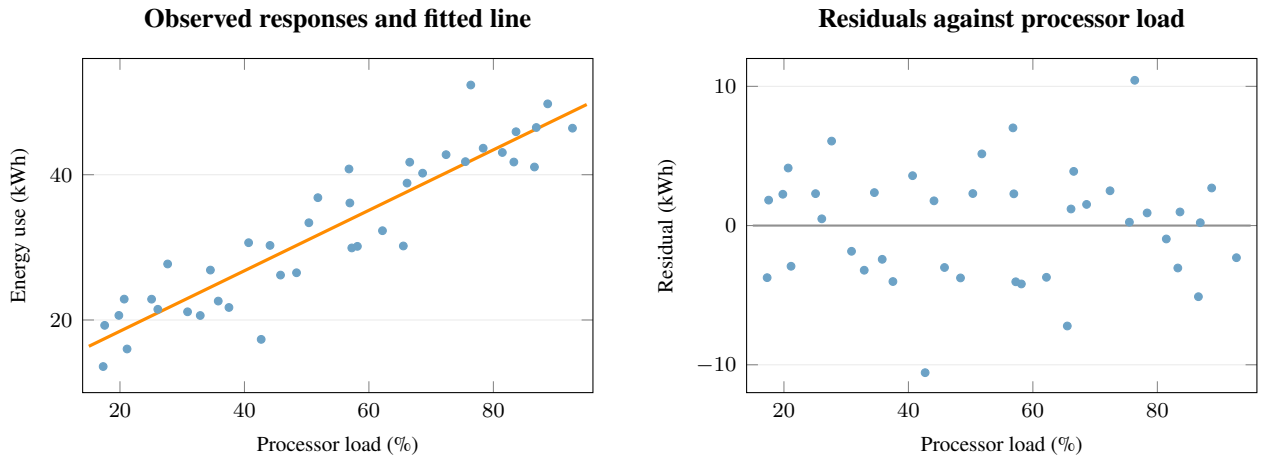


Figure 10.1: The fitted line summarises the increasing mean response. The residual plot displays the deviations on their original energy scale and makes model checking possible.

The residual sum of squares for the benchmark data is

$$SS_{\text{res}} = 653.365 \text{ kWh}^2.$$

Because two parameters have been estimated, the residual variance is estimated with $n - 2$ degrees of freedom:

$$s^2 = \frac{SS_{\text{res}}}{n-2}.$$

The residual standard deviation is therefore

$$s = \sqrt{\frac{653.365}{38}} \approx 4.147 \text{ kWh}.$$

It describes the typical vertical scale of unexplained variation around the fitted line. Unlike R^2, it retains the units of the response.

10.3 Is a Linear Model Credible?

Least squares always returns coefficients when the predictor varies, but statistical interpretation requires a credible model. Table 10.1 summarises the main conditions.

Table 10.1: Conditions to examine in a simple linear regression analysis.

Condition	Practical question
Linear conditional mean	Does a straight line adequately describe how the mean response changes across the observed predictor range?
Independent errors	Could repeated units, clusters, run order, process drift, or time dependence relate the errors?
Constant error variance	Is the vertical spread around the line reasonably similar across predictor values?
Approximately normal errors	For small-sample inference, are the errors reasonably compatible with a normal distribution?
Credible measurement	Are response and predictor measured consistently, without censoring, changing instruments, or substantial predictor error?
Representative or randomised design	Does the design support the intended population, operating range, and causal or associational interpretation?

Residual plots reveal model structure

A useful residual plot places residuals on the vertical axis and fitted values or predictor values on the horizontal axis. For a credible linear mean and constant variance, the residuals should form an approximately patternless horizontal cloud around zero. The plot does not need to look perfectly random, but systematic structure requires explanation.

Curvature indicates that the conditional mean is not adequately represented by one straight line. A funnel shape indicates that response variability changes with the predictor. A sequence of positive residuals followed by negative residuals in collection order may indicate drift or dependence even if a residual-versus-fitted plot looks acceptable.

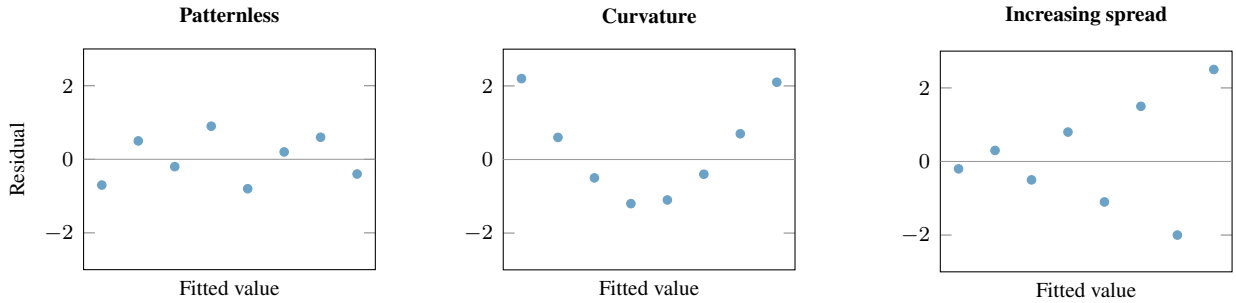


Figure 10.2: Illustrative residual patterns. A patternless cloud supports the linear constant-variance description; curvature and increasing spread indicate different model problems.

The benchmark residuals in Figure 10.1 show no obvious curvature or funnel shape, but two runs have residuals near -10.6 and 10.4 kWh. Their measurements and experimental logs should be checked. They should not be removed merely because they are unusual.

Normality and the role of a QQ-plot

Normality concerns the model errors around the line, not the marginal distributions of processor load or energy use. A normal quantile plot of the residuals can reveal strong skewness, heavy tails, and unusual observations. The normal assumption is needed for the exact small-sample t results developed later; it is not required to calculate the least-squares line itself.

Formal normality tests are not automatic gatekeepers. With small samples they may miss serious deviations, while large samples may make immaterial departures statistically detectable. Design information, residual plots, QQ-plots, and the influence of unusual observations should be considered together.

Outliers, leverage, and influence

Three related ideas must be distinguished.

A high-leverage observation can pull the line toward itself and consequently have a modest residual. Residual size alone therefore does not measure influence. Statistical software can report standardised residuals, leverage, and influence diagnostics, but the response is an investigation: verify data, understand the operating condition, compare analyses transparently, and retain scientifically valid observations unless there is a defensible reason to exclude them.

Table 10.2: Unusual observations in regression.

Term	Statistical meaning
Response outlier	An observation with a large vertical discrepancy from the fitted line, usually reflected in a large residual.
High leverage	An observation with a predictor value far from the centre of the observed predictor values, giving it unusual potential to affect the line.
Influential observation	An observation whose removal or modification would change fitted coefficients, predictions, or conclusions substantially.

10.4 Variation, Correlation, and the Coefficient of Determination

The fitted slope describes change in the original measurement units, but two unitless summaries are also commonly reported: Pearson's correlation r and the coefficient of determination R^2. They answer different questions. Correlation describes the direction and strength of a *linear association*; R^2 describes how much of the observed squared variation in the response is represented by the fitted line. Neither quantity can replace the scatterplot, residual diagnostics, or the slope with its units.

Correlation: direction and strength of linear association

Pearson's sample correlation is

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}, \qquad S_{yy} = \sum_{i=1}^{n}(y_i - \bar{y})^2.$$

The numerator S_{xy} compares paired deviations from the two sample means. It tends to be positive when above-average values of x occur with above-average values of y, and below-average values of x occur with below-average values of y. It tends to be negative when above-average values of one variable occur with below-average values of the other. Dividing by $\sqrt{S_{xx}S_{yy}}$ standardises this co-movement, so the result has no units and always satisfies

$$-1 \leq r \leq 1.$$

The sign gives the direction of the linear association:

- $r > 0$ indicates that larger predictor values tend to accompany larger response values;
- $r < 0$ indicates that larger predictor values tend to accompany smaller response values; and
- $r = 0$ indicates no sample *linear* association.

The magnitude $|r|$ describes how closely the observations follow some straight line. Values of $|r|$ near 1 indicate a tight linear pattern, while values near 0 indicate a weak linear pattern. The endpoints $r = 1$ and $r = -1$ occur only when every observation lies exactly on a straight line with positive or negative slope, respectively.

There is no universal boundary at which a correlation becomes "strong." Its practical meaning depends on measurement quality, natural variability, the engineering context, and the consequences of the decision. A correlation of 0.60 may be valuable in a noisy field study, while a correlation of 0.95 may be inadequate for a precision calibration system. Interpretation should therefore combine the numerical value with the scatterplot and the intended use.

For the benchmark experiment,

$$r = 0.920.$$

This indicates a strong positive linear association within the tested processor-load range: runs with greater load generally use more energy, and the observations lie reasonably close to an increasing straight line. It does *not* mean that energy use increases by 0.920 kWh for each percentage point of load. Correlation has no units; the fitted slope, 0.416 kWh per percentage point, measures that change.

Correlation is symmetric: the correlation between x and y equals the correlation between y and x. Regression is directional because it describes the conditional mean of a designated response using a designated predictor. Correlation is also sensitive to unusual observations, clusters, and the observed range of the variables. Restricting a study to a narrow predictor range can reduce $|r|$, while combining distinct groups can create or conceal an association. Finally, $r \approx 0$ does not imply independence or the absence of a relationship; a strong curved relationship can have correlation near zero.

Coefficient of determination: reduction in squared variation

Before using the predictor, a simple baseline prediction for every response is the sample mean $\bar{y}$. Its total squared prediction error is

$$SS_{\text{total}} = \sum_{i=1}^{n} (y_i - \bar{y})^2,$$

which measures the observed squared variation of the responses around their mean. Regression replaces the single baseline $\bar{y}$ with fitted values $\hat{y}_i$. The variation represented by the fitted line and the variation left in its residuals are

$$SS_{\text{model}} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2, \qquad SS_{\text{res}} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

Theorem 10.2 (Regression Sum-of-Squares Identity)

For an ordinary least-squares regression with an intercept,

$$SS_{\text{total}} = SS_{\text{model}} + SS_{\text{res}}.$$

The coefficient of determination is

$$R^2 = \frac{SS_{\text{model}}}{SS_{\text{total}}} = 1 - \frac{SS_{\text{res}}}{SS_{\text{total}}}.$$

♡

Thus R^2 compares the fitted regression with the mean-only baseline. It has two equivalent sample interpretations:

- R^2 is the proportion of the observed total squared variation in the response represented by the fitted line; and
- R^2 is the proportional reduction in squared prediction error obtained by using the fitted line instead of predicting $\bar{y}$ for every observation.

For ordinary least-squares regression with an intercept, $0 \le R^2 \le 1$. A value near 0 means that the fitted line improves little on the mean-only baseline, whereas a value near 1 means that the observations lie close to the fitted line relative to their total variation around $\bar{y}$.

For the benchmark experiment,

$$SS_{\text{total}} = 4256.930, \qquad SS_{\text{model}} = 3603.565, \qquad SS_{\text{res}} = 653.365,$$

so

$$R^2 = 1 - \frac{653.365}{4256.930} \approx 0.847.$$

Approximately 84.7% of the observed squared variation in energy use is represented by the fitted linear association with processor load. Equivalently, the fitted line reduces the squared errors by 84.7% relative to predicting the same mean energy use for every benchmark run. The remaining 15.3% is residual squared variation around the line.

For simple least-squares regression with one predictor and an intercept,

$$R^2 = r^2.$$

The identity explains why $0.920^2 \approx 0.847$. Squaring removes the sign, so R^2 alone does not reveal whether the association is increasing or decreasing. For example, correlations of 0.920 and -0.920 both give $R^2 \approx 0.847$; the slope or correlation must provide the direction.

The phrase "variation explained by the model" is often used for R^2, but *explained* means accounted for by the fitted values in this algebraic decomposition. It does not establish a causal mechanism. Nor does $R^2 = 0.847$ mean that the model is 84.7% correct, that 84.7% of individual outcomes are predicted correctly, or that processor load causally produces exactly 84.7% of the energy variation.

A high R^2 does not guarantee a credible model: curvature, changing variance, dependence, confounding, or one influential observation may remain. A low R^2 does not make a relationship useless: a precisely estimated slope can still describe an important average change when individual responses are naturally variable. Because R^2 is unitless and relative, it should be reported with the slope, its uncertainty, residual diagnostics, and the residual standard deviation. In the benchmark, $R^2 = 0.847$ describes relative fit, while $s = 4.15$ kWh describes the typical unexplained vertical variation in the response's original units.

Table 10.3: Distinct interpretations of three regression summaries.

Quantity	What it describes	What it does not establish
r	Direction and strength of sample linear association	Slope, prediction error, causation, or absence of nonlinear association
R^2	Proportion of sample squared response variation represented by the fitted line	Model correctness, causal explanation, or future predictive accuracy
s	Typical residual variation in response units	Proportion of variation represented or direction of association

Connection to the ANOVA F test

With one predictor, regression assigns one degree of freedom to the fitted linear change and $n-2$ degrees of freedom to residual variation. The regression F statistic is

$$F = \frac{SS_{\text{model}}/1}{SS_{\text{res}}/(n-2)}.$$

For the same two-sided null hypothesis $H_0 : \beta_1 = 0$, this statistic equals the square of the slope t statistic:

$$F = t^2.$$

The equality connects regression to the variance-comparison logic of the preceding chapter. In simple regression, a separate global F calculation adds no new decision beyond the two-sided slope test.

10.5 Inference for the Population Slope

Under the classical regression model, the errors are independent with

$$E(\varepsilon_i) = 0, \qquad \text{Var}(\varepsilon_i) = \sigma^2,$$

and are normally distributed for exact small-sample inference. The residual standard deviation s estimates the unknown error standard deviation σ.

The estimated standard error of the fitted slope is

$$\text{SE}(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}.$$

Greater residual variation increases slope uncertainty, while a wider spread of credible predictor values increases the information available about the slope.

Theorem 10.3 (Confidence Interval and Test for a Regression Slope)

Under the classical simple linear regression assumptions, a $100(1-\alpha)\%$ confidence interval for β_1 is

$$\hat{\beta}_1 \pm t_{n-2,1-\alpha/2}\,\text{SE}(\hat{\beta}_1).$$

To test

$$H_0 : \beta_1 = \beta_{1,0},$$

use

$$t_{\text{obs}} = \frac{\hat{\beta}_1 - \beta_{1,0}}{\text{SE}(\hat{\beta}_1)},$$

with $n-2$ degrees of freedom. The common no-linear-association claim has $\beta_{1,0} = 0$. ♡

The test asks how extreme the estimated slope would be if the stated population slope and model assumptions were true. Rejecting $H_0 : \beta_1 = 0$ provides evidence of a linear association. It does not prove a causal effect and does not rule out additional nonlinear structure.

Example 10.2 Inference for the Energy–Load Slope

For the benchmark experiment,

$$s = 4.1465 \text{ kWh}, \qquad S_{xx} = 20809.410,$$

so

$$\mathrm{SE}(\hat{\beta}_1) = \frac{4.1465}{\sqrt{20809.410}} \approx 0.02874$$

kWh per percentage point. Testing

$$H_0 : \beta_1 = 0 \qquad \text{against} \qquad H_1 : \beta_1 \neq 0$$

gives

$$t_{\text{obs}} = \frac{0.4161}{0.02874} \approx 14.48$$

with 38 degrees of freedom and $p < 0.0001$. With $t_{38,0.975} \approx 2.024$, the 95% confidence interval is

$$0.4161 \pm 2.024(0.02874) = [0.358, 0.474]$$

kWh per percentage point.

The data provide strong evidence of a positive linear association under the benchmark design. More importantly, the interval estimates that ten additional percentage points of load increase mean energy use by between

$$10(0.358) = 3.58 \text{ kWh}$$

and

$$10(0.474) = 4.74 \text{ kWh}$$

under the tested conditions. Operational importance depends on workload duration, energy price, thermal constraints, and whether the benchmark represents production operation.

10.6 Mean Response and Individual Prediction

At a predictor value x_0, the fitted mean response is

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0.$$

Two different uncertainties can be considered. A confidence interval estimates the population mean response $E(Y \mid x_0)$. A prediction interval describes a new individual response Y_{new} at x_0 under the same model and operating conditions.

Theorem 10.4 (Mean-Response and Prediction Intervals)

Under the classical simple linear regression assumptions, a $100(1-\alpha)\%$ confidence interval for the mean response at x_0 is

$$\hat{y}_0 \pm t_{n-2,1-\alpha/2} s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}.$$

A $100(1-\alpha)\%$ prediction interval for one new response at x_0 is

$$\hat{y}_0 \pm t_{n-2,1-\alpha/2} s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}.$$

♡

The additional 1 in the prediction standard error represents the variability of a new observation around its population mean. Consequently, a prediction interval is wider than a confidence interval for the mean response at the same x_0.

Both intervals are narrowest near $\bar{x}$ and widen as x_0 moves away from the centre of the observed predictor values. The formulas describe uncertainty under the fitted model; they do not make extrapolation far outside the observed range reliable.

Example 10.3 Mean Energy and a New Run at 70% Load

At $x_0 = 70\%$ processor load, the fitted energy use is

$$\hat{y}_0 = 10.134 + 0.4161(70) \approx 39.264 \text{ kWh}.$$

For the mean response, the estimated standard error is

$$4.1465\sqrt{\frac{1}{40} + \frac{(70 - 53.37)^2}{20809.410}} \approx 0.8114 \text{ kWh}.$$

The 95% confidence interval for mean energy use at 70% load is therefore

$$39.264 \pm 2.024(0.8114) = [37.62, 40.91] \text{ kWh}.$$

For a new individual run, the estimated prediction standard error is

$$4.1465\sqrt{1 + \frac{1}{40} + \frac{(70 - 53.37)^2}{20809.410}} \approx 4.2252 \text{ kWh},$$

giving the 95% prediction interval

$$39.264 \pm 2.024(4.2252) = [30.71, 47.82] \text{ kWh}.$$

The confidence interval addresses the mean over conceptually repeated runs at 70% load. The prediction interval is the relevant interval for one future run, whose energy use can vary substantially around that mean.

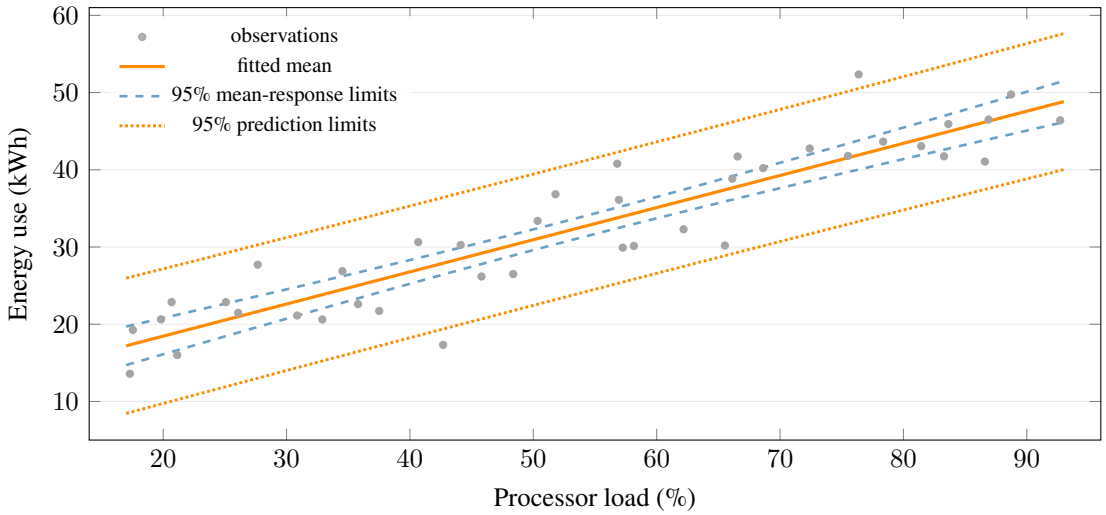


Figure 10.3: The mean-response limits quantify uncertainty about the regression line. The wider prediction limits include the additional variation of one new benchmark run.

10.7 Choosing and Reporting a Regression Analysis

A defensible analysis gives priority to the statistical question, the observational units, and the model diagnostics. Coefficients and p-values are interpreted only after the data and residuals have been examined.

Eight-step statistical regression workflow

A practical simple linear regression analysis proceeds in this order:

1. state the target population, observational unit, quantitative response, predictor, and intended operating range;
2. explain how observations were sampled or experimental settings assigned, including run order, repeated units, and possible confounding;
3. inspect the scatterplot for direction, linearity, unusual observations, clusters, and gaps in the predictor range;
4. fit the least-squares line and interpret slope and intercept with units, without assigning meaning to an implausible intercept;
5. examine residual-versus-fitted, residual-versus-order, and QQ-plots, together with leverage and influence diagnostics;
6. report the slope estimate, confidence interval, test statistic, p-value, residual standard deviation, and R^2;
7. when prediction is required, distinguish a confidence interval for the mean response from a prediction interval for a new observation and avoid unsupported extrapolation; and
8. communicate practical importance, uncertainty, model limitations, and whether the design supports association or causation.

For the benchmark experiment, a compact report could read:

> Across 40 controlled one-hour benchmark runs performed in random order, mean energy use increased linearly with processor load. The estimated increase was 0.416 kWh per percentage point, 95% CI $[0.358, 0.474]$, $t(38) = 14.48$, $p < 0.0001$. The fitted line accounted for 84.7% of the observed squared variation, and the residual standard deviation was 4.15 kWh. Residual plots showed no clear curvature or changing spread, although two runs with comparatively large residuals require audit. At 70% load, estimated mean energy use was 39.26 kWh, 95% CI $[37.62, 40.91]$, while the 95% prediction interval for one new run was $[30.71, 47.82]$ kWh. Conclusions are limited to systems and workloads adequately represented by the benchmark design.

Statistical software normally performs the calculations and produces diagnostic plots, but it cannot determine whether runs are independent, whether a predictor was controlled or merely observed, whether an unusual value is an error, or whether extrapolation is scientifically reasonable. Those decisions require knowledge of the study and engineering context.

Simple linear regression is therefore not merely a procedure for drawing a line. It is a model for a conditional mean, accompanied by residual variation, assumptions, diagnostics, and uncertainty. The next chapter turns from quantitative responses to categorical counts and develops chi-square methods for goodness of fit and association.

Chapter 11 Categorical Data and Chi-Square Tests

Engineering data are not always measurements on a numerical scale. A software incident may be classified by cause, a building component by condition, a production fault by type, and a shipment by destination and delivery status. Such categories are often summarised by counts and proportions rather than means and standard deviations.

Categorical analysis asks whether observed counts are compatible with a proposed probability model. A goodness-of-fit analysis compares one categorical distribution with specified probabilities. A contingency-table analysis compares two classifications and asks whether their distributions are associated. In both settings, Pearson's chi-square statistic measures discrepancies between observed and expected counts.

The arithmetic is compact, but a useful analysis requires more than a test statistic. The statistical question must match the sampling design, expected counts must support the chi-square approximation, and a significant result must be followed by an examination of where and how the counts differ. Effect magnitude and conditional proportions are essential because a small p-value does not by itself describe engineering importance.

This chapter develops goodness-of-fit tests with specified and estimated parameters, Monte Carlo assessment, tests of independence and homogeneity, residual interpretation, Cramér's V, sparse-table alternatives, and a practical reporting workflow. Statistical software normally performs the calculations; the objective here is to understand the quantities it reports and the conclusions they can support.

11.1 Categorical Counts and Statistical Questions

Definition 11.1 (Categorical Variable and Frequency)

A **categorical variable** assigns each observational unit to one of a set of categories. The **frequency** of a category is the number of observed units assigned to it. Dividing a frequency by the total number of units gives a sample proportion. ♣

Categories may be nominal, such as defect mechanism or material type, or ordered, such as low, medium, and high severity. The methods in this chapter use the category counts. They do not treat category labels as numerical measurements, even when the labels happen to be coded as $0, 1, 2, \ldots$.

A valid count table begins with clearly defined observational units. Each unit should contribute to exactly one category in a one-way table and exactly one cell in a two-way table. Categories should be mutually exclusive, collectively adequate for the stated population, and defined before discrepancies in the data are examined. Repeated observations from the same machine, person, site, or time series may be dependent and cannot automatically be treated as separate independent units.

Three related questions must be distinguished.

A test of independence and a test of homogeneity use the same Pearson calculation for a contingency table. Their interpretations differ because their data-generating designs differ. If one random sample of components is classified by both supplier and defect status, the question is independence. If separate samples are taken from several suppliers and their defect distributions are compared, the question is homogeneity.

Table 11.1: Three categorical questions and their data structures.

Question	Data structure	Null hypothesis
Goodness of fit	One sample classified into k categories	The population category probabilities equal a specified distribution
Independence	One sample; two categorical variables recorded on every unit	The two population classifications are independent
Homogeneity	Independent samples from several populations or groups	The populations have the same categorical distribution

Counts and proportions play complementary roles. Counts determine the sampling variation and enter the chi-square statistic. Proportions show the size and pattern of differences. Applying a chi-square test to a table containing percentages without the corresponding sample sizes loses the information needed to assess sampling variation.

11.2 Chi-Square Goodness of Fit

Suppose a sample of n independent units is classified into k categories. Let O_i be the observed count in category i. A fully specified null model gives category probabilities

$$p_{1,0}, p_{2,0}, \ldots, p_{k,0}, \qquad \sum_{i=1}^{k} p_{i,0} = 1.$$

The hypotheses are

$$H_0 : p_i = p_{i,0} \quad \text{for every } i$$

against

$$H_1 : \text{at least one population category probability differs from its specified value.}$$

Under H_0, the expected count in category i is

$$E_i = np_{i,0}.$$

Definition 11.2 (Pearson Chi-Square Statistic)

The **Pearson chi-square statistic** compares observed and expected counts:

$$\chi^2_{\text{obs}} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}.$$

Every term is nonnegative. A category contributes strongly when its observed count is far from its expected count relative to the size of that expected count.

Each summand is the square of a standardised count discrepancy,

$$\frac{(O_i - E_i)^2}{E_i} = \left(\frac{O_i - E_i}{\sqrt{E_i}}\right)^2.$$

Under H_0, category counts fluctuate around their expected counts. When the expected counts are sufficiently large, the vector of count discrepancies is approximately multivariate normal. The discrepancies are correlated

and are constrained because

$$\sum_{i=1}^{k}(O_i - E_i) = 0.$$

Consequently, only $k-1$ independent discrepancy directions remain. After accounting for the covariance among the counts, Pearson's quadratic form is asymptotically equivalent to a sum of squares of $k-1$ independent standard normal directions. Section 5.4 introduced exactly such a sum of squared standard normal variables as a chi-square distribution. This explains the $k-1$ degrees of freedom in the following approximation without requiring the categorical observations themselves to be normally distributed.

The category observations themselves are not chi-square distributed. It is the sampling distribution of Pearson's discrepancy statistic, calculated under H_0, that is approximately chi-square. The statistic equals zero only when every observed count equals its expected count. Increasing discrepancies produce larger values, so goodness-of-fit tests use the upper tail of the reference distribution.

Theorem 11.1 (Goodness-of-Fit Chi-Square Approximation)

If the null probabilities are fully specified, the observations are independent, and the expected counts are sufficiently large, then under H_0,

$$\chi^2_{\text{obs}} \quad \text{is approximately distributed as} \quad \chi^2_{k-1}.$$

The approximate p-value is

$$P\left(\chi^2_{k-1} \geq \chi^2_{\text{obs}} \mid H_0\right).$$

♡

The $k-1$ degrees of freedom reflect the fixed total count: after $k-1$ category counts are known, the final count is determined by their sum. A conservative introductory guideline is to require every expected count to be at least 5. This is a practical approximation guideline, not a mathematical law. It concerns expected counts, not observed counts.

Example 11.1 Automated Inspection Profile

Historical validation of an automated inspection system gives the following probabilities for four mutually exclusive defect classifications. A new production period contains 400 independently inspected defective components.

Category	Alignment	Contamination	Solder	Other
Specified probability $p_{i,0}$	0.45	0.30	0.15	0.10
Observed count O_i	212	96	58	34
Expected count E_i	180	120	60	40
$(O_i - E_i)^2/E_i$	5.689	4.800	0.067	0.900

All expected counts exceed 5. The test statistic is

$$\chi^2_{\text{obs}} = 5.689 + 4.800 + 0.067 + 0.900 = 11.456.$$

No parameters were estimated from the new data, so $df = 4 - 1 = 3$. The upper-tail probability is

$$p = P(\chi^2_3 \geq 11.456) \approx 0.0095.$$

At the 5% level, the specified historical profile is rejected. The largest contributions come from alignment and

contamination: alignment defects occur more often than expected and contamination defects less often. The test does not show why the profile changed, whether the inspection system or the production process changed, or whether every difference is operationally important.

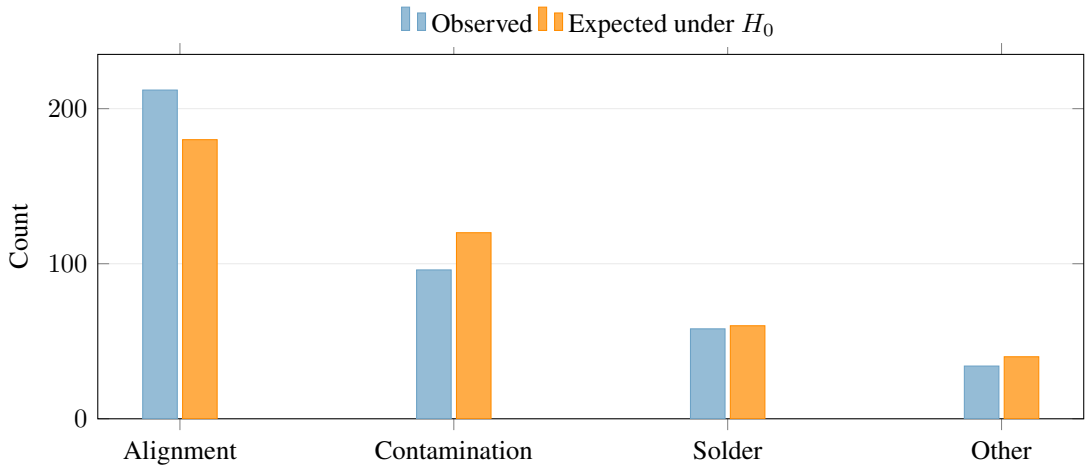


Figure 11.1: Observed and expected defect counts for the inspection-profile example. The chi-square statistic combines the category discrepancies after scaling them by their expected counts.

11.3 Estimated Parameters, Pooling, and Simulation

A goodness-of-fit null hypothesis may specify a family of distributions without specifying all its parameters. For example, the claim may be that event counts follow a Poisson distribution with an unknown rate. Estimating that rate from the same sample makes the fitted expected counts closer to the observations and consumes information that would otherwise contribute to the goodness-of-fit assessment.

Theorem 11.2 (Degrees of Freedom with Estimated Parameters)

Suppose a goodness-of-fit analysis uses k final categories and estimates m independent parameters from the same data under regular model conditions. The usual chi-square reference distribution has

$$df = k - 1 - m.$$

The parameter-estimation procedure and the expected-count calculation must correspond to the stated null model. ♡

The subtraction of m is not optional bookkeeping. Using $k - 1$ after fitting parameters from the data generally gives the wrong reference distribution and therefore the wrong p-value.

Pooling categories

Distributional models often contain tail categories with small expected counts. Adjacent or scientifically compatible categories may be pooled so that the approximation is more credible. Observed counts and null probabilities must be pooled in exactly the same way, and the number k refers to the final pooled categories.

Pooling should be based on the measurement scale and the expected counts, not on which observed discrepancies are inconvenient. The final category should include the entire remaining tail probability. Omitting rare outcomes makes the expected probabilities sum to less than one and changes the null model.

Example 11.2 Poisson Model for Service Alerts

A monitoring team records the number of service alerts in each of 200 independent 15-minute intervals. The ungrouped data give sample mean $\hat{\lambda} = 3.49$ alerts per interval. To assess a Poisson model, counts of 7 or more are pooled into one tail category.

Alerts	0	1	2	3	4	5	6	7+
Observed O_i	11	17	36	38	42	27	17	12
Expected E_i	6.10	21.29	37.15	43.22	37.71	26.32	15.31	12.90

For $i = 0, \ldots, 6$, the expected counts are

$$E_i = 200\, P(X = i)\,, \qquad X \sim \text{Pois}(3.49),$$

and the final expected count is

$$E_{7+} = 200\, P(X \geq 7).$$

The minimum expected count is 6.10. With $k = 8$ final categories and $m = 1$ estimated parameter,

$$df = 8 - 1 - 1 = 6.$$

The test gives

$$\chi^2_{\text{obs}} = 6.222, \qquad p = P(\chi^2_6 \geq 6.222) \approx 0.399.$$

The data do not provide evidence against the fitted Poisson model. This conclusion does not prove that alert counts are Poisson. Dependence across adjacent intervals, changing traffic levels, or unmodelled incident periods may still make the model unsuitable for operational use.

Monte Carlo assessment under the null model

When the chi-square approximation is doubtful, or when its adequacy is itself of interest, the null distribution of the statistic can be approximated by simulation. This preserves the same hypothesis-testing logic while replacing the theoretical reference curve with statistics generated under the null model.

Monte Carlo goodness-of-fit assessment

For an observed statistic T_{obs} and B simulated samples:

1. specify or fit the null distribution exactly as in the observed analysis;
2. simulate B samples of the same size under that null model;
3. if parameters were estimated from the observed data, repeat the parameter estimation and expected-count construction within every simulated sample;
4. calculate a statistic T_b for each simulated sample; and
5. estimate the upper-tail probability by
$$\hat{p}_{\text{MC}} = \frac{1 + \#\{b : T_b \geq T_{\text{obs}}\}}{B + 1}.$$

The added 1 prevents an estimated p-value of exactly zero and recognises that the observed dataset is one possible outcome under the null comparison. For the fully specified inspection profile in Figure 11.1, 200,000

multinomial simulations give an estimated upper-tail probability of about 0.0095, close to the chi-square approximation.

Simulation addresses the reference distribution, not defects in the study. It cannot repair dependent observations, categories chosen after inspecting the discrepancies, measurement errors, or a null model that fails to represent the scientific question.

11.4 Contingency Tables: Independence and Homogeneity

A two-way table cross-classifies observations using a row variable and a column variable. Let O_{ij} be the observed count in row i and column j, with r rows, c columns, and total sample size n.

Definition 11.3 (Contingency Table and Margins)

A **contingency table** displays counts for combinations of two categorical variables. Row totals and column totals are called the **margins**. Conditional proportions divide cell counts by the relevant row or column total and describe the categorical distribution within a group.

For an independence analysis, write p_{ij} for the population probability of cell (i, j), p_{i+} for the row probability, and p_{+j} for the column probability. The hypotheses are

$$H_0 : p_{ij} = p_{i+}p_{+j} \quad \text{for every cell}$$

against

$$H_1 : \text{the two population classifications are associated.}$$

Under independence, the expected cell count is estimated from the margins:

$$E_{ij} = \frac{(\text{row } i \text{ total})(\text{column } j \text{ total})}{n}.$$

For a homogeneity analysis, the null hypothesis states that all sampled populations have the same column-category probabilities. The expected-count formula and Pearson statistic are the same; only the sampling design and contextual wording differ.

Theorem 11.3 (Pearson Test for a Contingency Table)

For an $r \times c$ table with independent observational units and sufficiently large expected counts,

$$\chi^2_{\text{obs}} = \sum_{i=1}^{r}\sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

has, under the null hypothesis, an approximate chi-square distribution with

$$df = (r - 1)(c - 1).$$

The degrees of freedom account for the estimated row and column margins. The same conservative expected-count guideline can be used: every expected cell count should preferably be at least 5 for the introductory Pearson approximation.

Example 11.3 Defect Profiles Across Production Lines

A plant independently samples 120 defect records from each of three production lines. Each defect is classified as alignment, contamination, solder, or other.

Observed defect counts

Line	Alignment	Contamination	Solder	Other	Total
L1	25	40	15	40	120
L2	40	25	15	40	120
L3	55	10	10	45	120
Total	120	75	40	125	360

Because separate samples were taken from the three lines, the design asks whether the defect-type distributions are homogeneous. The null hypothesis states that all three lines have the same four category probabilities. For example,

$$E_{\text{L1,Alignment}} = \frac{120(120)}{360} = 40.$$

The expected table is

	Alignment	Contamination	Solder	Other
L1	40.00	25.00	13.33	41.67
L2	40.00	25.00	13.33	41.67
L3	40.00	25.00	13.33	41.67

and all expected counts exceed 5. The Pearson test gives

$$\chi^2_{\text{obs}} = 30.90, \qquad df = (3-1)(4-1) = 6, \qquad p < 0.001.$$

The data provide strong evidence that the defect-type distributions differ among the three lines. The global test does not yet identify which cells produce the discrepancy, how large the association is, or why the lines differ.

If the same 360 records instead formed one random sample from the plant and both line and defect type were recorded for every defect, the numerical test would be described as a test of independence. Its conclusion would be evidence of association between line and defect type. Neither design alone establishes that changing a component's line would change its defect mechanism; causal interpretation requires control of material, product mix, inspection, shift, and other systematic differences.

11.5 Residuals, Conditional Proportions, and Effect Magnitude

A significant global test is a starting point, not a complete interpretation. Conditional proportions display the practical pattern, residuals locate important deviations from independence or homogeneity, and an effect-size measure describes the overall strength of association.

For the production-line example, the row percentages are

	Alignment	Contamination	Solder	Other
L1	20.8%	33.3%	12.5%	33.3%
L2	33.3%	20.8%	12.5%	33.3%
L3	45.8%	8.3%	8.3%	37.5%

The alignment share rises from 20.8% on L1 to 45.8% on L3, while the contamination share falls from 33.3% to 8.3%. These conditional proportions are directly interpretable; the test statistic alone does not show this pattern.

Pearson and adjusted residuals

The signed Pearson residual for cell (i, j) is

$$r_{ij}^{\mathrm{P}} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}.$$

Its sign gives the direction of the discrepancy, and its square is the cell's contribution to the Pearson statistic:

$$\left(r_{ij}^{\mathrm{P}}\right)^2 = \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Positive residuals indicate more observations than expected; negative residuals indicate fewer.

For the production-line table, the Pearson residuals are

	Alignment	Contamination	Solder	Other
L1	−2.37	3.00	0.46	−0.26
L2	0.00	0.00	0.46	−0.26
L3	2.37	−3.00	−0.91	0.52

The largest absolute residuals occur for alignment and contamination on L1 and L3, agreeing with the conditional percentages.

Some software reports **adjusted standardised residuals**,

$$r_{ij}^{\mathrm{A}} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}(1 - \hat{p}_{i+})(1 - \hat{p}_{+j})}},$$

where $\hat{p}_{i+}$ and $\hat{p}_{+j}$ are the observed marginal proportions. Under adequate expected counts and the null model, these are approximately standard normal for an individual cell. Pearson residuals and adjusted residuals are therefore related but not identical. Software output and reports should name the quantity actually calculated.

Residual inspection is usually exploratory. The residuals in a table are correlated, and examining many cells creates a multiple-comparison issue. A value near 2 is not an automatic confirmatory discovery rule for every cell in a large table. The pattern, conditional proportions, design, and operational context should be considered together.

Cramér's V

The chi-square statistic generally increases with sample size, even when proportional differences remain similar. Cramér's V scales the statistic to a unitless descriptive measure:

$$V = \sqrt{\frac{\chi^2_{\mathrm{obs}}}{n \min(r - 1, c - 1)}}.$$

It lies between 0 and 1. A value of 0 corresponds to exact agreement with independence in the observed table, while larger values indicate stronger sample association. For a 2×2 table, Cramér's V equals the absolute value of the phi coefficient.

For the production-line example,

$$V = \sqrt{\frac{30.90}{360 \min(2,3)}} \approx 0.207.$$

This describes an overall association that is far from perfect even though the p-value is very small. There are no universal boundaries for weak, moderate, or strong values of V. Practical importance depends on the consequences of the defect shifts, baseline rates, costs, and whether the sampled records represent future production.

Cramér's V does not identify the cells involved and does not provide a causal explanation. It should be reported with conditional proportions and residual patterns, just as R^2 in regression should be accompanied by coefficients, residual diagnostics, and uncertainty.

11.6 Conditions, Alternatives, and Reporting

Sparse tables and exact alternatives

When expected counts are too small, the theoretical chi-square approximation may be inaccurate. Several responses are possible:

- combine categories only when the combined category is substantively meaningful and the rule is not chosen to hide an observed discrepancy;
- use a Monte Carlo null distribution that reproduces the complete analysis;
- for a sparse 2×2 table, use Fisher's exact test when its conditional sampling framework matches the design; or
- collect more information or report the table descriptively when no defensible inferential method is supported.

Fisher's exact test is not a licence to ignore dependence or data-dependent categories. It addresses small cell counts under a specific table model; it does not repair a weak sampling design.

For a 2×2 table comparing two independent proportions, the uncorrected Pearson chi-square statistic equals the square of the pooled two-proportion z statistic:

$$\chi^2_1 = z^2.$$

The corresponding two-sided tests therefore produce the same p-value. The two-proportion formulation emphasises a difference in proportions and its confidence interval; the contingency-table formulation emphasises association between two classifications. Running both does not create independent evidence.

Chi-square goodness of fit can also be applied to binned continuous measurements, but the result depends on the chosen bin boundaries and discards within-bin information. Quantile plots and appropriate unbinned goodness-of-fit methods are generally more informative for assessing a continuous distribution. Those procedures answer a related but distinct question and are not developed here.

Analysis and reporting order

Nine-step categorical analysis workflow

A defensible categorical analysis proceeds in this order:

1. state the target population, observational unit, categorical variables, and how units were sampled or assigned;
2. decide whether the question is goodness of fit, independence, or homogeneity;
3. present observed counts together with relevant sample or conditional proportions;
4. state H_0 and H_1 in both probability notation and contextual language;
5. calculate expected counts, identify any fitted parameters, and state the correct degrees of freedom;
6. check independence, category definitions, expected counts, and whether chi-square, simulation, or an exact alternative is justified;
7. report the test statistic, degrees of freedom, and p-value without describing the p-value as the probability that H_0 is true;
8. examine conditional proportions, cell contributions or residuals, and an effect magnitude such as Cramér's V; and
9. conclude in the language of the categories, including practical importance, design limitations, and the distinction between association and causation.

For the production-line example, a compact report could read:

> Defect-type distributions differed among the three sampled production lines, $\chi^2(6) = 30.90$, $p < 0.001$, with Cramér's $V = 0.207$. Alignment defects accounted for 20.8%, 33.3%, and 45.8% of sampled defects on L1, L2, and L3, respectively, while contamination defects accounted for 33.3%, 20.8%, and 8.3%. Pearson residuals located the largest discrepancies in those line-by-defect combinations. The analysis establishes heterogeneity of the sampled defect profiles, not its production mechanism; material mix, product type, shift, and inspection procedures require separate investigation.

Reporting only that "the chi-square test was significant" conceals the pattern and magnitude needed for a decision. Conversely, highlighting a large cell residual without the global table, conditional proportions, and design can exaggerate an isolated discrepancy. The test, follow-up summaries, and engineering context have distinct roles.

Categorical analysis therefore follows the same general discipline as the preceding methods. The study design defines the question, the null model defines expected behaviour, diagnostics and effect magnitudes interpret departures, and uncertainty limits the conclusion. The next chapter integrates these principles across quantitative and categorical analyses and develops a reusable method-selection and reporting workflow.

Chapter 12 Planning and Reporting a Statistical Analysis

A statistical method is only one link in a longer chain of reasoning. An engineering analysis begins with a decision or question, continues through study design and data collection, and ends with an interpretation that acknowledges uncertainty and limitations. A technically correct calculation cannot repair a vague question, dependent observations treated as independent, an invalid measurement process, or a conclusion that extends beyond the sampled conditions.

The preceding chapters developed individual tools for description, estimation, comparison, modelling, and categorical analysis. This chapter introduces no new probability distribution or testing procedure. Instead, it brings those tools together into a coherent workflow. The objective is to select analyses because they answer a well-defined question, to connect every result to the way the data were obtained, and to communicate evidence in a form that can support an engineering decision.

A complete analysis is not the analysis containing the greatest number of methods. It is the analysis in which every table, plot, estimate, interval, and test has a clear purpose. Often one primary method, supported by careful descriptive analysis and diagnostics, provides a stronger argument than a collection of unrelated calculations.

12.1 From an Engineering Question to a Statistical Question

Engineering questions are commonly expressed in terms of actions: whether a design should be adopted, whether a process is stable enough, whether a supplier meets a requirement, or whether two operating strategies perform differently. Statistical analysis requires those questions to be translated into quantities that could vary from sample to sample.

The translation begins by identifying six elements:

1. the **decision or purpose** that motivates the analysis;
2. the **target population or operating conditions** to which the conclusion should apply;
3. the **observational unit**, such as a component, building, shipment, user request, production run, or time period;
4. the **response and explanatory variables**, including their units and how they are measured;
5. the **population quantity** to be estimated or compared, such as a mean, proportion, mean difference, slope, or set of category probabilities; and
6. the **reference value or practical threshold** against which the result will be interpreted.

The population quantity must match the decision. If the concern is the expected lifetime of a component, the population mean may be relevant. If the concern is the proportion of components that fail before a warranty limit, a population proportion is more directly connected to the decision. If two processes are compared, the target is a difference rather than two unrelated summaries.

Question formulation before analysis

Before calculating a statistic, state the analysis in one sentence:

> For *[target population]*, estimate or test *[population quantity]* using *[observational units and variables]*, and interpret it relative to *[engineering reference or decision threshold]*.

This sentence does not determine the method by itself, but it exposes missing information before software produces an apparently precise answer.

Example 12.1 Converting a Design Objective into a Statistical Question

A production manager asks whether a new calibration procedure is "better." The word better is not yet measurable. If the primary concern is dimensional accuracy, a suitable statistical question could be:

> For components produced under normal operating conditions, estimate the change in mean absolute dimensional error when the same machine uses the new rather than the current calibration procedure.

The machine or production run must then be identified as the observational unit, the order of the two procedures must be controlled, and the size of a practically valuable reduction must be stated. A separate question would be required if "better" instead meant a lower defect probability, less variability, shorter setup time, or lower cost.

A statistical question should normally be chosen before detailed results are inspected. Questions created after seeing an unusual pattern can be useful for exploration, but they should be labelled as exploratory and investigated with new data when the decision is important.

12.2 Data Audit and Exploratory Analysis

The first analysis of a dataset is an audit rather than a hypothesis test. Its purpose is to determine what each row represents, whether the recorded values are credible, and whether the structure of the dataset agrees with the intended study design. This step often requires more engineering knowledge than statistical calculation.

Table 12.1: Core questions in a statistical data audit.

Audit element	Questions to answer
Provenance	Who or what generated each observation, under which protocol, and during which operating period?
Units and coding	Are numerical units consistent, and are category labels defined and mutually exclusive?
Observational units	Does each row represent an independent unit, a repeated measurement, a matched unit, or an aggregate?
Completeness	Which values are missing, censored, excluded, or unavailable, and could the mechanism be related to the outcome?
Validity	Are values within physically and logically possible ranges, and are duplicated identifiers or impossible combinations present?
Order and dependence	Could time order, spatial proximity, batches, users, machines, buildings, or repeated runs create dependence?
Comparability	Were groups measured with the same instruments, definitions, timing, and operating conditions?
Traceability	Can every transformation, exclusion, and derived variable be explained and reproduced?

An unusual value is not automatically an error. It may be a transcription mistake, a sensor failure, a rare but valid operating condition, or evidence that the assumed model is incomplete. The value should be investigated using the measurement record and engineering context. Deleting it solely because it weakens a desired conclusion changes the question after observing the answer.

Missing observations require similar care. Removing incomplete rows is convenient, but it can distort an analysis when missingness is related to the response. For example, failed sensors may be precisely the units with the most extreme measurements, and users experiencing the longest response times may abandon a request before the time is recorded. The number and pattern of missing values should therefore be reported, together with the rule used to handle them.

After the structural audit, descriptive statistics and plots reveal the empirical shape of the data. Frequencies and bar charts describe categorical variables. Means, medians, standard deviations, quartiles, histograms, boxplots, and quantile plots describe quantitative variables. Scatterplots examine relationships between quantitative variables, while side-by-side plots or plots of paired differences preserve the structure of comparisons. The methods of chapter 1 are not preliminary decoration; they determine whether later summaries are meaningful.

Exploration can reveal skewness, clusters, nonlinearity, changing variation, or possible dependence. It can also suggest new scientific questions. It cannot by itself distinguish a reproducible pattern from one found because many features were inspected. The analysis should therefore distinguish observations that were anticipated from patterns discovered during exploration.

Data audit before formal modelling

A defensible analysis should be able to answer the following questions:

1. What does one row represent?
2. Which observations can reasonably be treated as independent?
3. Are units, category definitions, and measurement procedures consistent?
4. How many values are missing or excluded, and why?
5. Which observations are unusual, and what does the source record show?
6. Do plots support the numerical summaries and the proposed model?
7. Which transformations or derived quantities were created before analysis?

12.3 Selecting an Appropriate Method

Method selection begins with the population quantity and the structure of the observations, not with a menu of procedures. The same dataset can support different questions, and the same question can require different methods under different designs. Two columns of measurements, for example, may represent independent groups, paired observations, or a response and predictor. Those structures lead to different analyses.

Table 12.2: A question-based guide to the core methods developed in this book.

Statistical objective	Data structure and target quantity	Core method
Describe observed quantitative data	One or more quantitative variables; empirical centre, spread, shape, or relationship	Descriptive statistics and plots, chapter 1
Describe uncertainty before data are observed	Specified random experiment or random variable	Probability models, chapter 2–chapter 4
Estimate one population mean or proportion	One quantitative or binary sample; μ or p	Confidence interval, chapter 6
Assess one population claim or threshold	One quantitative or binary sample; comparison with μ_0 or p_0	One-sample test, chapter 7
Compare two populations or conditions	Independent groups or paired observations; mean or proportion difference	Two-group procedures, chapter 8
Compare several population means	Quantitative response and one categorical factor with at least three levels	Analysis of variance, chapter 9
Describe a linear quantitative relationship	Quantitative response and predictor; slope, mean response, or prediction	Simple linear regression, chapter 10
Compare categorical counts	One categorical distribution or a table of two classifications	Chi-square procedures, chapter 11

The table identifies a starting point, not an automatic rule. A method is justified only when its assumptions match the design and the intended interpretation. A two-sample procedure cannot be used merely because two groups are displayed. Regression cannot establish causation merely because a slope differs from zero. Analysis of variance does not explain which groups differ without planned comparisons or suitable follow-up analysis. A chi-square test does not repair categories that were created after the counts were inspected.

Method-selection questions

Before choosing a formal procedure, determine:

1. whether the response is quantitative, binary, or categorical;
2. whether the objective is description, estimation, comparison, association, or prediction;
3. whether observations are independent, paired, repeated, clustered, or ordered in time;
4. how many populations, conditions, or explanatory variables are involved;
5. which population quantity directly answers the engineering question; and
6. whether the available design supports description, association, generalisation, or a causal conclusion.

One primary analysis should normally correspond to the primary question. Supplementary analyses may clarify mechanism, robustness, or practical relevance, but each needs its own purpose. Repeating several methods until one produces a small p-value weakens the evidential interpretation because the selected result no longer represents the originally stated question.

The direction of a difference must also be fixed. If improvement is defined as $\mu_1 - \mu_2 > 0$, the order should remain unchanged in estimates, confidence intervals, tests, tables, and prose. Switching the order halfway through an analysis is a common source of incorrect conclusions.

12.4 Assumptions and Analysis Strategy

Assumptions describe the conditions under which a statistical result has its stated interpretation. Some concern a probability distribution, but many concern the data-generating process and cannot be checked by a plot or a test.

Table 12.3: Assumptions connect statistical calculations to the study that generated the data.

Assumption family	Typical concern	Main source of evidence
Representation	The observations may not describe the intended population or operating range.	Sampling mechanism, inclusion rules, and engineering scope
Independence	Repeated units, batches, sites, users, or time order may create dependence.	Study design, identifiers, run order, and domain knowledge
Measurement	Calibration, resolution, censoring, or changed definitions may create systematic error.	Measurement protocol, instrument records, and audit trail
Distributional form	Skewness, heavy tails, outliers, nonlinearity, or unequal spread may conflict with a model.	Histograms, boxplots, quantile plots, scatterplots, and residual diagnostics
Approximation	A sample or expected count may be too small for an asymptotic reference distribution.	Sample sizes, expected counts, simulation, and method-specific conditions

No single diagnostic produces a certificate that all assumptions are true. A normal probability plot addresses distributional shape, not independence or representative sampling. A large sample can make some approximations accurate, but it does not remove bias, dependence, confounding, or invalid measurement. Conversely, a minor visible departure from a model need not make a procedure useless when the method is robust and the

resulting conclusion is not sensitive to that departure.

An analysis strategy should be stated before the final calculations when possible. A **confirmatory** analysis evaluates a question, direction, and method chosen independently of the observed result. An **exploratory** analysis searches for patterns that may motivate later investigation. Both are valuable, but they carry different evidential weight. Presenting an exploratory discovery as if it had been specified in advance exaggerates its reliability.

The distinction becomes especially important when many variables, subgroups, models, or tests are examined. Even under null models, some estimates will appear large and some p-values will appear small by chance. The remedy is not to hide the exploration. The report should identify the primary analysis, describe the extent of supplementary analysis, and treat unplanned patterns as hypotheses for future data unless an appropriate multiplicity strategy was specified.

Analysis strategy before calculation

A concise analysis strategy records:

1. the primary statistical question and population quantity;
2. the practical reference value or decision threshold;
3. the sampling or assignment structure and observational unit;
4. the descriptive displays and primary inferential method;
5. the assumptions and diagnostics that will be examined;
6. the treatment of missing values, unusual observations, and exclusions; and
7. any planned supplementary or sensitivity analyses.

Statistical software can execute this strategy, but it cannot supply its scientific justification. It can calculate every test available for a dataset without knowing which question was asked, which rows belong to the same unit, or whether a difference matters in practice.

12.5 Interpreting the Complete Result

A complete statistical result combines magnitude, uncertainty, model evidence, diagnostics, and practical context. None of these components should be interpreted in isolation.

Table 12.4: Complementary components of a statistical interpretation.

Component	Question it answers
Point estimate	What magnitude and direction are suggested by the observed sample?
Confidence interval	Which population values remain reasonably compatible with the estimate and its sampling uncertainty?
Test statistic and p-value	How extreme is the observed statistic, or one more extreme in the direction specified by H_1, if H_0 and the model assumptions are true?
Effect magnitude	Is the estimated difference, association, or variation important in engineering units or relative terms?
Diagnostics	Do the data reveal important conflict with the design or model assumptions?
Sensitivity	Does a defensible alternative treatment of assumptions or unusual observations materially change the conclusion?

A small p-value does not measure the probability that the null hypothesis is true, and it does not measure the size of an effect. With a sufficiently precise study, an operationally negligible difference can produce a small p-value. With a small or highly variable sample, an important difference can remain uncertain. Estimates and confidence intervals therefore carry information that a threshold decision based on the p-value alone conceals.

Practical importance should be stated in the units of the problem whenever possible. A reduction of 12 milliseconds, 0.8 percentage points, 1.4 kWh per cycle, or 3.2 MPa can be compared with service requirements, safety margins, cost, or tolerances. Standardised effects and relative measures may supplement that interpretation, but they should not replace a domain-relevant scale.

Uncertainty also constrains decisions. A confidence interval that contains both a negligible effect and a valuable effect indicates that the data do not yet distinguish between those possibilities. This is different from evidence that the effect is exactly zero. The appropriate response may be to collect more information, improve the design, or make a reversible decision while monitoring future performance.

The conclusion must remain within the scope of the study. Random assignment can support a causal comparison within the experimental conditions. Representative sampling can support generalisation to a target population. An observational association, however precisely estimated, does not by itself establish a causal mechanism. An experiment on one hardware platform does not automatically establish performance on every platform.

12.6 Communicating Conclusions and Limitations

A statistical report should allow another reader to reconstruct the argument from question to recommendation. The main result should not be hidden behind a sequence of software outputs, and the report should not require the reader to guess which analysis was primary.

Seven-step analysis-to-recommendation workflow

A clear report proceeds in this order:

1. state the engineering objective, target population, observational unit, and primary statistical question;
2. describe how observations were sampled, assigned, measured, paired, or grouped;
3. report sample sizes, missingness, exclusions, and the most relevant descriptive statistics and plots;
4. name and justify the primary method, including the direction of comparisons and the assumptions examined;
5. report estimates, confidence intervals, and any test statistic, degrees of freedom, and p-value required by the question;
6. interpret magnitude, uncertainty, diagnostics, and practical significance in context; and
7. state a proportionate recommendation together with specific limitations and any need for further evidence.

Tables and figures should each answer a question. A caption should state what is displayed and, when appropriate, identify the main pattern without claiming more than the display establishes. Decorative plots, duplicated summaries, and unlabelled software screenshots make the evidence harder to assess. Axes require quantities and units; tables require meaningful precision and sample sizes.

Limitations should be specific. Phrases such as “more data are needed” or “the sample may be biased” are too vague unless the report explains which population is under-represented, which dependency is unresolved, which operating conditions were absent, or which effect remains too imprecise. A limitation is most useful when its likely consequence for the conclusion is described.

Recommendations should use language proportional to the evidence. Data may *support* a process change, *suggest* a relationship, show that a result is *consistent with* a specified effect, or provide *insufficient evidence* to distinguish relevant possibilities. Words such as *prove*, *guarantee*, and *cause* require conditions that most observational studies cannot provide.

Example 12.2 From Software Output to an Engineering Statement

The statement “the test was significant, $p = 0.012$” omits the direction, magnitude, uncertainty, method, and population. A more informative statement is:

> For the 48 independently sampled production runs, the modified setting reduced mean cycle time by an estimated 3.6 seconds relative to the current setting, 95% CI $[0.8, 6.4]$ seconds, using Welch’s two-sample procedure. The interval excludes zero but includes reductions smaller than the pre-specified five-second operational target. The data therefore support a real average reduction under the sampled conditions, but do not establish that the target improvement is achieved.

The second statement separates statistical evidence from the engineering threshold and makes the remaining uncertainty visible.

12.7 Integrated Worked Example

This example illustrates how the preceding elements form one statistical argument. It deliberately uses only the methods needed for the question.

Engineering objective and study design

A building-services team is considering a new control algorithm for commercial heat-pump systems. Under a standard two-hour operating cycle, the new algorithm should reduce mean electrical energy use by more than 1.0 kWh per system. Twelve systems are sampled from the service portfolio. Each system completes the same controlled cycle once with the current algorithm and once with the proposed algorithm; the configuration order is randomised within each system.

Let

$$D_i = X_{i,\text{current}} - X_{i,\text{proposed}},$$

so that a positive difference represents an energy reduction. The population quantity is the mean paired reduction μ_D for systems and operating conditions represented by the sampling and test protocol. The primary hypotheses are

$$H_0 : \mu_D \leq 1.0 \qquad \text{against} \qquad H_1 : \mu_D > 1.0.$$

The direction and the 1.0 kWh threshold are determined by the engineering requirement, not by the observed data.

The same systems are measured under both algorithms, so the observations are paired. Treating the two columns as independent samples would discard the within-system connection and use the wrong sampling uncertainty. Randomising configuration order reduces the risk that warm-up, battery state, or environmental drift is systematically associated with one algorithm.

Audit and descriptive evidence

The recorded energy use is shown in Table 12.5. System identifiers are unique, the units are consistent, and no observations are missing. The protocol record confirms that each pair comes from one system under the two configurations.

Table 12.5: Energy use in kWh during the standard operating cycle.

System	Current	Proposed	Difference D
1	18.4	17.6	0.8
2	19.1	18.0	1.1
3	20.3	19.9	0.4
4	21.0	19.5	1.5
5	21.7	20.8	0.9
6	22.4	20.7	1.7
7	23.1	22.1	1.0
8	24.0	22.1	1.9
9	25.2	23.9	1.3
10	26.1	23.9	2.2
11	27.3	25.5	1.8
12	28.0	25.6	2.4

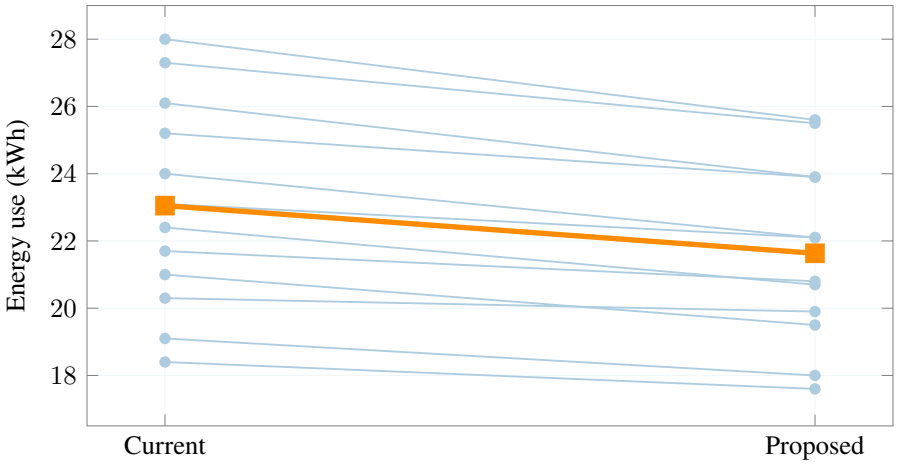


Figure 12.1: Paired energy measurements for the twelve systems. Thin blue lines preserve the within-system pairing; the thick orange line connects the two sample means.

All twelve differences are positive, ranging from 0.4 to 2.4 kWh. Their median is 1.4 kWh, and no difference is isolated from the remaining values. The sample summaries are

$$\overline{X}_{\text{current}} = 23.050, \qquad \overline{X}_{\text{proposed}} = 21.633,$$

and

$$\overline{D} = 1.4167, \qquad S_D = 0.6043, \qquad n = 12.$$

The raw means describe the two configurations, while the mean and standard deviation of the differences determine the paired inference.

Primary inference

The paired differences are observations from twelve different systems. Their ordered values show no extreme outlier or severe asymmetry, so a one-sample t procedure for the mean difference is a reasonable model under the sampling and protocol assumptions. The estimated standard error is

$$\mathrm{SE}(\overline{D}) = \frac{S_D}{\sqrt{n}} = \frac{0.6043}{\sqrt{12}} = 0.1744 \text{ kWh}.$$

The test statistic compares the observed mean reduction with the practical threshold of 1.0 kWh:

$$t_{\mathrm{obs}} = \frac{\overline{D} - 1.0}{S_D/\sqrt{n}} = \frac{1.4167 - 1.0}{0.1744} = 2.389.$$

With 11 degrees of freedom, the one-sided p-value is

$$p = P(T_{11} \geq 2.389) \approx 0.018.$$

At the boundary value $\mu_D = 1.0$ and under the model assumptions, a sample mean at least this far above the boundary, relative to its estimated standard error, would occur with probability about 0.018. For mean reductions below 1.0 kWh, that upper-tail probability would be smaller. The boundary therefore determines the p-value for the composite null hypothesis.

A two-sided 95% confidence interval for the mean reduction is

$$1.4167 \pm t_{0.975,11}(0.1744) = 1.4167 \pm 2.201(0.1744) = [1.033, 1.801] \text{ kWh}.$$

The interval lies above zero and, more importantly for this decision, above the pre-specified 1.0 kWh requirement. It also shows the remaining uncertainty: the population mean reduction could plausibly be only slightly above the requirement or as large as approximately 1.8 kWh under the model.

Interpretation, limitations, and recommendation

The analysis supports a mean energy reduction greater than 1.0 kWh per standard cycle for systems and conditions represented by the study. The evidence concerns a population mean. It does not imply that every individual system saves more than 1.0 kWh; three of the twelve observed reductions are below that value, and one equals it.

The paired and randomised-order design strengthens the comparison between algorithms under the controlled cycle. Important limitations remain. The sample contains only twelve systems, long-term seasonal operation was not observed, and the controlled cycle may not represent every building load or maintenance condition. The result also concerns energy use; reliability, comfort, installation cost, and control stability require separate evidence.

A compact final statement could read:

> Across twelve sampled heat-pump systems tested under both configurations in randomised order, the proposed control algorithm reduced mean energy use by 1.42 kWh per standard cycle, 95% CI $[1.03, 1.80]$ kWh. Relative to the pre-specified 1.0 kWh requirement, the paired test gave $t(11) = 2.39$, one-sided $p = 0.018$. The paired differences showed no extreme value or severe asymmetry. These results support a mean reduction exceeding the requirement under the tested conditions. A staged implementation with continued field monitoring is warranted because the sample is small and does not cover long-term seasonal operation, reliability, or every system type in the service portfolio.

No analysis of variance, chi-square test, or regression is added to this example because none is required by the primary question. Such methods would be justified only by additional questions and suitable data. The integration lies in the connection among the decision threshold, paired design, audit, descriptive evidence, model assumptions, estimate, uncertainty, and recommendation.

A strong statistical analysis therefore forms a traceable argument. The engineering question determines the population quantity; the study design determines which comparisons are valid; the data audit and plots reveal what was actually observed; the statistical method quantifies magnitude and uncertainty; and the conclusion states what the evidence supports, what it does not support, and which decision remains reasonable.

References

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